

Observational Signatures of Black Hole Shadows and Black Hole Phase Transitions in Different Spacetime Backgrounds



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Zhi Luo

Supervised by Prof. Jin Li

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Abstract

Black hole research is transitioning from an era dominated by purely theoretical derivations into a new epoch characterized by a profound integration of theoretical investigations and precise observational data. The successful detection of gravitational waves, along with the first observational imaging of shadows of the supermassive black hole M87* and Sagittarius A* by the Event Horizon Telescope, has provided crucial observational evidence for testing General Relativity and its potential modifications in extreme gravitational environments. In this thesis, we systematically investigate the observational appearances of black holes and wormholes in different spacetime backgrounds, the holographic Einstein rings associated with black holes, and the shadow features and dynamical properties related to black hole phase transitions, thus laying a robust theoretical foundation for future high-precision astronomical observations.

Firstly, we study the photon dynamics in black hole spacetimes within the framework of Horndeski theory. By utilizing observational data from the Event Horizon Telescope for M87* and Sagittarius A*, we tightly constrain the parameter space of Horndeski theory to a relatively small range of $-0.46 \lesssim \gamma \lesssim 0.31$. Through detailed analyses of the observational appearance of black holes surrounded by thin accretion disks, we find that the current resolving capability of the Event Horizon Telescope is only sufficient to capture direct emissions of black holes but inadequate for distinguishing fine structures such as lensing rings and photon rings. Given that black holes are not the only conceivable compact objects, this thesis further explores the observational signatures of asymmetric thin-shell wormholes in the Horndeski theory. The results indicate that, compared to black holes with identical mass parameters, wormholes exhibit distinctive additional observational features such as "lensing bands" and "photon ring groups," providing theoretical criteria to distinguish black holes from other compact astrophysical objects.

Secondly, employing the AdS/CFT correspondence and wave optics methodologies, we thoroughly examine the imaging characteristics of holographic Einstein rings in the AdS black hole spacetime in Horndeski theory. The results demonstrate that observational features of holographic Einstein rings depend significantly on the observer's location, black hole parameters, properties of the wave source, and configuration of the optical observation system. When the observer is positioned at the north pole on the AdS boundary, the holographic image exhibits a ring-like structure closely coincident with the black hole shadow. Additionally, we conduct a detailed analysis of how observational location, event horizon radius, temperature, and Horndeski theory parameters affect the

holographic Einstein ring. Variations in these parameters not only influence the radius and brightness of the ring but also induce distinct structural distortions, potentially transforming the ring into luminous spot-like features. By extensively comparing wave optics and geometric optics methods, we validate that these two approaches exhibit high consistency in predicting the positions and intensities of photon rings and Einstein rings. These findings not only enhance our understanding of black hole spacetime structure in Horndeski theory but also present a novel theoretical framework for exploring gravitational dualities in strongly coupled systems.

Furthermore, we investigate the phase structure of Hayward-AdS black holes by adopting the shadow radius as a characteristic parameter instead of the event horizon radius. The results reveal a clear monotonic relationship between the shadow radius and the event horizon radius, indicating that the shadow radius effectively describes the thermodynamic behavior of Hayward-AdS black holes. Specifically, we show that as magnetic charge increases, the black hole shadow radius grows substantially, whereas the coexistence temperature of the phase transition significantly decreases. By constructing thermal profile diagrams that combine temperature distributions and shadow radii, we demonstrate that under constant magnetic charge, the temperature of Hayward-AdS black holes rises with increasing pressure, while the corresponding thermal profile region shrinks. Below the critical temperature, the black hole temperature displays a typical N -type variation pattern. This finding clearly establishes the shadow radius as a novel and effective parameter for probing the thermodynamics of Hayward-AdS black holes.

Finally, to gain deeper insight into the mechanisms underlying black hole phase transitions, we utilize the Gibbs free energy landscape method to study the dynamics of the charged AdS black hole with global monopole, revealing the crucial influence of the global monopole on black hole phase transition dynamics. Due to thermal fluctuations, dynamical transitions occur between small and large black hole phases. By solving the Fokker-Planck equation with appropriate boundary conditions, we analyze the evolution characteristics of black hole state probabilities over time. Our results show that the presence of a global monopole markedly delays the phase transition dynamics for both small and large black hole phases. Furthermore, we obtain quantitative relations between the first passage time of black hole phases and the global monopole parameter, showing that increasing the global monopole parameter significantly prolongs the mean first passage times for both phases, quantitatively highlighting the regulatory role of the global monopole on black hole phase transition dynamics.

In summary, through systematic studies of black hole shadows, phase transitions, and dynamical behaviors in various gravitational backgrounds, this thesis proposes essential observational signatures and theoretical methods for deeply understanding black

hole properties in strong gravitational fields, providing a robust theoretical basis for future precision astronomical observations aimed at testing and extending General Relativity.

Keywords: Black hole shadow; Asymmetric thin-shell wormhole; Holographic Einstein ring; Thermodynamic phase transition

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Chapter 1 Introduction

Black holes teach us that space can be crumpled like a piece of paper into an infinitesimal dot, that time can be extinguished like a blown-out flame, and that the laws of physics we once regarded as sacrosanct are not beyond question.

— John Archibald Wheeler

1.1 Introduction

Black holes are among the most important and mysterious objects in modern physics. As compact objects predicted by general relativity, they possess gravitational fields so strong that even photons cannot escape once they cross the event horizon. During the past century, black holes have evolved from theoretical conjectures into firmly established astronomical objects and have become a central subject of research. Their significance lies not only in their remarkable intrinsic properties, but also in the fact that they provide natural laboratories for studying gravitational theory, spacetime structure, and physics under extreme conditions^[1-3].

In recent years, black-hole research has achieved a series of major breakthroughs. In 2015, the Laser Interferometer Gravitational-Wave Observatory (LIGO) Scientific Collaboration directly detected gravitational waves from a binary black-hole merger for the first time^[4-6], providing a powerful test of general relativity. Subsequent gravitational-wave detections^[4-10] marked the birth of gravitational-wave astronomy. In 2019, the Event Horizon Telescope (EHT) obtained the first direct image of the black-hole shadow of the supermassive black hole at the center of M87^[11-16]. In 2022, the EHT further released the image of the supermassive black hole at the center of the Milky Way^[17-22]. These images, showing dark central regions surrounded by bright rings, provide strong evidence for supermassive black holes in galactic centers. The dark region is usually called the black-hole shadow^[23]. Black-hole images provide information about accretion flows, jets, and matter near the event horizon, and allow us to infer key physical parameters such as mass, spin, and charge. Together with gravitational waves and electromagnetic observations, EHT imaging has moved black-hole research into a genuine multi-messenger era.

Many fundamental questions in black-hole physics remain unresolved. In general relativity, a spacetime singularity appears at the center of a black hole, where curvature diverges and the classical theory ceases to be adequate; a quantum theory of gravity is expected to be necessary^[24]. Hawking's prediction that black holes radiate thermally through quantum effects^[25] leads to the celebrated black-hole information paradox^[26]:

does black-hole evaporation violate the conservation of information in quantum mechanics? This question remains one of the central tensions between general relativity and quantum theory. In astrophysics, the rapid formation and growth of supermassive black holes in the early universe, and the abundance and origin of intermediate-mass black holes, are still open problems.

Against this background, this dissertation studies the observational appearance of black holes in Horndeski theory and compares it with the appearance of asymmetric thin-shell wormholes, which may act as black-hole mimickers. Going beyond conventional geometric optics, we also use wave optics to construct holographic Einstein-ring images of black holes and thus study black-hole imaging from a complementary perspective. Finally, we investigate how black-hole shadows can be connected to thermodynamic phase transitions and how nonequilibrium phase-transition dynamics can be characterized for black-hole systems.

1.2 The Birth of General Relativity

In 1687, Isaac Newton published *Philosophiae Naturalis Principia Mathematica*, in which he formulated the law of universal gravitation for massive bodies. The law can be written as

$$F = G \frac{m_1 m_2}{r^2}, \quad (1.1)$$

where G is Newton's gravitational constant, m_1 and m_2 are the masses of the two bodies, and r is their separation. Newtonian gravity successfully explained Kepler's laws of planetary motion and many other astronomical phenomena.

By the middle of the nineteenth century, however, astronomical observations posed a challenge to Newtonian gravity. The perihelion precession of Mercury differed from the classical prediction. Urbain Le Verrier's 1859 analysis found an excess precession of about 38 arcseconds per century, later refined to about 43 arcseconds per century. Attempts to explain the discrepancy by invoking an unseen planet inside Mercury's orbit failed, leaving the anomaly as a serious problem for classical gravity.

Newtonian mechanics assumes absolute space and absolute time: a fixed spatial background and a universal flow of time independent of matter. In the late nineteenth century, Ernst Mach criticized this notion of absolute space. Mach's principle suggested that inertia might not be defined relative to absolute space, but rather relative to the distribution of all matter in the universe. This idea influenced Einstein's search for a new understanding of gravitation, in which gravitational phenomena would arise from the interaction between matter and spacetime rather than from a force acting within an absolute background.

Classical mechanics also revealed the equivalence between gravitational mass and inertial mass. Galileo's free-fall experiments implied that bodies fall with the same acceleration in a given gravitational field, independent of their mass. Newton's pendulum experiments supported the same conclusion, and Lorand Eotvos later improved the precision of equivalence-principle tests to the level of 10^{-8} . These facts provided the empirical basis for Einstein's equivalence principle.

The equivalence principle is the cornerstone of general relativity. In 1907, Einstein realized that, within a sufficiently small region, the effects observed in a uniformly accelerated frame are indistinguishable from those in a uniform gravitational field^[27]. A freely falling elevator, for example, locally removes the sensation of gravity, while an accelerating elevator in empty space reproduces the effects of a gravitational field. Because all bodies acquire the same acceleration in a gravitational field, a local observer cannot distinguish gravity from acceleration by local mechanical experiments.

Special relativity is based on the principle that the laws of physics take the same form in all inertial frames. General relativity extends this idea to arbitrary coordinate systems and requires the laws of physics to be generally covariant^[28]. This requirement eliminates any privileged inertial frame and demands a geometric formulation of gravity. In this formulation, gravity is not a conventional force but a manifestation of curved spacetime.

Einstein expressed the relation between spacetime geometry and matter through the field equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}. \quad (1.2)$$

Here, $R_{\mu\nu}$ is the Ricci tensor, R is the Ricci scalar, $g_{\mu\nu}$ is the metric tensor, $T_{\mu\nu}$ is the energy-momentum tensor, G is Newton's constant, and c is the speed of light. The equation states that matter and energy determine the curvature of spacetime, while curved spacetime determines the motion of matter and light through geodesics.

General relativity immediately explained the anomalous perihelion precession of Mercury. The curvature of spacetime caused by the Sun yields an additional precession of about 43 arcseconds per century, in agreement with observation^[29]. The theory also predicts the deflection of light by gravity. Einstein's final calculation gave a deflection of about 1.75 arcseconds for starlight grazing the solar limb, and Eddington's 1919 solar-eclipse observations confirmed this prediction^[30]. These successes established general relativity as the modern theory of gravity.

1.3 Black Holes

1.3.1 The Origin of the Black-Hole Concept

A black hole is an astronomical object with such a strong gravitational field that, under certain conditions, even light cannot escape. Its boundary is called the event horizon; once matter or radiation crosses it from the outside, it can no longer return. Since Einstein proposed general relativity in 1915, black holes have been among the most fascinating and challenging solutions of the theory. Yet ideas resembling black holes had already appeared within Newtonian gravity.

Following Newton's law of gravitation and the corpuscular theory of light, John Michell proposed the idea of a "dark star" in 1783. He argued that if a star were sufficiently massive and compact, its gravitational field could prevent light from escaping. Pierre-Simon Laplace made a similar proposal in 1796, suggesting that some luminous bodies might be invisible because their gravity trapped their own light. These arguments anticipated the black-hole concept, although they were framed within Newtonian physics.

To estimate the condition for a dark star, let a star have mass M and radius r , and choose the gravitational potential energy to vanish at infinity. If a photon cannot escape, its kinetic energy must not exceed the gravitational binding energy:

$$\frac{1}{2}mc^2 \leq \frac{GMm}{r}. \quad (1.3)$$

This gives

$$r \leq \frac{2GM}{c^2}. \quad (1.4)$$

The derivation is not physically correct in a modern sense, because it treats light as a Newtonian particle and uses Newtonian gravity rather than general relativity. Nevertheless, the resulting radius coincides with the Schwarzschild radius.

The first exact black-hole solution of Einstein's field equations was obtained by Karl Schwarzschild in 1916. It describes the exterior gravitational field of a static, spherically symmetric, nonrotating mass. Assuming spherical symmetry, stationarity, and vacuum outside the source, the line element can be written as

$$ds^2 = -e^{\nu(r)}c^2dt^2 + e^{\lambda(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (1.5)$$

where $\nu(r)$ and $\lambda(r)$ are radial functions. Requiring asymptotic flatness as $r \rightarrow \infty$ and solving Eq. (1.2) gives

$$e^{-\lambda(r)} = 1 - \frac{2GM}{c^2r}, \quad e^{\nu(r)} = 1 - \frac{2GM}{c^2r}. \quad (1.6)$$

In natural units, $G = c = 1$, the Schwarzschild metric becomes

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (1.7)$$

The surface $r = 2M$ is the event horizon. Unless otherwise stated, natural units are used throughout this dissertation.

The Schwarzschild solution is one of the most important exact solutions of general relativity. The Schwarzschild radius of the Sun is about 3 km, while that of the Earth is about 9 mm, illustrating the enormous density required for ordinary astrophysical bodies to become black holes. As general relativity developed, black holes came to be understood not merely as mathematical curiosities, but as possible end states of stellar evolution. Oppenheimer and collaborators studied gravitational collapse in 1939^[31], and Penrose proved in 1965 that, under reasonable physical conditions, gravitational collapse generically leads to spacetime singularities^[32]. The term “black hole” was introduced by John Archibald Wheeler in 1967.

1.3.2 Types of Black Holes

In general relativity, black holes can be classified by three basic parameters: mass, angular momentum, and charge. According to the no-hair theorem, the spacetime geometry of a classical stationary black hole is completely characterized by these quantities. The four standard types are as follows.

First, a nonrotating and uncharged black hole is the Schwarzschild black hole, whose metric is given in Eq. (1.7). It is the simplest black-hole spacetime and depends only on the mass.

Second, a rotating but uncharged black hole is described by the Kerr metric,

$$ds^2 = - \left(1 - \frac{2Mr}{\Sigma}\right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{2Mra^2}{\Sigma} \sin^2 \theta\right) \sin^2 \theta d\phi^2 - \frac{4Mra \sin^2 \theta}{\Sigma} dt d\phi, \quad (1.8)$$

where

$$a = \frac{J}{M}, \quad \Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2. \quad (1.9)$$

Here M is the mass and J is the angular momentum. Since astrophysical black holes generally form from collapsing objects with angular momentum, Kerr black holes are expected to be highly relevant in nature.

Third, a nonrotating charged black hole is the Reissner-Nordström black hole,

$$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad (1.10)$$

where Q is the charge. Such black holes are mainly of theoretical interest, because astrophysical black holes are expected to be nearly neutral.

Fourth, a rotating charged black hole is described by the Kerr-Newman metric,

$$ds^2 = - \left(1 - \frac{2Mr - Q^2}{\Sigma^2} \right) dt^2 + \frac{\Sigma^2}{\Delta} dr^2 + \Sigma^2 d\theta^2 - \frac{2(2Mr - Q^2)a \sin^2 \theta}{\Sigma^2} dt d\phi + \left[(r^2 + a^2) \sin^2 \theta + \frac{(2Mr - Q^2)a^2 \sin^4 \theta}{\Sigma^2} \right] d\phi^2, \quad (1.11)$$

with

$$\rho^2 = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2. \quad (1.12)$$

This solution combines the effects of spin and charge.

Black holes can also be classified by their mass and formation mechanism. Stellar-mass black holes, with masses from a few to several tens of solar masses, are generally formed by the collapse of massive stellar cores. X-ray binaries such as Cygnus X-1 provide indirect evidence for such objects^[33], while gravitational-wave detections of binary black-hole mergers provide direct dynamical evidence^[4-6]. Intermediate-mass black holes have masses between roughly 10^2 and $10^5 M_\odot$; their origin may involve mergers of stellar-mass black holes or the collapse of dense stellar systems, and recent gravitational-wave observations have provided new candidates^[34]. Supermassive black holes, with masses from 10^6 to $10^{10} M_\odot$, reside in the centers of many large galaxies^[35]; Sgr A* at the center of the Milky Way is a prime example^[36]. Primordial black holes may have formed in the early universe from density fluctuations rather than stellar collapse, and they remain possible dark-matter candidates^[37].

1.3.3 Black-Hole Thermodynamics

In general relativity, the no-hair theorem implies that a black hole is completely described by mass, angular momentum, and charge. This apparent simplicity led to the view that black holes are among the simplest objects in the universe. Nevertheless, black holes also possess thermodynamic properties. Their temperature is^[25,38-41]

$$T = \frac{\kappa}{2\pi k_B}, \quad (1.13)$$

where T is the Hawking temperature, κ is the surface gravity, and k_B is Boltzmann's constant. The four laws of black-hole thermodynamics closely parallel the four laws of ordinary thermodynamics.

The zeroth law states that the surface gravity κ of a stationary black hole is constant over the event horizon. This is analogous to the uniform temperature of a system in thermal equilibrium. The first law reads

$$dM = \frac{\kappa}{8\pi} dA + \Omega dJ + \Phi dQ, \quad (1.14)$$

where M , A , Ω , J , Φ , and Q are the mass, horizon area, angular velocity, angular momentum, electric potential, and charge of the black hole. The second law states that the area of the event horizon never decreases in any classical process, analogous to the increase of entropy. The third law states that the surface gravity of a black hole cannot be reduced to zero by any finite sequence of physical operations.

The entropy of a black hole is given by the Bekenstein-Hawking formula

$$S = \frac{A}{4}. \quad (1.15)$$

Thus Eq. (1.14) can be rewritten as

$$\delta M = T\delta S + \Omega\delta J + \Phi\delta Q. \quad (1.16)$$

Comparing this expression with the ordinary first law,

$$\delta U = T\delta S - P\delta V + \sum_j \mu_j \delta N_j + \Phi\delta Q, \quad (1.17)$$

one finds that black-hole mass, angular velocity, and electric potential play roles analogous to internal energy, chemical potential, and electromagnetic potential. The corresponding Smarr relation is

$$M = 2(TS + \Omega J) + \Phi Q. \quad (1.18)$$

In recent studies of AdS black holes, the negative cosmological constant Λ is interpreted as thermodynamic pressure, thereby restoring the pressure-volume term in the first law^[42-44]. In Reissner-Nordström-AdS spacetime, the metric function is^[45]

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2} + \frac{r^2}{\ell^2}, \quad (1.19)$$

where

$$\ell^2 = -\frac{3}{\Lambda} = \frac{3}{8\pi P}. \quad (1.20)$$

The electric potential and temperature are

$$\Phi = -A_0 = \frac{Q}{r}, \quad (1.21)$$

$$T = \frac{f'(r_+)}{4\pi} = \frac{1}{4\pi r_+} \left(1 - \frac{Q^2}{r_+^2} + \frac{3r_+^2}{\ell^2} \right), \quad (1.22)$$

where the event horizon r_+ is determined by $f(r_+) = 0$. The entropy and thermodynamic volume are

$$S = \frac{A}{4} = \pi r_+^2, \quad (1.23)$$

$$V = \frac{4\pi r_+^3}{3}. \quad (1.24)$$

In the extended phase space, the first law and Smarr relation become

$$dM = T dS + \Phi dQ + V dP, \quad (1.25)$$

$$M = 2TS + \Phi Q - 2VP. \quad (1.26)$$

From Eq. (1.22), the equation of state is

$$P = \frac{T}{2r_+} - \frac{1}{8\pi r_+^2} + \frac{Q^2}{8\pi r_+^4}. \quad (1.27)$$

Introducing $v = 2r_+$, it becomes

$$P = \frac{T}{v} - \frac{1}{2\pi v^2} + \frac{2Q^2}{\pi v^4}. \quad (1.28)$$

This resembles the van der Waals equation,

$$\left(P + \frac{a}{v^2} \right) (v - b) = T, \quad (1.29)$$

and indeed the Reissner-Nordström-AdS black hole exhibits van der Waals-like critical behavior. The critical point is obtained from

$$\frac{\partial P}{\partial v} = 0, \quad \frac{\partial^2 P}{\partial v^2} = 0, \quad (1.30)$$

which yields

$$T_c = \frac{\sqrt{6}}{18\pi Q}, \quad v_c = 2\sqrt{6}Q, \quad P_c = \frac{1}{96\pi Q^2}, \quad (1.31)$$

and

$$\frac{P_c v_c}{T_c} = \frac{3}{8}. \quad (1.32)$$

Using $f(r_+) = 0$, the mass can be written as

$$M = \frac{v}{4} + \frac{Q^2}{v} + \frac{1}{6}\pi P v^3, \quad (1.33)$$

and the Gibbs free energy is

$$G(T, P) = M - TS = \frac{v}{4} + \frac{Q^2}{v} + \frac{1}{6}\pi P v^3 - \frac{1}{4}\pi v^2 T. \quad (1.34)$$

At the critical point,

$$G_c = \frac{\sqrt{6}}{3}Q. \quad (1.35)$$

With the dimensionless variables

$$p = \frac{P}{P_c}, \quad \nu = \frac{v}{v_c}, \quad \tau = \frac{T}{T_c}, \quad \mathcal{G}(\tau, p) = \frac{G(T, P)}{G_c}, \quad (1.36)$$

the equation of state and Gibbs free energy become

$$\tau = \frac{3}{8} \left[\nu \left(p + \frac{2}{\nu^2} \right) - \frac{1}{3\nu^3} \right], \quad (1.37)$$

$$\mathcal{G}(\tau, p) = \nu + \frac{1}{3\nu} - \frac{1}{3}\nu^2 \tau. \quad (1.38)$$

For $T < T_c$, the black hole undergoes a first-order phase transition between small and large black-hole phases. At $T = T_c$, the transition becomes second order. Near the critical point, expanding Eq. (1.37) gives

$$p = 1 + \frac{8}{3}t - \frac{8}{9}t\omega - \frac{4}{81}\omega^3 + \mathcal{O}(t\omega^2, \omega^4), \quad (1.39)$$

where

$$t = \frac{T}{T_c} - 1 = \tau - 1, \quad \omega = \frac{v}{v_c} - 1 = \nu - 1. \quad (1.40)$$

The critical exponents are then

$$C_{r_+} \sim \left(\frac{|T - T_c|}{T_c} \right)^{-\alpha}, \quad (1.41)$$

$$r_{+l} - r_{+s} \propto (T - T_c)^\beta, \quad (1.42)$$

$$\kappa_T \sim \left(\frac{|T - T_c|}{T_c} \right)^{-\gamma}, \quad (1.43)$$

$$P - P_c \sim (r_+ - r_c)^\delta, \quad (1.44)$$

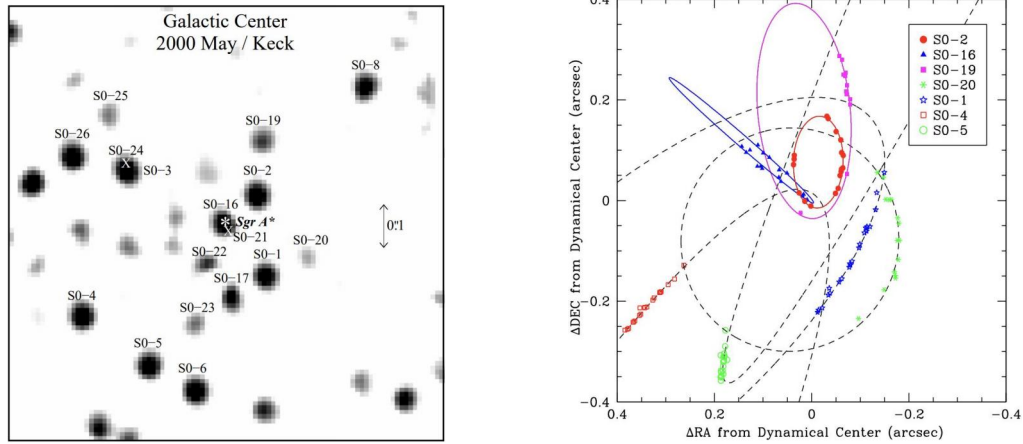
with

$$\alpha = 0, \quad \beta = 1/2, \quad \gamma = 1, \quad \delta = 3. \quad (1.45)$$

Hawking and Page first discovered a phase transition between Schwarzschild-AdS black holes and thermal AdS space in 1982^[41]. Since then, black-hole thermodynamics has become a major topic, especially in connection with the AdS/CFT correspondence^[46-52]. Extensive work on black-hole phase transitions can be found in Refs.^[53-79] and in the review^[80].

The significance of black-hole thermodynamics is fivefold. It probes physics under extreme conditions, where general relativity, quantum mechanics, and thermodynamics interact. It reveals deep analogies between black holes and ordinary matter, including van der Waals-like phase transitions. Through holographic duality, it connects gravitational systems to strongly coupled quantum field theories. It provides thermodynamic criteria for distinguishing modified gravity theories. Finally, it sheds light on quantum effects and black-hole evolution, including Hawking radiation and the information problem.

1.3.4 Observational Evidence for Black Holes



(a) Central 1 arcsec $\times$ 1 arcsec region of the cross-correlation map of stars around Sgr A*.

(b) Measured positions and orbital fits for seven stars with significant nonlinear motion.

Figure 1.1: Observations of stars in the vicinity of Sgr A* and their astrophysical implications^[81].

The 2020 Nobel Prize in Physics was awarded to Roger Penrose, Reinhard Genzel,

and Andrea Ghez^①. Penrose was recognized for showing that black-hole formation is a robust prediction of general relativity. Genzel and Ghez were recognized for developing techniques to observe stellar motion near the Galactic center and for discovering a supermassive compact object at Sgr A*. As shown in Fig. 1.1, stellar orbital dynamics, density constraints, and the association with the position and motion of Sgr A* provide strong evidence that the Galactic center hosts a supermassive black hole with mass of order $3.7 \times 10^6 M_\odot$.

It should be emphasized that the work of Genzel and Ghez established the existence of the central compact object mainly through stellar dynamics; it did not directly image the black hole. The direct image of Sgr A* was obtained later by the EHT. These two lines of evidence are complementary.

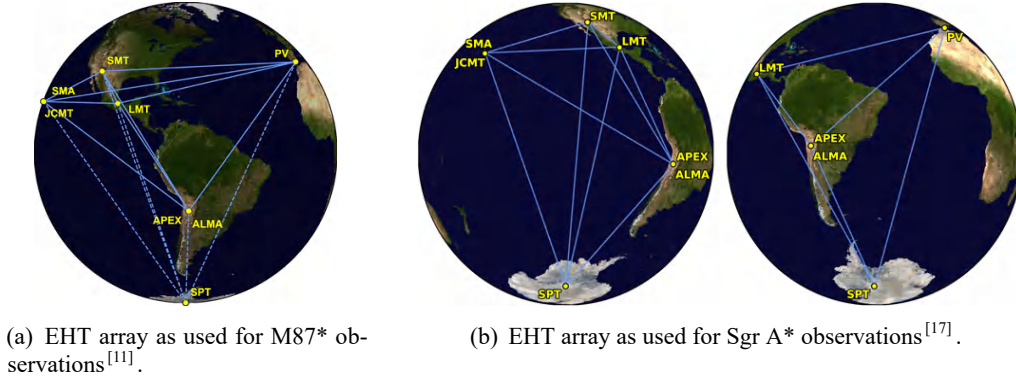


Figure 1.2: The EHT array comprises eight observatories distributed across six geographical locations.

Supermassive black holes are believed to reside at the centers of many galaxies, making them ideal observational targets. When matter surrounds a black hole, strong gravitational lensing and photon capture can produce a black-hole shadow. The EHT is a global very-long-baseline interferometry (VLBI) array that combines synchronized observations from telescopes across the world, producing a virtual telescope with an Earth-sized aperture, as illustrated in Fig. 1.2.

For M87*, the EHT image shows an asymmetric bright ring with diameter $42 \pm 3 \mu\text{as}$ and a central brightness depression, interpreted as the black-hole shadow. At a distance of $16.8 \pm 0.8 \text{ Mpc}$, the inferred black-hole mass is $(6.5 \pm 0.7) \times 10^9 M_\odot$. For Sgr A*, the bright ring diameter is $51.8 \pm 2.3 \mu\text{as}$ and the inferred mass is $(4.0 \pm 0.6) \times 10^6 M_\odot$. These images provide direct visual evidence for black holes and open a new observational route to studying strong gravity, accretion physics, and jet formation.

Black holes have also been confirmed through gravitational-wave observations. Gen-

① <https://www.nobelprize.org/prizes/physics/2020/summary/>

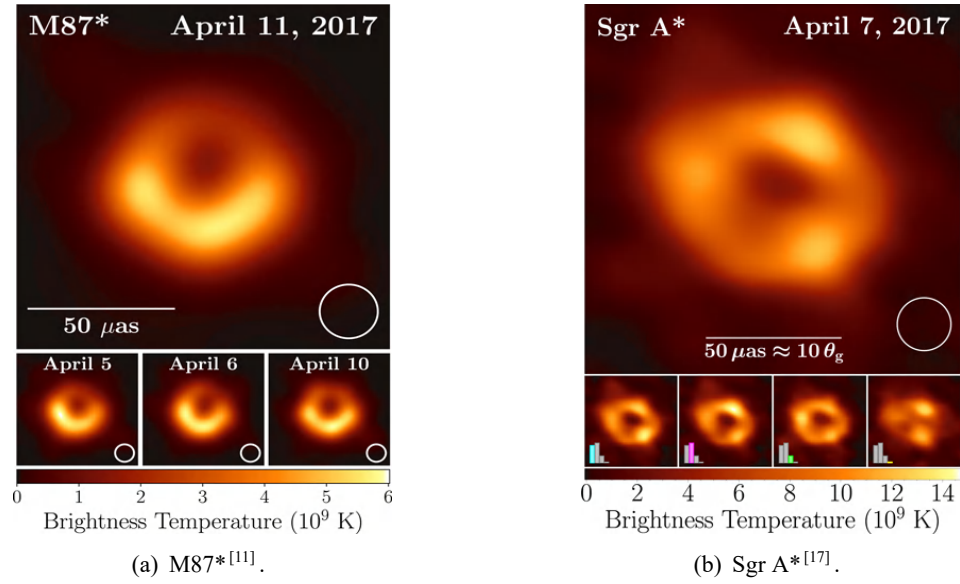


Figure 1.3: The EHT images of black holes are displayed in units of brightness temperature.

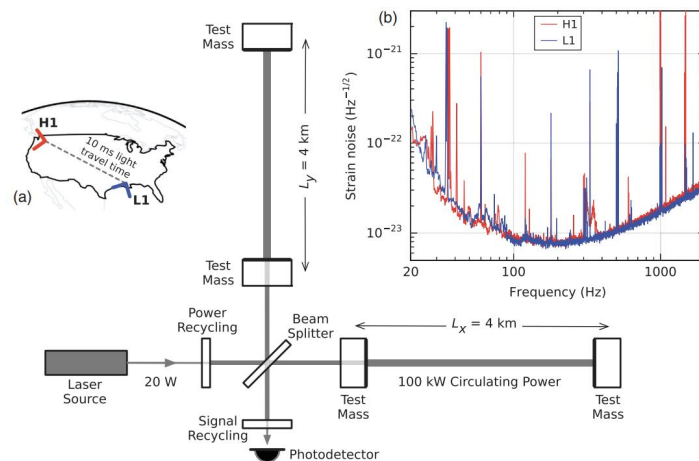


Figure 1.4: Simplified diagram of an Advanced LIGO detector (not to scale)^[6].

eral relativity predicts gravitational waves as ripples in spacetime produced by accelerated masses. LIGO uses laser interferometry to measure tiny changes in arm length caused by passing gravitational waves. In 2015, LIGO detected a gravitational-wave signal from the merger of two stellar-mass black holes^[4-6]. The 2017 Nobel Prize in Physics was awarded to Rainer Weiss, Barry C. Barish, and Kip S. Thorne for their contributions to this discovery^②.

Figure 1.4 shows the simplified layout of Advanced LIGO. Its core components include a laser source, power-recycling mirror, beam splitter, 4 km optical cavities, and test masses. A linearly polarized gravitational wave propagating perpendicular to the detector plane stretches one arm and compresses the other during half a cycle, and the photodetector records the resulting change in optical path length. The figure also indicates the locations and orientations of the Hanford (H1) and Livingston (L1) detectors and illustrates major noise sources affecting detector sensitivity.

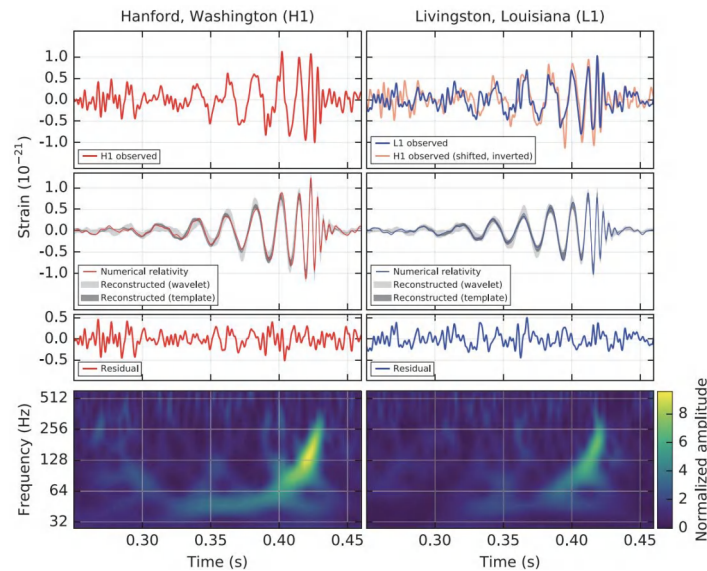


Figure 1.5: The gravitational-wave event GW150914 observed by LIGO Hanford (H1, left panels) and Livingston (L1, right panels) detectors. The time is shown relative to Coordinated Universal Time (UTC) 09:50:45 on September 14, 2015^[6].

Figure 1.5 shows the LIGO data for GW150914. The time series were band-pass filtered between 35 and 350 Hz to suppress large fluctuations outside the sensitive band and to remove strong instrumental lines. The top panels show the measured strain in the two detectors; L1 observed the signal first, followed by H1 after 6.9 ms. The middle panels compare the data with numerical-relativity and reconstructed waveforms, while the bottom panels show the time-frequency evolution. The event involved two black holes with initial masses of approximately $36M_{\odot}$ and $29M_{\odot}$, forming a remnant of about $62M_{\odot}$.

② <https://www.nobelprize.org/prizes/physics/2017/summary/>

and radiating roughly $3M_{\odot}$ in gravitational waves.

After the first detection, the LIGO-Virgo Collaboration also observed GW170817 from a binary neutron-star merger^[9]. The event was associated with the gamma-ray burst GRB 170817A and with electromagnetic counterparts across the spectrum, inaugurating multi-messenger astronomy and providing a new way to study compact objects, gravitational waves, and cosmology.

1.4 Organization of the Dissertation

The remainder of this dissertation is organized as follows.

Chapter 2 introduces the formation mechanism and research status of black-hole shadows, with quantitative derivations for the Schwarzschild and Kerr black-hole shadows.

Chapter 3 studies photon motion in Horndeski black-hole spacetimes, derives the effective potential, critical impact parameter, and photon-sphere radius, and constrains the Horndeski parameter using EHT observations of M87* and Sgr A*. It also analyzes the optical appearances of black holes and asymmetric thin-shell wormholes surrounded by thin accretion disks. Finally, it turns to AdS black holes in Horndeski theory and studies holographic Einstein rings through wave optics and geometric optics.

Chapter 4 studies the shadow signatures and thermodynamic phase transitions of Hayward-AdS black holes, using the shadow radius as an alternative state parameter. It then investigates the dynamics of phase transitions for charged AdS black holes with a global monopole by using the Gibbs free-energy landscape and the Fokker-Planck equation.

Chapter 5 summarizes the dissertation and discusses future research directions.

Chapter 2 Black-Hole Shadows

Once an object crosses the event horizon of a black hole, it can never escape; even light cannot break free from that relentless gravitational pull.

— Stephen William Hawking

2.1 Concept of the Black-Hole Shadow

In everyday life, a shadow is the dark region formed behind an opaque object when light propagating through a uniform medium is blocked. It is essentially a region lacking light energy. Figure 2.1 illustrates the umbra, penumbra, and antumbra produced when sunlight illuminates Earth^①. The umbra is the region in which the light source is completely blocked, the penumbra is a partially shaded transition region, and the antumbra appears when the center of the light source is blocked while its edge remains visible.

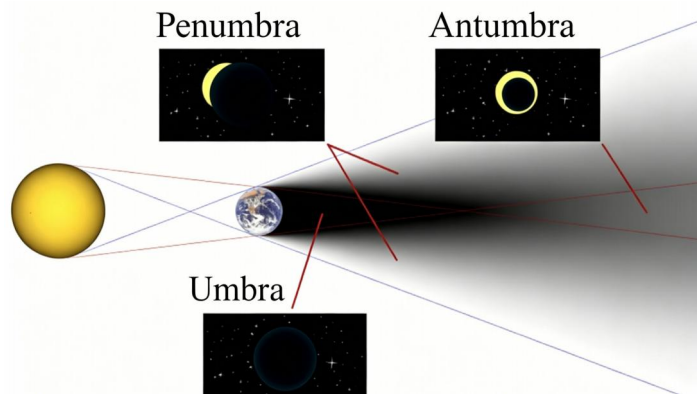


Figure 2.1: Umbra, penumbra and antumbra of Earth and images that could be seen at some points in these areas.

The shape of an ordinary shadow depends on the type of light source, the relative distances among the source, the object, and the screen, and the geometry of the object. Point sources produce sharp boundaries, whereas extended sources create penumbral regions. Such shadow phenomena are important in basic optics, astronomy, engineering, and medical imaging.

A black-hole shadow is fundamentally different from an ordinary optical shadow. It is not produced by a solid object blocking light, but by the capture and strong deflection of photons in curved spacetime. Near a black hole, photons with sufficiently small impact parameters fall into the event horizon and never reach the observer, while photons that

^① en.wikipedia.org/wiki/Umbra

avoid capture form bright lensed structures around the dark region. The resulting image is therefore a macroscopic manifestation of spacetime curvature in a strong gravitational field. The EHT images of M87* and Sgr A* have made black-hole shadows a central observational probe of strong gravity.

The theoretical basis of black-hole shadows dates back to Synge's 1966 study of photon escape from an extremely compact star^[82]. He showed that, as the stellar surface approaches the Schwarzschild radius, the escape cone shrinks and only photons emitted nearly normal to the surface can escape. Bardeen later studied photon motion in Kerr spacetime and introduced the critical curve, or apparent boundary, of the shadow, showing that spin deforms the shadow into a D-shaped contour. The optical appearance of stars around extreme Kerr black holes was also studied by Bardeen and collaborators^[83].

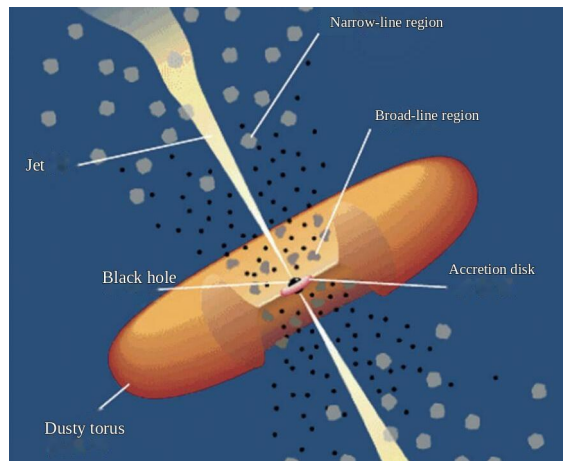


Figure 2.2: The standard model of quasars centered around a supermassive black hole^[84-85].

Because black holes do not directly emit electromagnetic radiation, they are observed through their surroundings. Quasars are among the brightest objects in the universe and are powered by accretion onto supermassive black holes. As shown in Fig. 2.2, matter captured by the black hole forms a rotating accretion disk because of its angular momentum. Viscous processes redistribute angular momentum, heat the gas to extremely high temperatures, and generate a broad continuum spectrum. Accretion disks are therefore the key source that illuminates black holes. Shakura and collaborators studied the structure and radiation of black-hole accretion disks^[86], while Page and Thorne derived the energy flux of thin disks around stationary axisymmetric black holes^[87]. In 1979, Luminet computed the appearance of a Schwarzschild black hole with a thin accretion disk^[88], as shown in Fig. 2.3.

In the early 2000s, Falcke and collaborators emphasized the observability of Sgr A* and the importance of very-long-baseline interferometry, laying theoretical groundwork

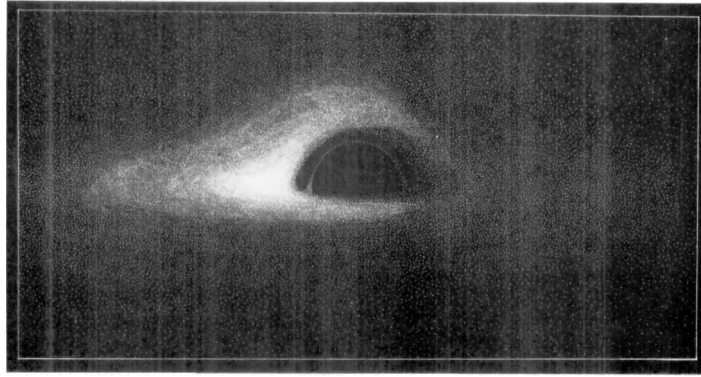


Figure 2.3: The simulated image of a Schwarzschild black hole with a thin accretion disk^[88].

for the EHT. Since the first black-hole images were obtained, black-hole shadows and optical appearances have become an active field. Studies have explored shadow substructure, photon rings, lensing rings, accretion models, and the effects of modified gravity^[23,89-139].



Figure 2.4: The visuals observed by observers near Gargantua in *Interstellar*^[140].

Black-hole images have also attracted public attention. In Christopher Nolan's 2014 film *Interstellar*, the black hole Gargantua became widely known. As shown in Fig. 2.4, Thorne and collaborators developed ray-tracing tools to model the appearance of a rotating black hole, bringing black-hole shadow calculations into cinematic visual effects while revealing the complex lensing of star fields and accretion disks^[140].

2.2 Formation of Black-Hole Shadows

2.2.1 Schwarzschild Black-Hole Shadow

The Schwarzschild black hole is the first exact black-hole solution in general relativity and describes a static, spherically symmetric, uncharged, nonrotating black hole. Figure 2.5 illustrates the formation of its shadow^[106]. The left panel shows a black disk representing the event horizon with radius $2M$, an orange ring representing the photon sphere, and a distant observer. Since only photons received by the observer contribute to the observed image, one may trace light rays backward from the observer toward the

black hole. According to the impact parameter $b = L/E$ relative to the critical impact parameter b_c , photon trajectories fall into three classes.

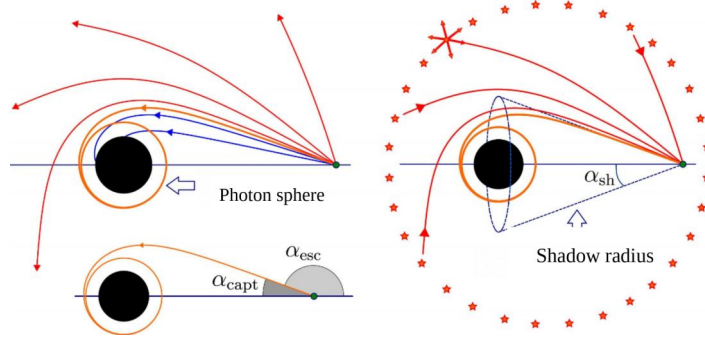


Figure 2.5: The formation mechanism of a black-hole shadow within the spacetime geometry of a Schwarzschild black hole^[106].

If $b > b_c$, the photon is deflected by the black hole but eventually reaches a turning point and escapes to the observer. If $b = b_c$, the photon asymptotically approaches the unstable circular photon orbit at $r = 3M$, which determines the shadow boundary. If $b < b_c$, the photon is captured by the black hole and cannot be detected by an external observer; the corresponding missing light forms the dark shadow region.

The Schwarzschild metric is^[82]

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 (\sin^2 \theta d\phi^2 + d\theta^2), \quad (2.1)$$

where

$$f(r) = 1 - \frac{2M}{r}. \quad (2.2)$$

Photon motion can be described by the Hamiltonian

$$H = \frac{1}{2}g^{\mu\nu}p_\mu p_\nu, \quad (2.3)$$

with

$$\dot{p}_\mu = -\frac{\partial H}{\partial x^\mu}, \quad \dot{x}_\mu = -\frac{\partial H}{\partial p^\mu}. \quad (2.4)$$

For photons moving in the equatorial plane, $\theta = \pi/2$ and $\dot{\theta} = 0$, the conserved quantities are associated with t and ϕ translations. The photon four-velocity is

$$\dot{t} = -\frac{p_t}{f(r)}, \quad \dot{r} = \frac{p_r}{f(r)}, \quad \dot{\phi} = -\frac{p_\phi}{f(r)}. \quad (2.5)$$

The null condition $H = 0$ gives

$$-\frac{p_t^2}{f(r)} + \frac{p_r^2}{f(r)} + \frac{p_\phi^2}{f(r)} = 0. \quad (2.6)$$

The photon-sphere conditions $\dot{r} = 0$ and $\ddot{r} = 0$ yield $r_p = 3M$. For a static observer at $r_O > 3M$, let α be the angle between the photon trajectory and the radial direction. Then

$$\cot \alpha = \frac{\sqrt{f(r)} dr}{r d\phi} \Big|_{r=r_O}. \quad (2.7)$$

Combining Eqs. (2.5) and (2.6), one obtains

$$\frac{dr}{d\phi} = \pm \frac{r}{\sqrt{f(r)}} \sqrt{\frac{r^2}{f(r)} \frac{p_t^2}{p_\phi^2} - 1}. \quad (2.8)$$

At the turning point $r = d$,

$$\frac{r^2}{f(r)} \Big|_{r=d} = \frac{p_\phi^2}{p_t^2} \Big|_{r=d}. \quad (2.9)$$

For photons on the photon sphere,

$$\sin^2 \alpha_s = \frac{f(r_O) r_p^2}{r_O^2 f(r_p)}. \quad (2.10)$$

Substitution of the Schwarzschild metric gives

$$\sin^2 \alpha_{sh} = \frac{27M^2 (1 - 2M/r_O)}{r_O^2}. \quad (2.11)$$

The escape cone α_{esc} and gravitational capture cone α_{capt} are^[82]

$$\sin^2 \alpha_{\text{esc}} = \frac{27M^2 (1 - 2M/r_O)}{r_O^2}, \quad (2.12)$$

$$|\tan \alpha_{\text{capt}}| = \frac{\left(1 - \frac{2M}{r_O}\right)^{1/2}}{\left(\frac{2M}{r_O} - 1 + \frac{r_O^2}{27M^2}\right)^{1/2}}. \quad (2.13)$$

For $2M < r_O < 3M$ and $r_O > 3M$, the angular radius of the shadow can be written as^[82]

$$\alpha_{\text{sh}}(r_O) = \begin{cases} \pi - \arcsin\left(3\sqrt{3}M\sqrt{1 - 2M/r_O}/r_O\right), & 2M < r_O \leq 3M, \\ \arcsin\left(3\sqrt{3}M\sqrt{1 - 2M/r_O}/r_O\right), & r_O \geq 3M. \end{cases} \quad (2.14)$$

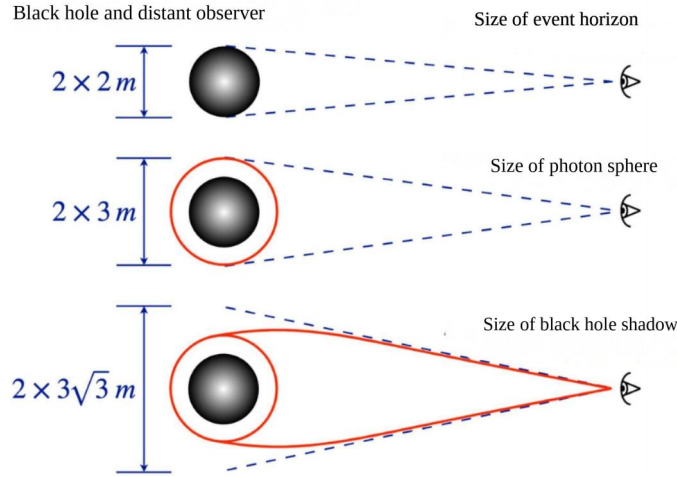


Figure 2.6: The sizes of the event horizon, photon sphere, and black-hole shadow in a Schwarzschild black hole^[106].

Thus, the event-horizon radius, photon-sphere radius, and shadow radius all scale with the black-hole mass. However, the visual radius of the shadow differs from the physical radius of the photon sphere. For a Schwarzschild black hole, the horizon and photon-sphere radii are $2M$ and $3M$, respectively, while gravitational lensing enlarges the apparent shadow radius to $3\sqrt{3}M$, as shown in Fig. 2.6.

2.2.2 Kerr Black-Hole Shadow

Astrophysical black holes are generally expected to rotate, so the Kerr black-hole shadow is more relevant than the Schwarzschild case. As shown in Fig. 2.7, photons in the equatorial plane of a Kerr black hole have prograde and retrograde circular orbits. In the Schwarzschild limit both have radius $3M$, while for a Kerr black hole with spin increasing from 0 to M , the prograde and retrograde radii range from $3M$ to M and from $3M$ to $4M$, respectively. This difference deforms the shadow into a D-shaped contour.

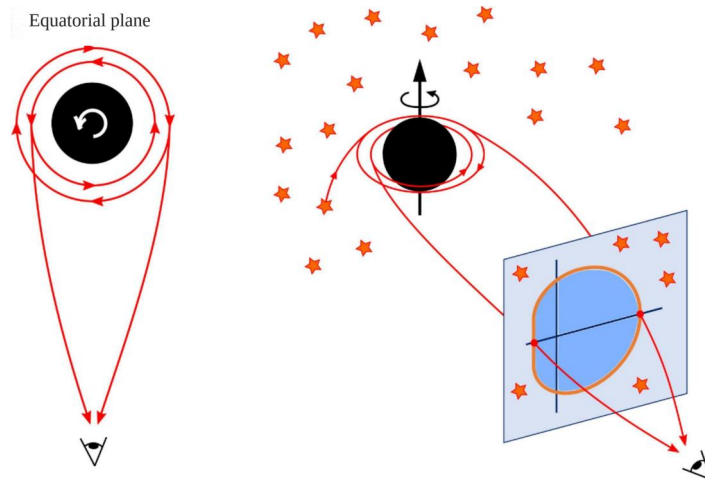


Figure 2.7: The shadow of a Kerr black hole^[106].

The Kerr metric in Boyer-Lindquist coordinates (t, r, θ, ϕ) is

$$\begin{aligned} ds^2 = & -\frac{\Delta - a^2 \sin^2 \theta}{\rho^2} dt^2 + \frac{\rho^2}{\Delta} dr^2 + \sin^2 \theta \left(\frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\rho^2} \right) d\phi^2 \\ & - 2a \sin^2 \theta \left(\frac{r^2 + a^2 - \Delta}{\rho^2} \right) dt d\phi + \rho^2 d\theta^2, \end{aligned} \quad (2.15)$$

where

$$\rho^2 = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2. \quad (2.16)$$

Here M and a are the mass and spin parameter. The photon Hamiltonian is

$$H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu. \quad (2.17)$$

The Killing vectors $\xi_{t,\phi} = \partial_{t,\phi}$ imply conserved energy E and angular momentum L ,

$$E := -g_{tt} \frac{dt}{d\lambda} - g_{t\phi} \frac{d\phi}{d\lambda}, \quad L := g_{\phi\phi} \frac{d\phi}{d\lambda} + g_{t\phi} \frac{dt}{d\lambda}. \quad (2.18)$$

The Hamilton-Jacobi equation separates as

$$\begin{aligned} \rho^2 \frac{dt}{d\lambda} &= a (L - aE \sin^2 \theta) + \frac{r^2 + a^2}{\Delta} ((r^2 + a^2) E - aL), \\ \rho^2 \frac{dr}{d\lambda} &= \sqrt{\mathcal{R}}, \\ \rho^2 \frac{d\theta}{d\lambda} &= \sqrt{\Theta}, \\ \rho^2 \frac{d\phi}{d\lambda} &= \frac{L}{\sin^2 \theta} - aE + \frac{a}{\Delta} ((r^2 + a^2) E - aL), \end{aligned} \quad (2.19)$$

where

$$\begin{aligned} \mathcal{R} &= ((r^2 + a^2) E - aL)^2 - \Delta (\mathcal{K} + (L - aE)^2), \\ \Theta &= \mathcal{K} + \cos^2 \theta \left(a^2 E^2 - \frac{L^2}{\sin^2 \theta} \right). \end{aligned} \quad (2.20)$$

Here $\mathcal{K}$ is the Carter constant^[141]. Critical photon motion satisfies

$$\dot{r} = 0, \quad \ddot{r} = 0, \quad (2.21)$$

or equivalently

$$\mathcal{R} = 0, \quad \mathcal{R}' = 0. \quad (2.22)$$

Introducing

$$\xi = L/E, \quad \eta = \mathcal{K}/E^2, \quad (2.23)$$

one obtains

$$\xi = \frac{M(a^2 - r^2) + \Delta r}{a(M - r)}, \quad (2.24)$$

$$\eta = \frac{r^3(2a^2M - r(5M^2 - 4Mr + r^2) + 2\Delta M)}{a^2(M - r)^2}. \quad (2.25)$$

These parameters determine the photon motion and the Kerr shadow.

To connect photon orbits with the image seen by a distant observer, one introduces a local orthonormal tetrad $(e_{\bar{t}}, e_{\bar{r}}, e_{\bar{\theta}}, e_{\bar{\varphi}})$ related to the coordinate basis by $e_{\bar{\mu}} = e_{\bar{\mu}}^{\nu} \partial_{\nu}$ and satisfying^[142-148]

$$g_{\mu\nu} e_{\bar{\alpha}}^{\mu} e_{\bar{\beta}}^{\nu} = \eta_{\bar{\alpha}\bar{\beta}}. \quad (2.26)$$

For the zero-angular-momentum observer frame, the locally measured photon momentum components are

$$p^{\bar{t}} = \sqrt{\frac{g_{\varphi\varphi}}{g_{t\varphi}^2 - g_{tt}g_{\varphi\varphi}}} E + \frac{g_{t\varphi}}{g_{\varphi\varphi}} \sqrt{\frac{g_{\varphi\varphi}}{g_{t\varphi}^2 - g_{tt}g_{\varphi\varphi}}} p_{\varphi}, \quad (2.27)$$

$$p^{\bar{r}} = \frac{1}{\sqrt{g_{rr}}} p_r, \quad (2.28)$$

$$p^{\bar{\theta}} = \frac{1}{\sqrt{g_{\theta\theta}}} p_{\theta}, \quad (2.29)$$

$$p^{\bar{\varphi}} = \frac{1}{\sqrt{g_{\varphi\varphi}}} p_{\varphi}. \quad (2.30)$$

The spatial momentum is

$$\bar{p} = p^{\bar{r}} e_{\bar{r}} + p^{\bar{\theta}} e_{\bar{\theta}} + p^{\bar{\varphi}} e_{\bar{\varphi}}. \quad (2.31)$$

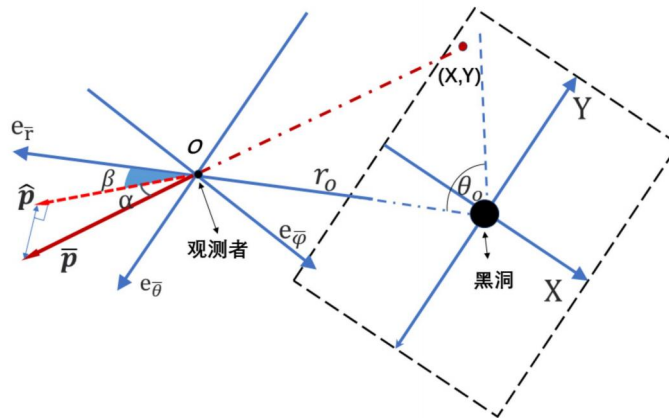


Figure 2.8: A perspective drawing of the geometric projection of the photon's momentum $\bar{p}$ in the observer's coordinate system $\{e_{\bar{r}}, e_{\bar{\theta}}, e_{\bar{\varphi}}\}$ ^[149-150].

As shown in Fig. 2.8, the local photon momentum can be projected onto the celestial

coordinates (X, Y) . The components satisfy

$$p^{\bar{r}} = |\bar{p}| \cos \alpha \cos \beta, \quad (2.32)$$

$$p^{\bar{\theta}} = |\bar{p}| \sin \alpha, \quad (2.33)$$

$$p^{\bar{\varphi}} = |\bar{p}| \cos \alpha \sin \beta. \quad (2.34)$$

Thus^[120,151]

$$\begin{aligned} X &= -r \tan \beta \Big|_{(r_0, \theta_0)} = -r \frac{p^{\bar{\varphi}}}{p^{\bar{r}}} \Big|_{(r_0, \theta_0)}, \\ Y &= r \frac{\tan \alpha}{\cos \beta} \Big|_{(r_0, \theta_0)} = -r \frac{p^{\bar{\theta}}}{p^{\bar{r}}} \Big|_{(r_0, \theta_0)}. \end{aligned} \quad (2.35)$$

For an observer at infinity, the Kerr shadow boundary is

$$X = -\xi \csc \theta_0, \quad Y = \pm \sqrt{\eta + a^2 \cos^2 \theta_0 - \xi^2 \cot^2 \theta_0}. \quad (2.36)$$

Figure 2.9 illustrates the main properties of Kerr shadows. For a fixed inclination $\theta_0 = \pi/2$, increasing the spin deforms the shadow from a circle into a D-shaped contour and shifts it along the X direction, while leaving its Y -direction size almost unchanged. For fixed spin $a = 0.999M$, increasing the inclination from $\pi/36$ to $\pi/2$ makes the shadow more deformed and larger in the Y direction. Thus an equatorial observer sees both the largest apparent size and the strongest deformation.

2.3 Significance of Black-Hole Shadow Studies

The study of black-hole appearances connects gravitational theory, astrophysics, and cosmology. By observing shadows, accretion-disk radiation, and the surrounding space-time structure, one can test general relativity, explore modified gravity, probe extreme astrophysical processes, and search for new compact objects. The main motivations include the following.

First, black-hole shadows provide a direct test of general relativity in the strong-field regime. General relativity predicts photon rings and central brightness depressions around black holes, and EHT images can be compared with these predictions.

Second, precise shadow measurements can constrain alternative theories of gravity. Different models, such as Gauss-Bonnet gravity, Horndeski theory, and Rastall gravity, may predict slightly different shadow sizes and shapes. Comparing observations with such predictions can restrict the parameter spaces of these theories.

Third, shadow observations can search for exotic compact objects. Wormholes, boson stars, and other non-black-hole compact objects may produce shadow-like images

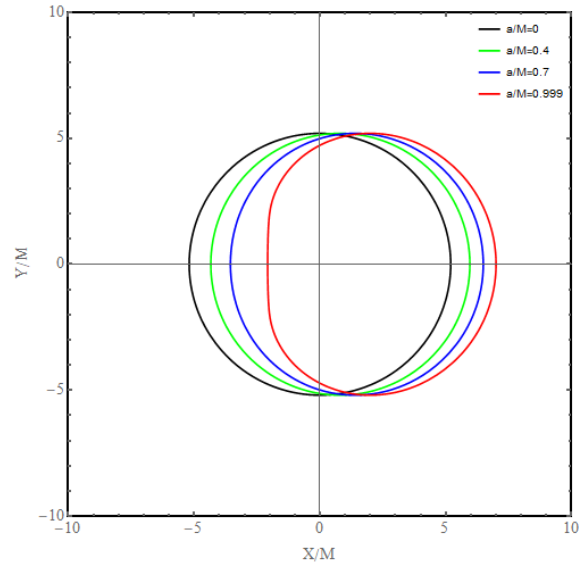
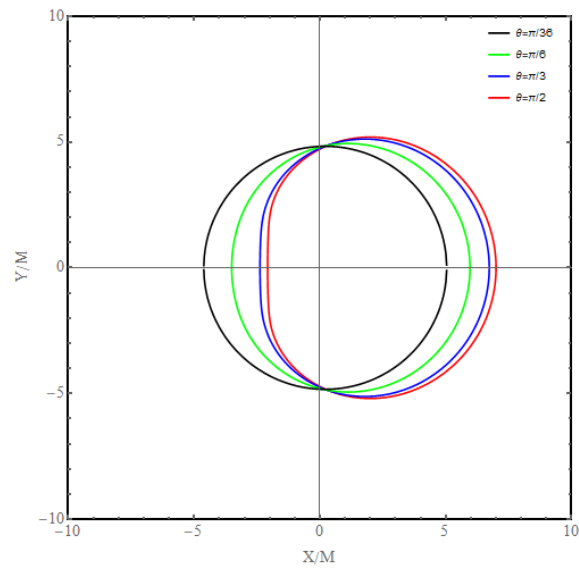
(a) $\theta = \pi/2$ (b) $a/M = 0.999$

Figure 2.9: The shadows of a Kerr black hole as seen by observers in the equatorial plane under two scenarios: fixed observational inclination angles and fixed spin parameters, respectively.

that differ from those of classical black holes. Identifying such deviations would provide evidence for new physics.

Fourth, black-hole imaging helps us understand accretion disks and jets. The distribution of brightness around the shadow encodes the geometry and dynamics of the accretion flow, plasma behavior in strong gravity, and the role of magnetic fields in jet formation.

Fifth, the demand for black-hole imaging drives observational technology. The EHT demonstrated the power of global VLBI, and future instruments with higher sensitivity and angular resolution will allow more stringent tests of gravity and may reveal finer structures such as photon rings and time-dependent shadow features.

Chapter 3 Shadow Features and Related Studies of Black Holes in Horndeski Theory

Imagine a black hole, its dark outline surrounded by a ring of hot rotating gas. This is one of the most striking scenes predicted by modern physics.

— Kip Stephen Thorne

In this chapter, based on our works^[152-154], we investigate the observational appearance of black holes in Horndeski theory by using ray-tracing techniques in the geometric-optics framework. We then compare these signatures with those of asymmetric thin-shell wormholes, which may act as black-hole mimickers. Finally, going beyond conventional geometric optics, we use wave optics based on the holographic principle and the AdS/CFT correspondence to study holographic Einstein-ring images, and we compare the wave-optics and geometric-optics descriptions of black-hole appearance.

3.1 Research Background

Breakthroughs in gravitational-wave astronomy and black-hole imaging have opened a new era in the study of compact objects in strong gravitational fields. In 2015, LIGO directly detected gravitational waves from a binary black-hole merger, providing compelling evidence for the existence of black holes^[4-6]. Shortly afterward, the Event Horizon Telescope released the first image of the supermassive black hole at the center of M87^[11-16], and later the image of Sgr A* at the center of the Milky Way^[17-22]. These observations make black-hole shadows a direct probe of spacetime geometry, accretion physics, and possible deviations from general relativity.

Although the observed images are consistent with general relativity, they do not exclude compact objects or black-hole solutions arising in alternative theories of gravity. Horndeski theory, the most general scalar-tensor theory with second-order field equations^[155], provides a natural arena in which additional degrees of freedom can modify photon dynamics and hence the shadow. Since astrophysical black holes are generally surrounded by accretion flows, the theoretical prediction of black-hole images also requires modeling the emission from accretion disks. The combination of spacetime geometry, photon motion, and emission models therefore offers a direct way to constrain modified gravity with EHT data.

At the same time, compact objects with photon spheres are not necessarily black holes. Boson stars, wormholes, and other ultracompact objects may produce black-hole-like shadows while retaining distinguishable optical features^[156-166]. Motivated by recent

studies of asymmetric thin-shell wormholes, we examine whether additional lensing bands and photon-ring groups can separate wormholes from black holes with the same mass parameters.

Finally, conventional ray tracing is not the only possible description of black-hole imaging. Hashimoto and collaborators proposed a wave-optics method based on the holographic principle and the AdS/CFT correspondence^[167-168]. By introducing a localized oscillating source on the AdS boundary and computing the boundary response function, one can construct holographic images of the dual black hole. This method produces an Einstein-ring-like structure and provides a bridge between black-hole imaging and gravitational duality. We use this approach to study holographic Einstein rings of AdS black holes in Horndeski theory and compare the results with the geometric-optics picture.

3.2 Black-Hole Shadows in Horndeski Theory

We start from the Horndeski action^[155]

$$S = \int d^4x \sqrt{-g} (\mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5), \quad (3.1)$$

where

$$\mathcal{L}_2 = G_2(X), \quad \mathcal{L}_3 = -G_3(X)\Box\phi, \quad (3.2)$$

$$\mathcal{L}_4 = G_4(X)\mathcal{R} + G_{4X} [(\Box\phi)^2 - (\nabla_\mu \nabla_\nu \phi)^2], \quad (3.3)$$

$$\mathcal{L}_5 = G_5(X)G_{\mu\nu}\nabla^\mu\nabla^\nu\phi - \frac{G_{5X}}{6} [(\Box\phi)^3 - 3\Box\phi(\nabla_\mu\nabla_\nu\phi)^2 + 2(\nabla_\mu\nabla_\nu\phi)^3]. \quad (3.4)$$

Within this framework, $\mathcal{R}$ represents the Ricci scalar, $G_{\mu\nu}$ is the Einstein tensor, and $G_i = G_i(\phi, X)$ is the function of the scalar field ϕ and its kinetic term $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$. The subscript X denotes the derivative with respect to X . Following Ref^[169], we consider a special case of the Horndeski action (3.1), which is described as

$$G_2 = \eta X, \quad G_4 = \zeta + \beta\sqrt{-X}, \quad G_3 = G_5 = 0. \quad (3.5)$$

Here, we have $\zeta = \frac{1}{2}$ in natural units ($8\pi G = 1 = c$). Note that η and β are dimensionless parameters, whose values currently can only be determined through observations or experiments. These two parameters mean that in Horndeski theory, the coefficients of X and $\sqrt{-X}$ can be modified. Therefore, the action (3.1) is reduced to

$$S = \int d^4x \sqrt{-g} \left\{ \left[\frac{1}{2} + \beta\sqrt{(\partial\phi)^2/2} \right] \mathcal{R} - \frac{\eta}{2}(\partial\phi)^2 - \frac{\beta}{\sqrt{2}(\partial\phi)^2} [(\Box\phi)^2 - (\nabla_\mu \nabla_\nu \phi)^2] \right\}. \quad (3.6)$$

For the spherically symmetric solution of the black hole in Horndeski theory, the corresponding line element takes the form^[169]

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2), \quad (3.7)$$

where

$$f(r) = 1 - \frac{2M}{r} - \frac{\beta^2}{\eta r^2}. \quad (3.8)$$

Here, M is an integration constant that represents the black hole mass. Since both parameters η and β are dimensionless and independent, one can regard β^2/η as a whole. In this work, we define a new parameter $\gamma = \beta^2/\eta$ to simplify the following discussion. Then, we can reformulate Eq. (3.8) as

$$f(r) = 1 - \frac{2M}{r} - \frac{\gamma}{r^2}. \quad (3.9)$$

When the parameter γ takes a negative value, $\sqrt{-\gamma}$ behaves similarly to the electric charge of the Reissner-Nordström black hole. The radius of the outer event horizon can be determined by solving the equation $f(r_h) = 0$, yielding $r_h = M + \sqrt{M^2 + \gamma}$. To avoid the naked singularity, we need $\gamma \geq -M^2$. For $\gamma = -M^2$, the solution (3.7) corresponds to an extremal black hole. The metric function (3.9) is illustrated in Fig. 3.1. It is found that for $\gamma > 0$ and $\gamma = 0$ (the Schwarzschild case), there is only one event horizon. However, for $-M^2 < \gamma < 0$, both inner and outer horizons are present.

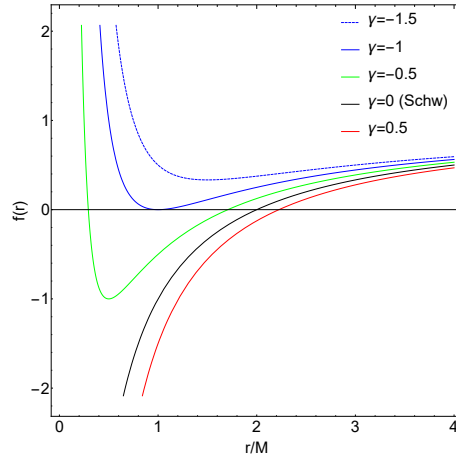


Figure 3.1: The metric function $f(r)$ for different values of the parameter γ . We set $\gamma = -1.5, -1, -0.5, 0, 0.5$ and $M = 1$.

To study the black hole shadow in Horndeski theory, we use the Lagrangian of pho-

tons to derive the equation of motion of photons, which is given by

$$\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \frac{1}{2}\left(-f(r)\dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2\left(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2\right)\right). \quad (3.10)$$

Here, $\dot{x}^\mu \equiv dx^\mu/d\lambda$ is the four-velocity of photons and λ is the affine parameter. Considering a photon moving on the equatorial plane (i.e. $\theta = \pi/2$ and $\dot{\theta} = 0$), since the metric function $f(r)$ does not depend on the time coordinate t and the azimuthal angle φ , there are two conserved quantities:

$$p_t = \frac{\partial\mathcal{L}}{\partial\dot{t}} = f(r)\dot{t} = E, \quad p_\varphi = -\frac{\partial\mathcal{L}}{\partial\dot{\varphi}} = r^2\dot{\varphi} = L. \quad (3.11)$$

In addition, we can obtain

$$\dot{t} = \frac{1}{b}f(r)^{-1}, \quad \dot{r} = \sqrt{\frac{1}{b^2} - \frac{1}{r^2}f(r)}, \quad \dot{\varphi} = \pm\frac{1}{r^2}, \quad (3.12)$$

where “ $\pm$ ” indicates the direction of the photon’s motion (“+” represents clockwise and “−” represents counterclockwise). The parameter $b = |L|/E$ is called the impact parameter. Using Eq. (3.12), we can obtain

$$\dot{r}^2 + V_{\text{eff}} = \frac{1}{b^2}, \quad (3.13)$$

where $V_{\text{eff}} = \frac{f(r)}{r^2}$ denotes the effective potential of the photon. The critical conditions for the photon’s unstable circular orbit are

$$V_{\text{eff}}(r_{\text{ph}}) = \frac{1}{b_c^2}, \quad V'_{\text{eff}}(r_{\text{ph}}) = 0, \quad (3.14)$$

where r_{ph} is the radius of the photon sphere and b_c is the critical impact parameter. With the solution (3.7), we can obtain the analytical expressions of r_{ph} and b_c :

$$r_{\text{ph}} = \frac{1}{2}\left(3M + \sqrt{9M^2 + 8\gamma}\right), \quad (3.15)$$

$$b_c = \frac{\sqrt{27M^4 + 36M^2\gamma + 8\gamma^2 + 9M^3\sqrt{9M^2 + 8\gamma} + 8M\gamma\sqrt{9M^2 + 8\gamma}}}{\sqrt{2}\sqrt{M^2 + \gamma}}. \quad (3.16)$$

For a given M , the photon sphere radius r_{ph} and the critical impact parameter b_c only depend on the parameter γ . From Fig. 3.2, one can find that the event horizon radius, photon sphere radius, and critical impact parameter all increase with the parameter γ . We plot the effective potential of the photon for different values of the parameter γ in

Fig. 3.3. It is found that the effective potential (e.g. $\gamma = -0.2$) rises from the event horizon radius, reaches a peak at the photon sphere radius, and then declines. Additionally, as the parameter γ increases, the peak value of the effective potential will decrease.

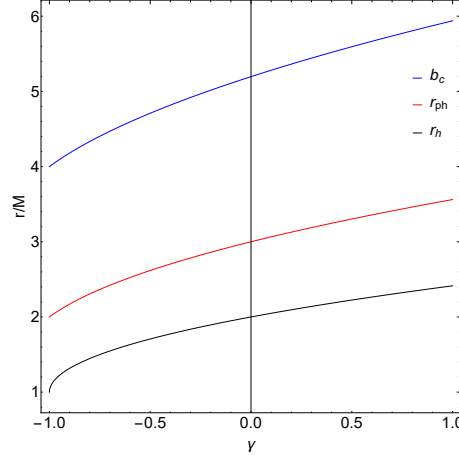


Figure 3.2: The event horizon radius, photon sphere radius, and critical impact parameter vary with the parameter γ . We set $M = 1$.

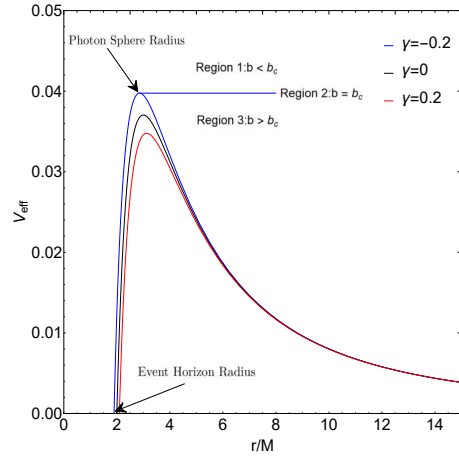


Figure 3.3: The effective potential of the photon for different values of the parameters γ . We set $\gamma = -0.2, 0, 0.2$ and $M = 1$.

To study the trajectory of the photon, we need to rewrite the photon's equation of motion using Eq. (3.12), which can be written as

$$\frac{dr}{d\varphi} = \pm r^2 \sqrt{\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2M}{r} - \frac{\gamma}{r^2} \right)}. \quad (3.17)$$

Through a coordinate transformation $x \equiv 1/r$, Eq. (3.17) can be further expressed as

$$\mathcal{G}(x, b) \equiv \frac{dx}{d\varphi} = \sqrt{\frac{1}{b^2} - x^2 (1 - 2Mx - \gamma x^2)}. \quad (3.18)$$

When the photon with an impact parameter b approaches the black hole from infinity, there are three possible scenarios based on the value of b ^[23]:

- $b < b_c$ The photon will finally fall into the black hole. If $b = 0$, the trajectory of the photon is a straight line. If $0 < b < b_c$, there is a deflection angle for the trajectory of the photon. For the event horizon r_h , the total deflection angle of the photon is given by

$$\Delta\varphi = \int_0^{1/r_h} \frac{dx}{\sqrt{\mathcal{G}(x, b)}}. \quad (3.19)$$

- $b = b_c$ The photon will reach a critical point x_c and then orbit the black hole along a circular path with a radius of r_{ph} . In this scenario, the motion of the photon is in a critical state, which delineates the boundary of the black hole shadow.
- $b > b_c$ The photon will reach a nearest point x_{min} (the smallest positive real root of $\mathcal{G}(x, b) = 0$) and then escape to infinity. The corresponding total deflection angle is given by

$$\Delta\varphi = 2 \int_0^{x_{min}} \frac{dx}{\sqrt{\mathcal{G}(x, b)}}. \quad (3.20)$$

In Fig. 3.4, we plot the trajectories of photons with different impact parameters. Here, the impact parameter b can be considered as the vertical distance between the photon's geodesic and the parallel line intersecting the origin. When $b < b_c$ (see Region 1 in Fig. 3.3 and the black lines in Fig. 3.4), photons will be captured by the black hole. When $b = b_c$ (see Region 2 in Fig. 3.3 and the red rings in Fig. 3.4), photons will continue to orbit the black hole. When $b > b_c$ (see Region 3 in Fig. 3.3 and the blue lines in Fig. 3.4), photons will be reflected back and move to infinity.

With the calculations above, we can constrain the parameter γ using the observed shadows of the supermassive black hole M87* and SgrA* captured by the EHT collaboration^[11-22]. For a static observer, the radius of the black hole shadow can be expressed as^[22,170-171]

$$\mathcal{R}_{sh} = \frac{r_{ph}}{\sqrt{f(r_{ph})}}. \quad (3.21)$$

In units of the black hole mass, the diameter of the black hole shadow in the practical observations is given by

$$d_{sh} = 2\mathcal{R}_{sh} = \frac{D\theta}{M}, \quad (3.22)$$

where θ is the angular diameter of the black hole shadow and D is the distance between the black hole and the observer. According to Refs.^[11,19], the angular diameter of the shadow for M87* is $\theta_{M87*} = (42 \pm 3) \mu\text{as}$. The distance from the Earth is $D_{M87*} = 16.8^{+0.8}_{-0.7}$

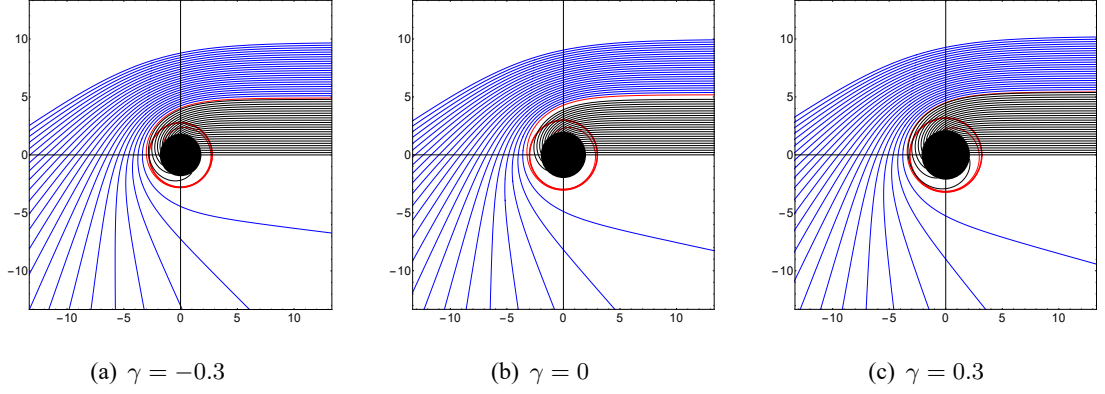


Figure 3.4: The trajectories of photons moving on the equatorial plane of the black hole for different values of the parameter γ in polar coordinates (r, φ) . The black hole is depicted as a black disk in the center. The red rings represent the critical curve. The black lines and blue lines correspond to $b < b_c$ and $b > b_c$, respectively. We set $\gamma = -0.3, 0, 0.3$ and $M = 1$. The spacing of the impact parameter is set to $1/5$.

Mpc and the mass is $M_{\text{M87}^*} = (6.5 \pm 0.9) \times 10^9 M_\odot$. For SgrA*, the angular diameter is $\theta_{\text{SgrA}^*} = (48.7 \pm 7) \mu\text{as}$. The distance is $D_{\text{SgrA}^*} = 8277 \pm 33 \text{ pc}$ and the mass is $M_{\text{SgrA}^*} = (4.3 \pm 0.013) \times 10^6 M_\odot$. Hence, the diameters of the shadow images of M87* and SgrA* are $d_{\text{sh}}^{\text{M87}^*} = (11 \pm 1.5)$ and $d_{\text{sh}}^{\text{SgrA}^*} = (9.5 \pm 1.4)$, respectively. Taking these results into Eq. (3.21), then we can constrain the parameter λ (see Fig. 3.5 for details). Based on the confidence intervals for $d_{\text{sh}}^{\text{M87}^*}$, the parameter γ is constrained to $-0.46 \lesssim \gamma \lesssim 1.50$ at the 1σ confidence level and $-1.00 \lesssim \gamma \lesssim 2.92$ at the 2σ confidence level. For $d_{\text{sh}}^{\text{SgrA}^*}$, the parameter γ is constrained to $-0.97 \lesssim \gamma \lesssim 0.31$ at the 1σ confidence level and $-1.12 \lesssim \gamma \lesssim 1.34$ at the 2σ confidence level. Apparently, the constraint on the upper limit of the parameter γ obtained from SgrA* is stricter than that obtained from M87*. But the constraint on the lower limit of the parameter γ from M87* is more stringent than that from SgrA*. Therefore, combining the observations of M87* and SgrA*, and considering the 1σ confidence level, the parameter γ can be strictly constrained within the range $-0.46 \lesssim \gamma \lesssim 0.31$. To simplify the problem, our subsequent discussions regarding γ will be confined to this range.

3.3 Optical appearances of the black hole in Horndeski theory

In this section, we investigate the optical appearances of the black hole encircled by a geometrically thin accretion disk in Horndeski theory. We analyze the impact of the parameter γ on these optical appearances.

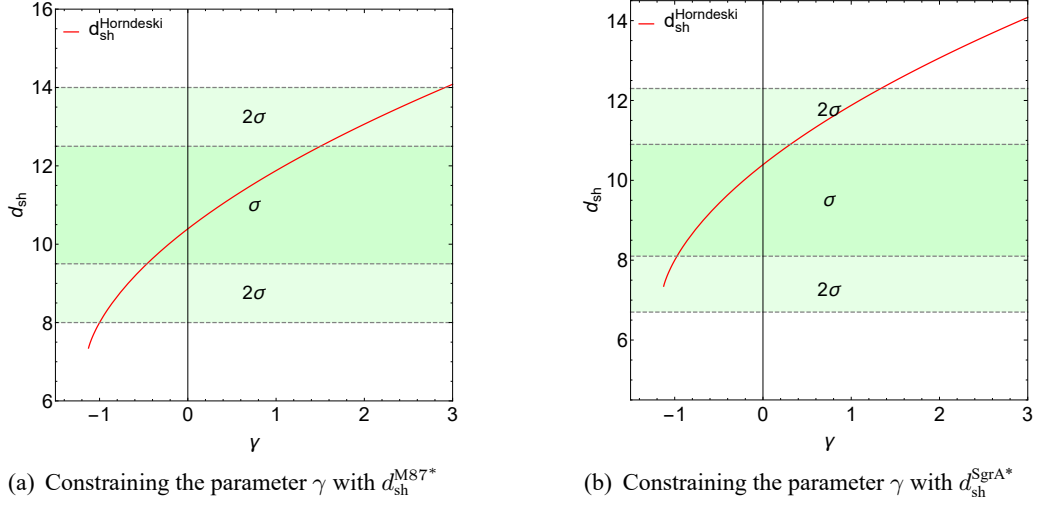


Figure 3.5: The diameter $d_{\text{sh}}^{\text{Horndeski}}$ (the red lines) of the black hole shadow varies with the parameter γ . The shaded areas represent the value of $d_{\text{sh}}^{\text{Horndeski}}$ that are consistent with the observed shadow diameters of M87* (the left panel) and SgrA* (the right panel) given by the EHT. The green shaded areas indicate the 1σ confidence level for $d_{\text{sh}}^{\text{Horndeski}}$, while the light green shaded areas indicate the 2σ confidence level. We set $M = 1$.

3.3.1 Direct emission, photon ring, and lensing ring

To investigate the optical appearances of the black hole in Horndeski theory, we suppose that the equatorial plane of the black hole is fixed. The static observer and the light source are positioned at infinity in the north pole and south pole directions, respectively. We define the total number of photon orbits around the black hole as $N \equiv \Delta\varphi/(2\pi)$, where $\Delta\varphi$ is the total deflection angle of the photon given by Eq. (3.19). Based on the value of N , one can classify the light rays into the following three categories^[23]:

- *Direct Emission:* $N < 3/4$. Light rays and the equatorial plane intersect at most once;
- *Lensing Ring:* $3/4 < N < 5/4$. Light rays and the equatorial plane intersect twice;
- *Photon Ring:* $N > 5/4$. Light rays and the equatorial plane intersect at least three times.

For the black hole in Horndeski theory, the total number N of photon orbits is only dependent on the impact parameter b/M and the parameter γ . We plot the relationship between the total number N of photon orbits and the impact parameter b/M for different values of the parameter γ ($= -0.3, 0, 0.3$), as shown in Fig. 3.6. The graph is divided into three regions: “Direct Emission”, “Lensing Ring”, and “Photon Ring” corresponding to the three categories of light rays mentioned above. These regions are separated by

two horizontal dashed lines at $N = 3/4$ and $N = 5/4$. In Tab 3.1, we present the ranges of the impact parameter corresponding to the direct emission, lensing ring, and photon ring, respectively. For the three cases of γ , the evolution of N with respect to the impact parameter b/M is consistent, i.e., N increases with b/M , reaches a peak, and then decreases. Comparing the peaks corresponding to different values of the parameter γ , we find that the peak point shifts to the right along the horizontal axis as the parameter γ increases. This indicates that the size of the photon ring increases with the parameter γ .

Parameter γ	Direct Emission	Lensing Ring	Photon Ring
-0.3	$b/M \notin (4.720, 5.948)$	$b/M \in (4.720, 4.908)$ or $b/M \in (4.950, 5.948)$	$b/M \in (4.908, 5.948)$
0	$b/M \notin (5.014, 6.181)$	$b/M \in (5.014, 5.187)$ or $b/M \in (5.231, 6.181)$	$b/M \in (5.187, 6.181)$
0.3	$b/M \notin (5.272, 6.422)$	$b/M \in (5.272, 5.435)$ or $b/M \in (5.461, 6.422)$	$b/M \in (5.435, 6.422)$

Table 3.1: The ranges of the impact parameter b/M corresponding to the direct emission, lensing ring, and photon ring, respectively. We set $\gamma = -0.3, 0, 0.3$ and $M = 1$.

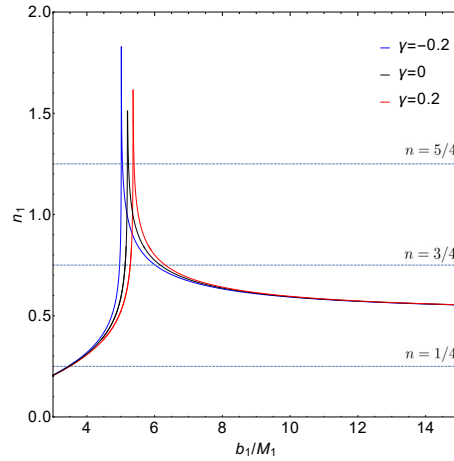


Figure 3.6: The relationship between the total number N of photon orbits and the impact parameter b/M for the black hole in Horndeski theory. The blue, black, and red lines correspond to $\gamma = -0.3, \gamma = 0$, and $\gamma = 0.3$, respectively. We set $M = 1$.

Based on the above classification of light rays, we plot the trajectories of photons for different values of the parameter γ in polar coordinates (r, φ) (see Fig. 3.7). In the figure, the black disk represents the black hole in Horndeski theory, the red ring indicates the photon sphere, and the vertical black line passing through the origin denotes the equatorial plane. The observer is positioned at infinity in the north polar direction, corresponding to the right side of the panel. Taking $\gamma = -0.3$ as an example (see Fig. 3.7(a)), one can find that for the observer, the direct emission occupies a significant portion of the observed appearances of the black hole (see the red lines). In the figure, the direct emission can be divided into upper and lower sections based on the value of the impact parameter b/M . In the upper section ($1/4 < N < 3/4$), photons with larger impact parameters intersect the

equatorial plane only once and then escape the black hole. In the lower section ($N < 1/4$), photons with smaller impact parameters fall into the black hole. The lensing ring occupies a smaller central region of the observed appearances of the black hole (see the blue lines). For the lensing ring, all photons intersect the equatorial plane twice, and the distance between the two intersections increases with the impact parameter. Similarly, the lensing ring could also be divided into upper and lower sections based on the value of the impact parameter b/M . The upper section (photons with larger impact parameters) could escape the black hole, while the lower section (photons with smaller impact parameters) will fall into the black hole. As for the photon ring, it occupies only a very narrow region of the observed appearance of the black hole (see the green lines). The corresponding photons intersect the equatorial plane more than three times. Comparing the three cases corresponding to different γ 's, it is observed that the trajectories of photons in the three pictures exhibit similarity. The influence of γ on the trajectories of photons is visually evident in the radii of the black disk and the photon sphere. Both of them expand as γ increases (see Figs. 3.7(a), 3.7(b), and 3.7(c), click to zoom in).

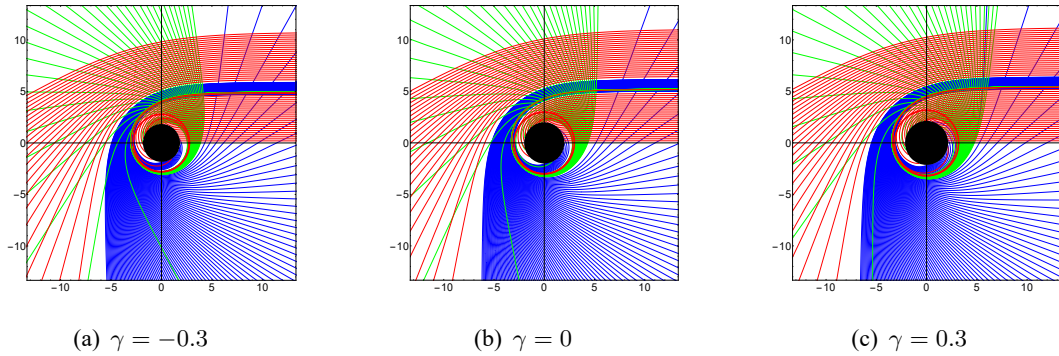


Figure 3.7: The trajectories of photons for different values of the parameter γ in polar coordinates (r, φ) . The red lines, blue lines and green lines represent the direct emission, lensing ring, and photon ring, respectively. We set $\gamma = -0.3, 0, 0.3$ and $M = 1$. The spacing of the impact parameter for the direct emission, lensing ring, and photon ring is set to $1/5, 1/100$, and $1/1000$, respectively.

3.3.2 Transfer functions

Now, we study the transfer functions of the black hole surrounded by a geometrically thin accretion disk in Horndeski theory. The disk is still located on the equatorial plane and emits radiation isotropically for a static observer at infinity in the north pole. The emitted specific intensity of the disk can be expressed as

$$I_{\nu}^{\text{em}} = I(r), \quad (3.23)$$

where ν represents the emission frequency in the static frame. Considering that I_ν/ν^3 remains conserved along a ray, the specific intensity of the radiation emitted from a radius r and received at any frequency ν' can be given by^[23]

$$I_{\nu'}^{\text{obs}} = f(r)^{3/2} I(r), \quad (3.24)$$

where $f(r) = 1 - \frac{2M}{r} - \frac{\gamma}{r^2}$. Then, we can obtain the total observed specific intensity as follows^[23]:

$$I^{\text{obs}} = \int I_{\nu'}^{\text{obs}} d\nu' = \int f(r)^2 I_\nu^{\text{em}} d\nu = f(r)^2 I(r). \quad (3.25)$$

Since we are considering a geometrically thin accretion disk located on the equatorial plane, each time a photon intersects this plane, it acquires brightness from the disk's emission, as shown in Fig. 3.7. Consequently, the observed intensity is given by^[23]

$$I^{\text{obs}}(b) = \sum_n f(r)^2 I \Big|_{r=r_n(b)}, \quad (3.26)$$

which is a sum of the intensities from each intersection. Here, $r_n(b)$ ($n = 1, 2, 3, \dots$) is the transfer function, which represents the radial coordinate of the n -th intersection point between the photon and the geometrically thin accretion disk. The slope of the function $r_n(b)$ at each b yields the demagnification factor at the point, denoted as dr/db . In Fig. 3.8, we plot the first three transfer functions for different values of the parameter γ . One can find that for a given γ , there exists a critical impact parameter. When an impact parameter is less than the critical impact parameter, the transfer functions do not support it. Taking $\gamma = -0.3$ as an example (see Fig. 3.8(a)), one can observe that the transfer functions do not support $b \lesssim 2.63M$. This critical value is greater than the event horizon radius ($r_h \approx 1.84M$), but smaller than the photon sphere radius ($r_{\text{ph}} \approx 2.78M$). The first transfer function with a slope close to 1 ($n = 1$) corresponds to the direct emission (see the red line in Fig. 3.8(a)), which represents the redshifted source profile of the geometrically thin accretion disk. The second transfer function with a steeper slope ($n = 2$) pertains to the lensed ring (see the blue line in Fig. 3.8(a)), which, for the observer, manifests a highly demagnified image of the back side of the disk. The third transfer function with the steepest slope ($n = 3$) is associated with the photon ring (see the green line in Fig. 3.8(a)), which corresponds to an extremely demagnified image of the front side of the disk. This implies that the first transfer function significantly contributes to the total flux, while the second transfer function contributes to a lesser extent, and the third transfer function makes very little contribution to the total flux. For $\gamma = 0$ (see Fig. 3.8(b)) and $\gamma = 0.3$ (see Fig. 3.8(c)), the transfer functions exhibit similarities to the case when $\gamma = -0.3$.

However, a larger value of the parameter γ results in an increased initial b/M value for the transfer functions. But, when the parameter γ is larger, the transfer function has an increased critical impact parameter.

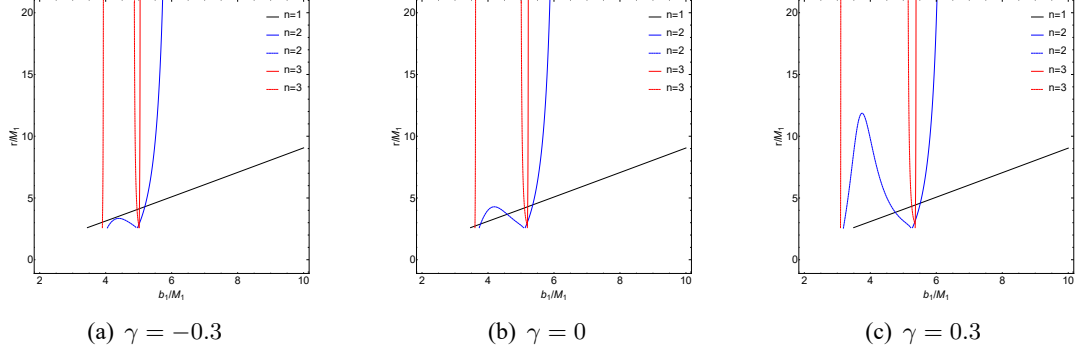


Figure 3.8: The first three transfer functions of the black hole in Horndeski theory for different values of the parameter γ . The red, blue, and green lines represent the radial coordinates of photons at their first, second, and third intersections with the geometrically thin accretion disk, respectively. We set $\gamma = -0.3, 0, 0.3$ and $M = 1$.

3.3.3 Optical appearances for three emission models

In this section, we investigate the optical appearances of the black hole in Horndeski theory using three specific emission models of geometrically thin accretion disks, as illustrated in Fig. 3.9.

The first emission model can be expressed as

$$I_1^{\text{em}}(r) = \begin{cases} I_0 \left[\frac{1}{r - (r_{\text{isco}} - 1)} \right]^2, & r > r_{\text{isco}}, \\ 0, & r \leq r_{\text{isco}}. \end{cases} \quad (3.27)$$

Here, we assume that the emission starts from the innermost stable circular orbit r_{isco} . Then, it suddenly increases, reaches a peak, and rapidly declines, as shown in Fig. 3.9(a).

The second emission model assumes that the emission starts from the event horizon and decays towards the innermost stable circular orbit at a slower rate compared to the first emission model. As shown in Fig. 3.9(b), the emitted specific intensity of the model is given by

$$I_2^{\text{em}}(r) = \begin{cases} I_0 \frac{\frac{\pi}{2} - \arctan[r - (r_{\text{isco}} - 1)]}{\frac{\pi}{2} - \arctan[r_h - (r_{\text{isco}} - 1)]}, & r > r_h, \\ 0, & r \leq r_h. \end{cases} \quad (3.28)$$

In the third emission model, the emission is assumed to occur at the photon sphere radius r_{ph} and then rapidly decay at a cubic rate, as shown in Fig. 3.9(c). The emitted

specific intensity is given by

$$I_3^{\text{em}}(r) = \begin{cases} I_0 \left[\frac{1}{r - (r_{\text{ph}} - 1)} \right]^3, & r > r_{\text{ph}}, \\ 0, & r \leq r_{\text{ph}}. \end{cases} \quad (3.29)$$

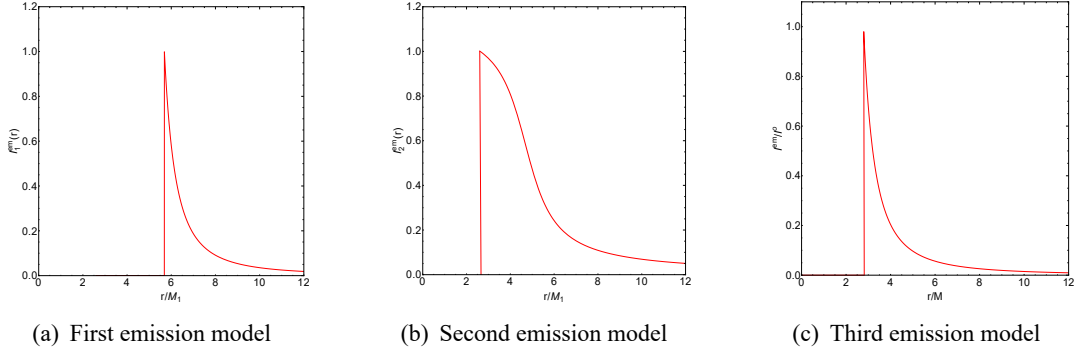


Figure 3.9: The emitted specific intensities of geometrically thin accretion disks for three emission models.

Combining the observed intensity (3.26) and the three emission models (3.27-3.29), we plot the optical appearances of the black hole in Horndeski theory with $\gamma = -0.3$, see Fig. 3.10. For other valid values of γ (e.g., $\gamma = 0$ and $\gamma = 0.3$), it is challenging to make intuitive comparisons between their results and the results obtained for $\gamma = -0.3$. Therefore, we will not present additional results here and focus solely on $\gamma = -0.3$ as an example for further discussion.

For the first emission model, the observed specific intensity has a peak at $b/M \approx 6.43$ (see Figs. 3.10(a)), corresponding to the direct emission. At $b/M \approx 5.24$, there is a narrow and slightly lower peak, which corresponds to the lensing ring. At the far left of Fig. 3.10(a), there is an extremely narrow peak at $b/M \approx 4.95$, which corresponds to the photon ring. Based on the first emission model, we present the two-dimensional image of the black hole as observed by the observer in Fig. 3.10(d). It is found that there is a bright ring that gradually fades in intensity as it extends outward from the outer edge of the dark region, which represents the direct emission. Since the demagnification factor for the first transfer function corresponding to the direct emission is close to 1, it contributes the majority of the total flux. In the dark region of the two-dimensional image, there is a narrow bright ring representing the lensing ring. The demagnification factor for the second transfer function corresponding to the lensing ring is relatively large, so it contributes a smaller portion of the total flux. Moreover, clicking to zoom in Fig. 3.10(d), at the innermost part of the two-dimensional image, one can see that there is a very narrow

and faint ring (i.e., the photon ring) near the lensing ring. The demagnification factor for the third transfer function corresponding to the photon ring is extremely large. As a result, it makes a minimal contribution to the total flux, causing the photon ring to appear less prominent in the two-dimensional image.

The observed specific intensity for the second emission model is shown in Fig. 3.10(b). We find that the direct emission appears at $b/M \approx 2.66$, and the lensed ring overlaps with the direct emission at $b/M \approx 4.74$. From the two-dimensional image of the black hole shown in Fig. 3.10(e), we can also clearly see that the brightness of the two-dimensional image appears at $b/M \approx 2.66$ and increases with the radius.

The observed specific intensity for the third emission model is shown in Fig. 3.10(c). We find that the direct emission appears at $b/M \approx 3.61$, and the lensing ring appears at $b/M \approx 4.94$. The lensing ring has a higher peak intensity compared to the direct emission. As seen in Fig. 3.10(f), the brightness of the two-dimensional image suddenly appears at $b/M \approx 3.61$ and decreases with the radius. Then, the bright light appears again at $b/M \approx 4.94$ and decreases with the radius again. Compared to the first and second emission models, the emitted specific intensity of the third emission model (which starts at the event horizon) results in a dimmer two-dimensional image due to the redshift effect.

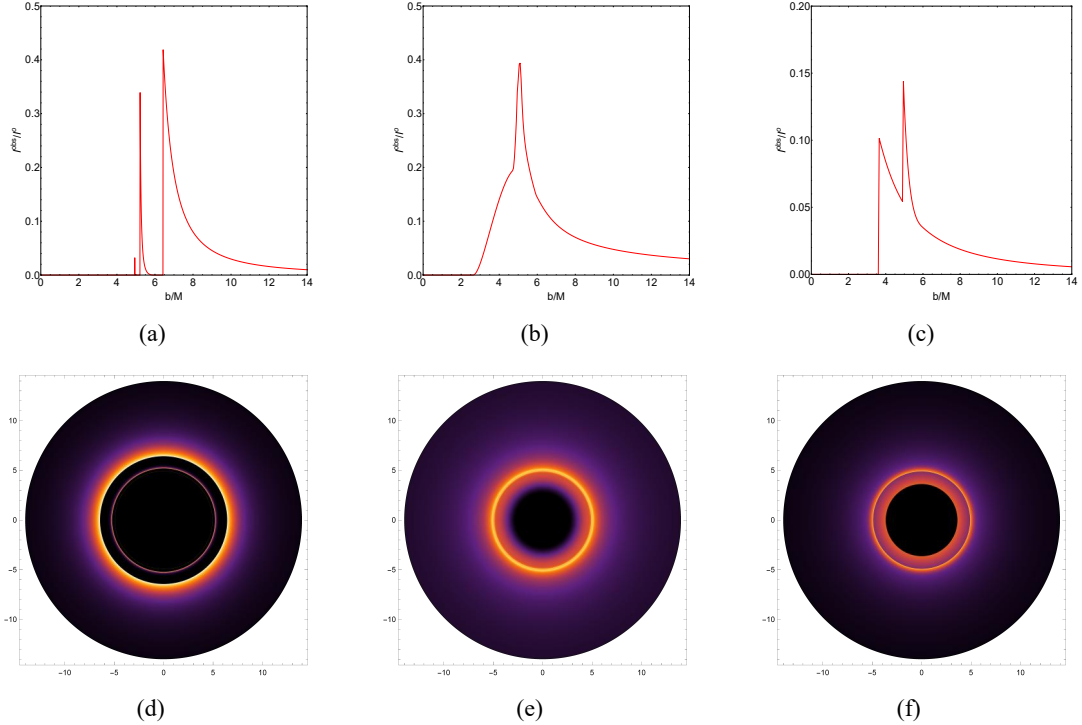


Figure 3.10: The observed specific intensities and two-dimensional images of the black hole in Horndeski theory for three emission models of geometrically thin accretion disks. We set $\gamma = -0.3$ and $M = 1$.

To enable a direct comparison with the black hole images provided by the EHT,

we apply a Gaussian filter with a width of $20\mu\text{as}$ to smooth the images presented in Figs. 3.10(d), 3.10(e), and 3.10(f)^[11,172-173], and then we generate Fig. 3.11. It can be observed that the application of the filter results in the removal of the distinct features associated with the lensing ring and the photon ring. Consequently, only a blurred image of the direct emission remains, with no change in its size. This suggests that, for the current resolution of the EHT, it is challenging to obtain more information about the lensing ring and photon ring in the black hole images within the framework of Horndeski theory.

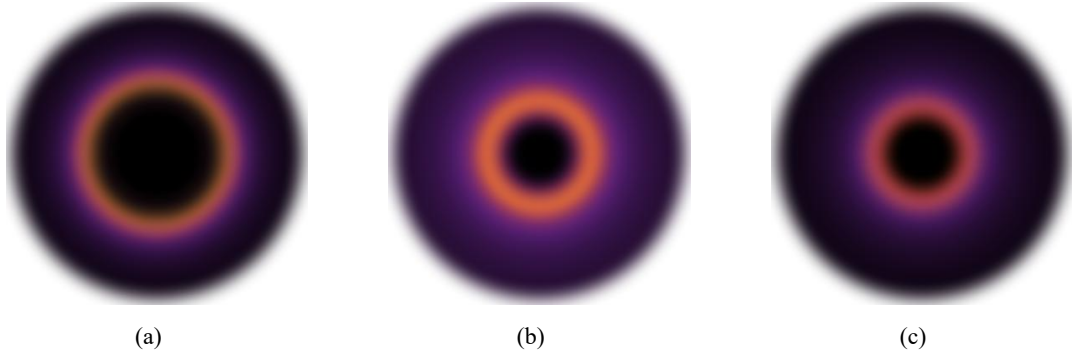


Figure 3.11: The blurred two-dimensional images with a Gaussian filter with a width of $20\mu\text{as}$.

3.4 Effective potential and null geodesic of the asymmetric thin-shell wormhole in Horndeski theory

In this section, inspired by the works^[174-176], we employ the cut-and-paste method to construct an ATW in Horndeski theory. The ATW consists of two distinct spacetimes, $\mathcal{M}_1$ and $\mathcal{M}_2$, with different mass parameters. They are glued by a thin shell (i.e., “throat”) and the whole manifold can be labeled as $\mathcal{M} \equiv \mathcal{M}_1 \cup \mathcal{M}_2$.

According to Ref.^[169], we consider a special case of the action (3.1), i.e., $G_2 = \eta X$, $G_4 = \zeta + \beta\sqrt{-X}$, and $G_3 = G_5 = 0$. Note that η and β are dimensionless parameters, but $\zeta = \frac{1}{2\kappa}$ (κ is the Einstein gravitational constant) is just a fixed constant. Hence, the action (3.1) can be reduced to

$$S = \int d^4x \sqrt{-g} \left\{ \left[\zeta + \beta\sqrt{(\partial\phi)^2/2} \right] \mathcal{R} - \frac{\eta}{2}(\partial\phi)^2 - \frac{\beta}{\sqrt{2(\partial\phi)^2}} [(\Box\phi)^2 - (\nabla_\mu \nabla_\nu \phi)^2] \right\}. \quad (3.30)$$

The field equations derived from the action (3.30) can give rise to the following black hole solution^[169]:

$$ds^2 = -f(r)dt^2 + \frac{1}{h(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (3.31)$$

with

$$f(r) = h(r) = 1 - \frac{2M}{r} - \frac{\beta^2}{2\zeta\eta r^2}, \quad (3.32)$$

where M is a free integration constant representing the mass of the black hole. It is worth noting that both parameters β and η share the same sign, and so the scalar field can be denoted as

$$\phi(r) = \pm 2\sqrt{\frac{\zeta}{\eta}} \left\{ \text{Arctan} \left[\frac{\beta^2 + 2M\zeta\eta r}{\beta\sqrt{2\zeta\eta r(r-2M) - \beta^2}} \right] - \text{Arctan} \left(\frac{2M}{\beta} \sqrt{\frac{\zeta\eta}{2}} \right) \right\} \text{ for } \beta > 0 \text{ \& } \eta > 0; \quad (3.33)$$

$$\phi(r) = \pm 2\sqrt{\frac{\zeta}{-\eta}} \left\{ \text{Argth} \left[\frac{\beta^2 + 2M\zeta\eta r}{\beta\sqrt{\beta^2 - 2\zeta\eta r(r-2M)}} \right] + \text{Argth} \left(\frac{2M}{\beta} \sqrt{\frac{-\zeta\eta}{2}} \right) \right\} \text{ for } \beta < 0 \text{ \& } \eta < 0. \quad (3.34)$$

Since $\beta^2/(2\zeta\eta)$ appears as a whole, for simplicity, we can define the parameter $\gamma = \beta^2/(2\zeta\eta)$. In this case, Eq. (3.32) can be re-written as

$$f(r) = h(r) = 1 - \frac{2M}{r} - \frac{\gamma}{r^2}. \quad (3.35)$$

It can be found that when γ is negative, the scalar field $\phi(r)$ exhibits behavior similar to an electric contribution, and $\sqrt{-\gamma}$ resembles an electric charge. By solving the equation $f(r_h) = 0$, the event horizon radius is given by $r_h = M + \sqrt{M^2 + \gamma}$. In order to prevent the presence of the naked singularity, it is crucial to ensure that $\gamma \geq -M^2$. When $\gamma = -M^2$, the solution describes an extremal black hole.

Utilizing the cut-and-paste method^[176], two distinct Horndeski spacetimes $\mathcal{M}_1$ and $\mathcal{M}_2$ with different mass parameters can be connected through a thin shell yielding an ATW in Horndeski theory. The metric on the whole manifold $\mathcal{M} \equiv \mathcal{M}_1 \cup \mathcal{M}_2$ (i.e., the ATW spacetime) is given by

$$ds_i^2 = -f_i(r_i) dt_i^2 + \frac{1}{f_i(r_i)} dr_i^2 + r_i^2 (d\theta_i^2 + \sin^2 \theta_i d\phi_i^2), \quad (3.36)$$

where

$$f_i(r_i) = 1 - \frac{2M_i}{r_i} - \frac{\gamma}{r_i^2}, \quad r_i \geq R. \quad (3.37)$$

Here, $i = 1, 2$ denotes distinct Horndeski spacetimes. The parameter R represents the position of the throat, which satisfies

$$R > \max \left\{ M_1 + \sqrt{M_1^2 + \gamma}, M_2 + \sqrt{M_2^2 + \gamma} \right\}. \quad (3.38)$$

Considering an incoming photon originating from the spacetime $\mathcal{M}_1$ and passing

through the throat, we assume that there is only gravitational interaction between the photon and the throat. This implies that the 4-momentum p^a ($a = t, r, \theta, \phi$) of the photon remains constant when passing through the throat. Furthermore, due to the continuity of the metric in the spacetime $\mathcal{M}$, it follows that $g_{\mu\nu}^{\mathcal{M}_1}(R) = g_{\mu\nu}^{\mathcal{M}_2}(R)$ ^[174]. Since the metric (3.32) is static and spherically symmetric, there are two conserved quantities for the photon: $E_i = -p_{t_i}$ (energy) and $L_i = p_{\phi_i}$ (orbital angular momentum). Taking them into the equation of motion of the photon:

$$\frac{(p_i^{r_i})^2}{f_i(r_i)} = \frac{p_{t_i}^2}{f_i(r_i)} - \frac{p_{\phi_i}^2}{r_i^2}, \quad (3.39)$$

where $p_i^{a_i}$ represents the 4-momentum of the photon in the spacetime $\mathcal{M}_i$, the radial motion of the photon can be expressed as

$$p_i^{r_i} = \pm E_i \sqrt{1 - \frac{b_i^2}{r_i^2} f_i(r_i)}. \quad (3.40)$$

The symbol $\pm$ corresponds to the directions of the outgoing and incoming photons, respectively. The parameter $b_i = L_i/E_i$ denotes the impact parameter of the photon in the spacetime $\mathcal{M}_i$. With Eq. (3.40), the effective potential $V_i(r_i)$ of the ATW can be given by

$$V_i(r_i) = \frac{f_i(r_i)}{r_i^2} = \frac{1}{r_i^2} \left(1 - \frac{2M_i}{r_i} - \frac{\gamma}{r_i^2} \right). \quad (3.41)$$

For the unstable circular orbit of the photon, the effective potential satisfies the following conditions:

$$V_i(r_{ph_i}) = \frac{1}{b_{c_i}^2}, \quad V_i'(r_{ph_i}) = 0, \quad (3.42)$$

where r_{ph_i} is the radius of the photon sphere and b_{c_i} is the critical impact parameter. Note that these two parameters only depend on M_i and γ .

In this study, we focus on how to distinguish between ATWs and black holes in Horndeski theory when the event horizon is obscured by a photon sphere. We can suppose that the observer is situated in the spacetime $\mathcal{M}_1$ with $M_1 = 1$ and the mass parameter of the spacetime $\mathcal{M}_2$ can be set to $M_2 = k$. According to Refs.^[174,177], k and R should satisfy

$$1 < k < \frac{R}{2} \leq \frac{r_{ph_1}}{2}. \quad (3.43)$$

In fact, for any $1 < k < \frac{R}{2}$, our subsequent discussion and results are similar. Therefore, we follow the practice adopted in Refs.^[175,177] and let $R = 2.6$ for simplicity. The impact parameters b_1 and b_2 in the spacetimes $\mathcal{M}_1$ and $\mathcal{M}_2$ can be connected through the

following equation^[174]:

$$\frac{b_1}{b_2} = \sqrt{\frac{f_2(R)}{f_1(R)}} = \sqrt{\frac{1 - \frac{2M_2}{R} - \frac{\gamma}{R^2}}{1 - \frac{2M_1}{R} - \frac{\gamma}{R^2}}} \equiv Z. \quad (3.44)$$

For the given $M_1 = 1$ and $M_2 = 1.2$, to study the influence of the parameter γ on the values of r_{ph_i} and b_{c_i} , we use Eq. (3.42) to calculate r_{ph_i} and b_{c_i} for various positive and negative values of γ , as shown in Eqs. (3.45)-(3.48) and Table 3.2. It is worth noting that $\gamma = 0$ and $\gamma < 0$ correspond to the Schwarzschild case and the Reissner-Nordström case, respectively. The results demonstrate a direct correlation between an increase in the parameter γ and increases in both r_{ph_i} and b_{c_i} .

$$r_{ph_1} = \frac{1}{2}(3 + \sqrt{9 + 8\gamma}), \quad (3.45)$$

$$b_{c_1} = \frac{\sqrt{27 + 36\gamma + 8\gamma^2 + 9\sqrt{9 + 8\gamma} + 8\gamma\sqrt{9 + 8\gamma}}}{\sqrt{2}\sqrt{1 + \gamma}}, \quad (3.46)$$

$$r_{ph_2} = \frac{1}{5}(9 + \sqrt{81 + 50\gamma}), \quad (3.47)$$

$$b_{c_2} = \frac{2\sqrt{486(9 + \sqrt{81 + 50\gamma}) + 25\gamma(162 + 25\gamma + 12\sqrt{81 + 50\gamma})}}{5\sqrt{36 + 25\gamma}}. \quad (3.48)$$

γ	-0.2	-0.1	0	0.1	0.2
b_{c_1}	5.01561	5.10779	5.19615	5.28114	5.36311
r_{ph_1}	2.86015	2.93178	3	3.06525	3.12788
b_{c_2}	6.08691	6.16221	6.23538	6.30659	6.37602
r_{ph_2}	3.48523	3.54356	3.6	3.65472	3.70788

Table 3.2: The radius r_{ph_i} of the photon sphere and the critical impact parameter b_{c_i} . We consider five specific values for the parameter γ : -0.2, -0.1, 0, 0.1, 0.2. For r_{ph_1} and b_{c_1} , the mass parameter is $M_1 = 1$, while for r_{ph_2} and b_{c_2} , the mass parameter is $M_2 = 1.2$.

In addition, changing the mass parameters will affect the geometric structure of the ATW spacetime (3.36). Without loss of generality, we still set $M_1 = 1$ and then consider five specific values for the mass parameter M_2 in the ATW spacetime, as shown in Eqs. (3.49)-(3.52) and Table 3.3. We focus on the effect of changes in the mass parameter M_2 on the ATW spacetime. In Table 3.3, the radius r_{ph_2} of the photon sphere and the critical impact parameter b_{c_2} show an increasing trend with the mass parameter M_2 . Furthermore, the critical impact parameter b_{c_2} in Table 3.3 is consistent with that in Table 3.2, but the corresponding radius r_{ph_2} of the photon sphere in Table 3.3 is larger than that in Table 3.2. The results indicate that, although the critical impact parameters b_{c_2} (i.e., the

shadow radii) in these two ATW spacetimes are equal, the observational appearances of these two ATWs could be different.

$$r_{ph_1} = \frac{1}{10}(15 + \sqrt{265}), \quad (3.49)$$

$$b_{c_1} = \sqrt{\frac{1}{30} \left(34 + \frac{53}{2}(15 + \sqrt{265}) \right)}, \quad (3.50)$$

$$r_{ph_2} = \frac{1}{10} \left(15M_2 + \sqrt{5} \sqrt{8 + 45M_2^2} \right), \quad (3.51)$$

$$b_{c_2} = \frac{\sqrt{8 + 180M_2^2 + 675M_2^4 + 8\sqrt{5}M_2\sqrt{8 + 45M_2^2} + 45\sqrt{5}M_2^3\sqrt{8 + 45M_2^2}}}{\sqrt{10}\sqrt{1 + 5M_2^2}}. \quad (3.52)$$

M_2	1.14308	1.15791	1.17232	1.18634	1.2
b_{c_2}	6.08691	6.16221	6.23538	6.30659	6.37602
r_{ph_2}	3.54218	3.58532	3.62726	3.66807	3.70788

Table 3.3: The radius r_{ph_2} of the photon sphere and the critical impact parameter b_{c_2} . We consider five specific values for the mass parameter M_2 : 1.14308, 1.15791, 1.17232, 1.18634, 1.2. We set $\gamma = 0.2$ and $M_1 = 1$, then we can get $b_{c_1} \approx 5.36311$ and $r_{ph_1} \approx 3.12788$.

In Fig. 3.12, we plot the effective potentials of ATWs and black holes for different values of γ . According to Eq. (3.44), we rescale the effective potential $V_2(r_2)$ in the spacetime $\mathcal{M}_2$ by a factor of Z^2 , as shown in Fig. 3.12(a). It is found that for a given γ , the trajectories of photons can be classified into three scenarios. For example, when $\gamma = -0.2$ (see the blue and green lines), if $b_1 > b_{c_1} = 5.01561$, the photons in spacetime $\mathcal{M}_1$ will approach turning points in the spacetime $\mathcal{M}_1$ from infinity and then return to infinity in spacetime $\mathcal{M}_1$, i.e., the photons remain within the spacetime $\mathcal{M}_1$ throughout their trajectories. If $Zb_{c_2} = 3.89321 < b_1 < b_{c_1}^c$, the photons in the spacetime $\mathcal{M}_1$ will reach turning points in the spacetime $\mathcal{M}_2$ and then return to the spacetime $\mathcal{M}_1$. If $b_1 < Zb_{c_2}$, the photons in spacetime $\mathcal{M}_1$ will fall into the spacetime $\mathcal{M}_2$ and then go to infinity in the spacetime $\mathcal{M}_2$.

For the effective potentials of black holes [see Fig. 3.12(b)], one can find that a light ray's trajectory can also be classified into three distinct cases^[23] [see the illustration positioned in the bottom right corner in Fig. 3.12(b)]. When the impact parameter of a photon is less than the critical impact parameter b_c (for example see Region 3 for $\gamma = -0.2$ and $b_c = 5.01561$), it will fall into the black hole. When the impact parameter of a photon

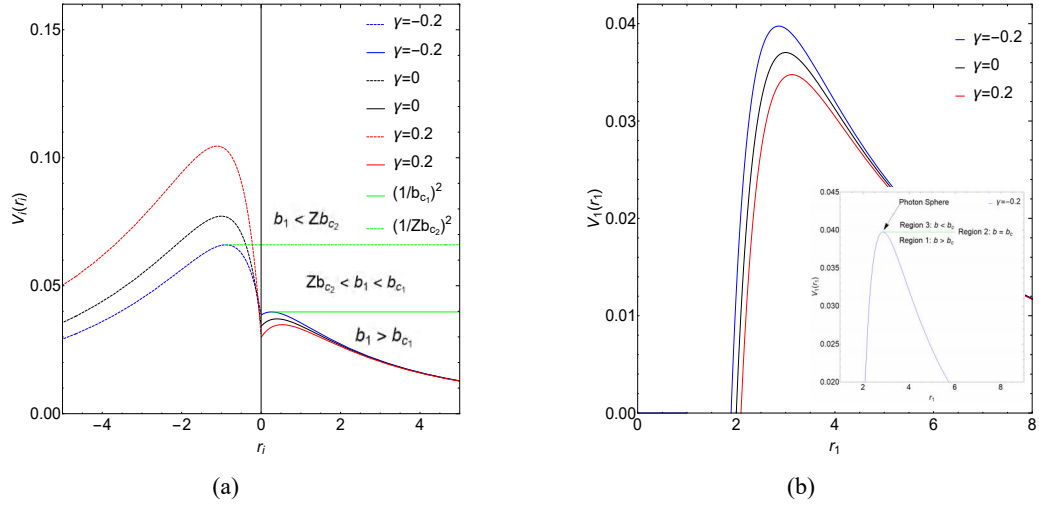


Figure 3.12: The effective potentials of ATWs and black holes. We consider three specific values for the parameter γ : -0.2 , 0 , 0.2 . The left panel is for ATWs, where the solid and dashed lines represent the effective potentials of the spacetime $\mathcal{M}_1$ and the spacetime $\mathcal{M}_2$, respectively. When $\gamma = -0.2$, we have $b_{c1} = 5.01561$ and $Zb_{c2} = 3.89321$ (see the green lines). The right panel is for black holes. We set $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

is equal to the critical value b_c (for example see Region 2 for $\gamma = -0.2$ and $b_c = 5.01561$), it can undergo periodic circular motion around the black hole. When the impact parameter of a photon is larger than b_c (for example see Region 1 for $\gamma = -0.2$ and $b_c = 5.01561$), the gravitational force will cause the deflection of the light ray's trajectory. Moreover, Fig. 3.12(b) illustrates that the peak of the effective potential increases with the parameter γ .

To get more information about the observational appearance of the ATW in Horndeski theory, it is necessary to study the trajectories and deflection angles of photons as they propagate in the ATW spacetime. With Eq. (3.39), the trajectory of a photon in the ATW spacetime can be formulated as

$$\frac{1}{b_i^2} - \frac{f_i(r_i)}{r_i^2} = \frac{1}{r_i^4} \left(\frac{dr_i}{d\phi_i} \right)^2. \quad (3.53)$$

By implementing the coordinate transformation $x \equiv 1/r$, one can obtain

$$G_i(x_i) = \left(\frac{dx_i}{d\phi} \right)^2 = \frac{1}{b_i^2} - x_i^2 (1 - 2M_i x_i - x_i^2 \gamma). \quad (3.54)$$

When $b_1 > b_{c1}$, the photon remains in the spacetime $\mathcal{M}_1$. The turning point relies on the smallest positive root of $G_1(x_1) = 0$, which is denoted as x_1^{min} . Referring to Eq. (3.54), the deflection angle (the total change in the azimuth angle) of the photon in the spacetime

$\mathcal{M}_1$ is

$$\phi_1(b_1) = 2 \int_0^{x_1^{min}} \frac{dx_1}{\sqrt{G_1(x_1)}}, \quad b_1 > b_{c_1}. \quad (3.55)$$

When $Zb_{c_2} < b_1 < b_{c_1}$, the photon could reach the turning point in the spacetime $\mathcal{M}_2$ and return to the spacetime $\mathcal{M}_1$. The deflection angle of the photon in the spacetime $\mathcal{M}_1$ is

$$\phi_1^*(b_1) = \int_0^{1/R} \frac{dx_1}{\sqrt{G_1(x_1)}}, \quad b_1 < b_{c_1}. \quad (3.56)$$

The turning point in the spacetime $\mathcal{M}_2$ is determined by the largest positive root of $G_2(x_2) = 0$, which is denoted as x_2^{max} . In this case, the impact parameter b_2 of the photon in the spacetime $\mathcal{M}_2$ is given by Eq. (3.44). Therefore, the deflection angle of the photon in the spacetime $\mathcal{M}_2$ is

$$\phi_2(b_2) = 2 \int_{x_2^{max}}^{1/R} \frac{dx_2}{\sqrt{G_2(x_2)}}, \quad b_2 > b_{c_2}. \quad (3.57)$$

When $b_1 < Zb_{c_2}$, since the photon will drop into the spacetime $\mathcal{M}_2$ and then move to infinity in the spacetime $\mathcal{M}_2$, we do not need to consider the deflection angle of the photon.

Using Eqs. (3.55)-(3.57), we can plot the trajectories of photons in the ATW spacetime for different values of the parameter γ and the impact parameter b_1 . The results are illustrated in Fig. 3.13. Here, we take the case of $\gamma = -0.2$ [see Figs. 3.13(a), 3.13(b), and 3.13(c)] as an example of the trajectories of photons when $Zb_{c_2} < b_1 < b_{c_1}$. For a light ray originating from infinity in the spacetime $\mathcal{M}_1$, it can be seen that when the impact parameter b_1 decreases, the light trajectory within the spacetime $\mathcal{M}_2$ will be longer. From the left column and the right column in Fig. 3.13, one can find that b_{c_1} increases with the parameter γ , while Zb_{c_2} decreases with it.

3.5 Observational appearance of the asymmetric thin-shell wormhole

Since the reflection mechanism of ATWs differentiates them from that of black holes, the observable appearance of ATWs is different from that of black holes. In this section, we consider the emission originating from an optically and geometrically thin disk encircling an ATW in Horndeski theory. By comparing the observational appearances of the ATW and a black hole with the same mass parameter, we can find that there exist distinct properties in the observational appearance of the ATW. Without loss of generality, we consider an observer located at the North Pole in the spacetime $\mathcal{M}_1$ for an ATW with a thin accretion disk.

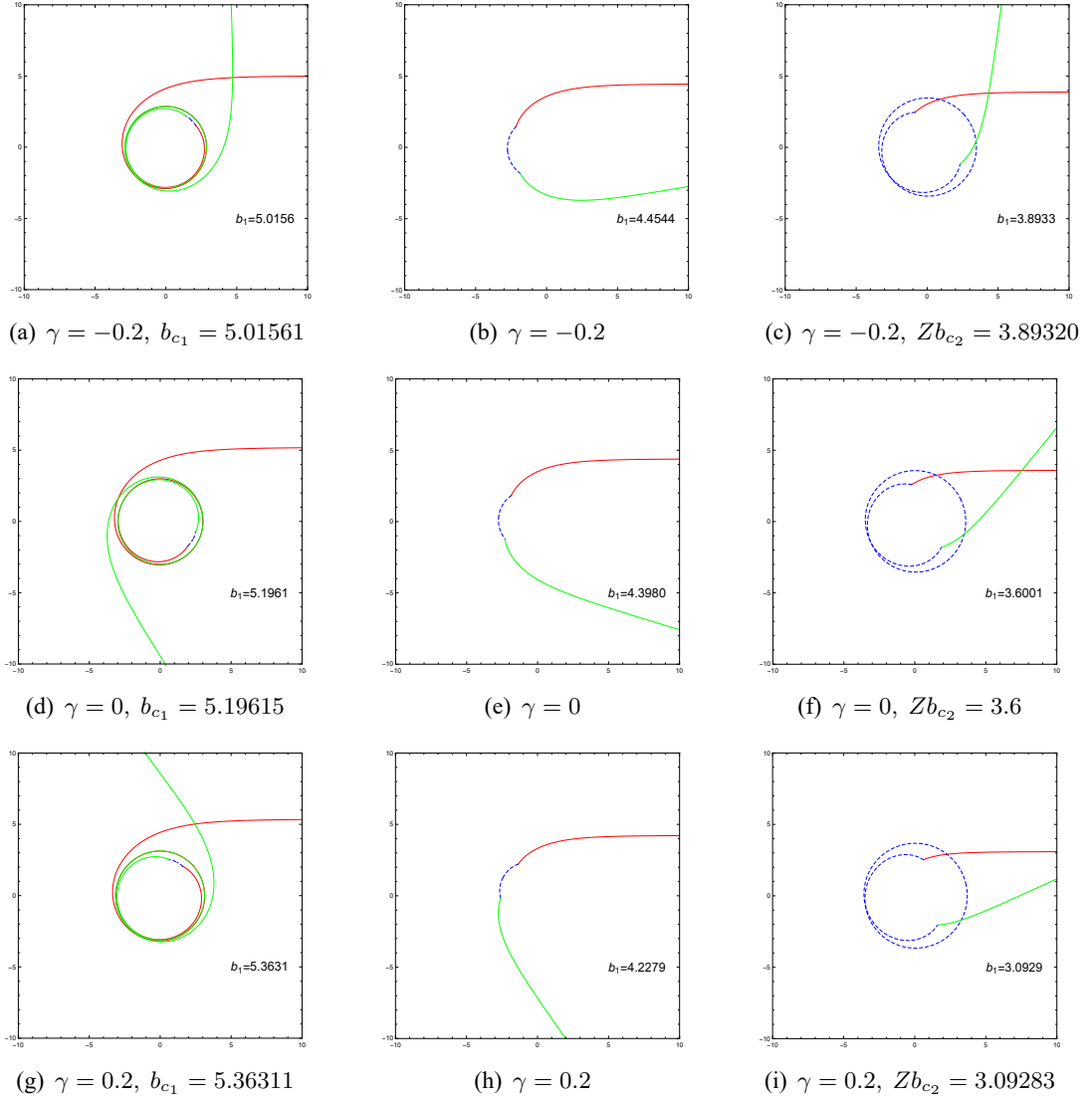


Figure 3.13: The trajectories of photons in the polar coordinate (r_1, ϕ) with the impact parameter $Zb_{c_2} < b_1 < b_{c_1}$. The top, middle, and bottom panels correspond to $\gamma = -0.2$, $\gamma = 0$, and $\gamma = 0.2$, respectively. The incoming and outgoing light trajectories in the spacetime $\mathcal{M}_1$ are represented by the red solid lines and the green solid lines, respectively, while the light trajectories in the spacetime $\mathcal{M}_2$ indicated by the blue dashed lines. We set $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

3.5.1 Classification of light trajectories

In Ref.^[23], Gralla *et al.* investigated the observational appearance of a black hole surrounded by a thin accretion disk by considering the orbit number ($n = \phi/2\pi$) and the number of intersections between light rays and the accretion disk. Based on the orbit number, the light trajectories can be classified into three categories:

- *Direct emission:* $n < 3/4$. A light ray intersects with the equatorial plane only once;

- *Lensing ring:* $3/4 < n < 5/4$. A Light rays crosses the equatorial plane twice;
- *Photon ring:* $n > 5/4$. A light ray crosses the equatorial plane at least three times.

When a light ray originates from the spacetime $\mathcal{M}_1$ and subsequently drops into the spacetime $\mathcal{M}_2$ through the throat, we can redefine the orbit number as follows^[177]:

$$n_1(b_1) = \frac{\phi_1(b_1)}{2\pi}, \quad (3.58)$$

$$n_2(b_2) = \frac{\phi_1^*(b_1) + \phi_2(b_1/Z)}{2\pi}, \quad (3.59)$$

$$n_3(b_1) = \frac{2\phi_1^*(b_1) + \phi_2(b_1/Z)}{2\pi}. \quad (3.60)$$

Here, n_2 and n_3 are extra orbit functions for the ATW, which correspond to extra photon rings. From Eqs. (3.58)-(3.60), it can be found that the orbit number is a function of the impact parameter, as shown in Fig. 3.14 .

Here n_1 is similar to the orbit number of a black hole because the light ray remains in the spacetime $\mathcal{M}_1$ [see Fig. 3.14(a)]. Taking the parameter $\gamma = -0.2$ as an example [see the solid blue lines in Fig. 3.14(a)], with the increase in the impact parameter b_1 , the orbit number n_1 gradually increases and reaches a peak value near $b_1 = 5.01561M_1$, then decreases and gradually approaches $n_1 = 1/2$. Fig. 3.14(a) shows that as the orbit number n_1 reaches its peak value, with an increase of the impact parameter b_1 , the light deflection becomes less obvious and the trajectory of the photon tends towards a straight line. When $\gamma = -0.2$, our calculations show that the direct emission ($n < 3/4$) corresponds to two large regions of the impact parameter: $b_1 < 4.96588M_1$ and $b_1 > 6.01417M_1$. The lensing ring ($3/4 < n < 5/4$) corresponds to two regions: $4.96588M_1 < b_1 < 5.01349M_1$ and $5.05061M_1 < b_1 < 6.01417M_1$. The photon ring ($n > 5/4$) corresponds to a narrow region: $5.01349M_1 < b_1 < 5.05061M_1$. In addition, we also consider the parameter $\gamma = 0$ and $\gamma = 0.2$ [see the solid black and red lines in Fig. 3.14(a)] for the orbit number n_1 . The results show that as the parameter γ increases, the impact parameter b_1 corresponding to the peak value of the orbit number n_1 becomes larger. It is shown that the size of the ATW shadow increases with the parameter γ .

For the ATW spacetime, the light ray can drop into the spacetime $\mathcal{M}_2$ and then return to the spacetime $\mathcal{M}_1$. To further study the light trajectory around the ATW, the orbit numbers n_2 [the solid lines] and n_3 [the dashed lines] are plotted in Fig. 3.14(b) . Taking the parameter $\gamma = -0.2$ as an example, the orbit numbers $n_2 < 3/4$ and $n_3 > 3/4$ correspond to the new second transfer function^[177], while the light ray finally intersects

with the accretion disk on the back side of the accretion disk. Furthermore, our calculations show that the region of the impact parameter for new second transfer function is $4.05311M_1 < b_1 < 4.92555M_1$. The orbit numbers $n_2 < 5/4$ and $n_3 > 5/4$ correspond to the new third transfer function^[177], while the light ray finally intersects with the accretion disk on the front side of the accretion disk. The regions of the impact parameter for new third transfer function are $3.89952M_1 < b_1 < 3.93779M_1$ and $4.84363M_1 < b_1 < 5.01255M_1$. Moreover, we also consider the parameter $\gamma = 0$ and $\gamma = 0.2$ [see the black and red lines in Fig. 3.14(b)] for the orbit numbers n_2 and n_3 . It is shown that the range of the impact parameter b_1 increases with the value of the parameter γ , which aligns with the results shown in Table 3.2 and Fig. 3.13. This implies that as γ increases, the size of these extra lensing rings and photon rings corresponding to new transfer functions will increase.

3.5.2 Observed intensity and transfer functions

In this part, we consider that the emission originates from an optically and geometrically thin accretion disk around an ATW located in the spacetime $\mathcal{M}_1$. The observational appearance of the ATW could be different for different observers. Without loss of generality, we consider that the static observer in the spacetime $\mathcal{M}_1$ is at the North Pole, and the disk is situated in the equatorial plane. The four-velocity of the static observer is $u_{\text{obs}}^\mu = (1, 0, 0, 0)$. In the rest frame of the static observer, the disk emits light rays isotropically. Due to the spherical symmetry of the spacetime, the emitted specific intensity is only related to the radial coordinate, represented as $I_\nu^{\text{em}}(r)$, where ν is the emission

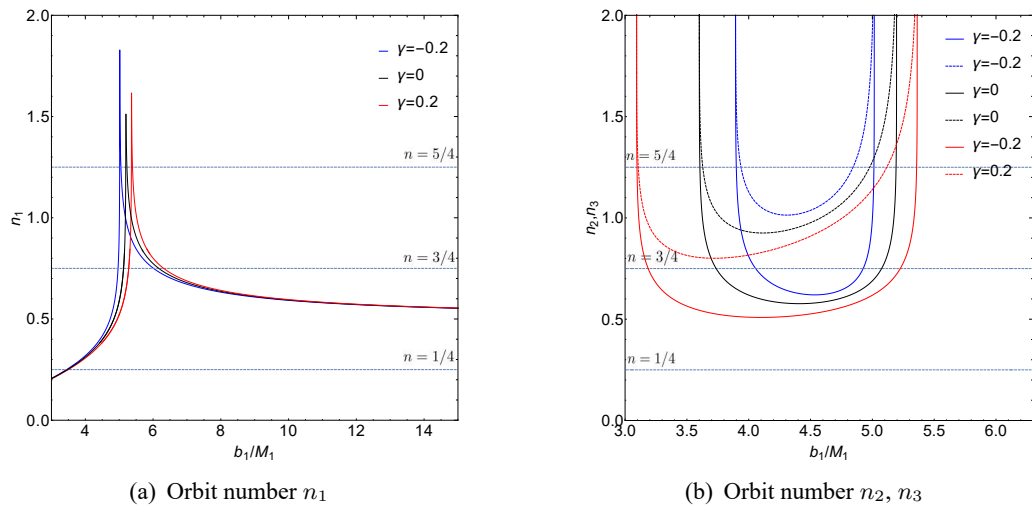


Figure 3.14: The orbit number of the light rays around ATWs. The blue, black, and red lines correspond to $\gamma = -0.2$, $\gamma = 0$, and $\gamma = 0.2$, respectively. In the right panel, the solid lines represent n_2 , while the dashed lines represent n_3 . We set $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

frequency in the static frame. If the static observer at infinity receives a specific intensity $I_{\nu'}^{\text{obs}}$ with a redshifted frequency $\nu' = \sqrt{f}\nu$, then we have

$$\frac{I_{\nu'}^{\text{obs}}}{\nu'^3} = \frac{I_{\nu}^{\text{em}}}{\nu^3}. \quad (3.61)$$

Then the observed specific intensity can be rewritten as

$$I_{\nu'}^{\text{obs}} = f^{3/2}(r) I_{\nu}^{\text{em}}(r). \quad (3.62)$$

By integrating Eq. (3.62) over all frequencies, one can obtain the total observed intensity for each intersection:

$$I^{\text{obs}} = \int I_{\nu'}^{\text{obs}} d\nu' = \int f^2 I_{\nu}^{\text{em}} d\nu = f^2(r) I^{\text{em}}(r), \quad (3.63)$$

in which $I^{\text{em}}(r) = \int I_{\nu}^{\text{em}} d\nu$ represents the total emitted intensity of the accretion disk. Therefore, the total observed intensity for all intersections can be calculated as follows:

$$I^{\text{obs}}(b) = \sum_n I^{\text{em}}(r) f^2(r) \Big|_{r=r_n(b)}, \quad (3.64)$$

where $b = L/E$ is the impact parameter of the photon. The transfer function $r_n(b)$ determines the radial location of the n -th intersection between the accretion disk and the light ray with the impact parameter b . According to Ref.^[23], the demagnification factor can be defined as the derivative of the transfer function, i.e., dr/db_1 . The first transfer function ($n = 1$) provides the “direct emission” of the accretion disk, while the second transfer function ($n = 2$) and the third transfer function ($n = 3$) correspond to the “lensing ring” and the “photon ring”, respectively.

The transfer functions with respect to the impact parameter b_1 are plotted in Fig. 3.15. The top and bottom panels correspond to $M_2 = 1.2$ and $M_2 = 1.15$, respectively. In addition, we consider three specific values for the parameter γ : $-0.2, 0, 0.2$. Taking the mass parameter $M_2 = 1.2$ as an example [see the top panels in Fig. 3.15], one can find that the “direct image” (the redshift of the source profile) of the accretion disk is given by the first transfer function ($n = 1$, the black lines), which has a small demagnification factor. The second transfer function ($n = 2$, the blue lines) produces a “lensing ring” with a large demagnification factor, which represents a reduced image of the back side of the accretion disk. The third transfer function ($n = 3$, the red lines) exhibits the largest demagnification factor and generates a highly reduced image known as the “photon ring” on the front side of the accretion disk. Compared to the black hole case, the blue dashed line ($n = 2$)

represents the new second transfer function of the ATW, which corresponds to the “lensing band”^[177]. The red dashed lines near Zb_{c_2} and b_{c_1} are the new third transfer functions that correspond to the “photon ring group”^[175]. Furthermore, comparing the demagnification factors in the three cases of γ [see the dashed blue lines in the top panels of Fig. 3.15], it is evident that an increase in the parameter γ leads to a higher demagnification factor and a large region of the impact parameter for the new second transfer function. Moreover, we also consider the mass parameter $M_2 = 1.15$ [see the bottom panels in Fig. 3.15] for the transfer functions. The results imply that the size of the additional lensing band increases with the parameter γ . Taking the parameter $\gamma = -0.2$ as an example [see Fig. 3.15(a) and 3.15(d)], it is shown that as the mass parameter M_2 increases, the demagnification factor and region of the new second transfer function increase. Additionally, the bottom panels in Fig. 3.15 also show that the demagnification factor and region of the new second transfer function increase with the parameter γ , which is consistent with the case of the mass parameter $M_2 = 1.2$. The results show that the size of the additional lensing band corresponding to the new second transfer function could increase with the parameter γ and the mass parameter M_2 .

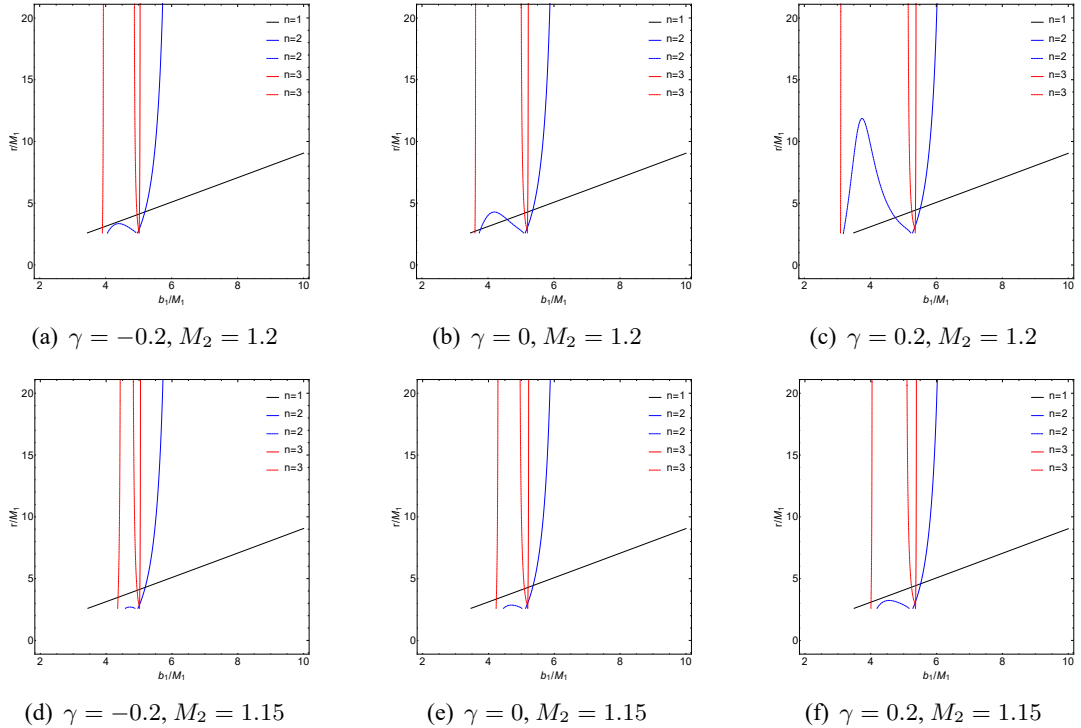


Figure 3.15: The first (black lines), second (blue lines), and third (red lines) transfer functions of the ATW. Note that all the dashed lines indicate the new transfer functions of the ATW. The left, middle, and right panels correspond to $\gamma = -0.2$, $\gamma = 0$, and $\gamma = 0.2$, respectively. The top and bottom panels correspond to $M_2 = 1.2$, and $M_2 = 1.15$, respectively. We set $M_1 = 1$ and $R = 2.6$.

3.5.3 Observational appearance of the asymmetric thin-shell wormhole with two emission models of the thin accretion disk

Now, we investigate the observational appearance of the ATW by considering two specific emission models of the thin accretion disk. The emission from the thin accretion disk can be approximated by a Gaussian function^[178]. Here, the mass parameter M_1 is still assumed to be 1. The innermost stable circular orbit is labeled as r_{isco} . Next, we consider two typical emission models of the thin accretion disk to study the observational appearance of the ATW.

In emission model I, the radiation function is given by

$$I_1^{\text{em}}(r) = \begin{cases} 0 & r < r_{isco}, \\ \left(\frac{1}{r - (r_{isco} - 1)} \right)^2 & r \geq r_{isco}, \end{cases} \quad (3.65)$$

where r_{isco} denotes the inner edge of the accretion disk. Therefore, no radiation is emitted within the region smaller than the inner edge. The radiation function (3.65) is plotted in Fig. 3.16(a). With the radiation function, we plot the observed intensity, density plot, and local density plot of the ATW in the top panels of Fig. 3.17. To compare with a black hole with the same mass parameter and radiation function, we also plot the observational appearance of the black hole in the bottom panel of Fig. 3.17. From Figs. 3.17(a) and 3.17(d), we find that the spatial separations between the direct emission, lensing band and photon rings are distinct. For the observed intensity of the ATW [see Fig. 3.17(a)], the direct emission appears near the critical curve $b_1 \simeq 6.60M_1$ with an initial intensity of 0.42, followed by a subsequent decrease. The range of the lensing band is confined within a narrow interval spanning from the critical curve $b_1 \simeq 5.30M_1$ to the critical curve $b_1 \simeq 5.85M_1$. The photon rings manifest sequentially near the critical curves $b_1 \simeq 3.91M_1$, $b_1 \simeq 4.85M_1$, and $b_1 \simeq 5.03M_1$. It is noteworthy that the ATW shadow exhibits two additional photon rings (near the critical curves $b_1 \simeq 3.91M_1$ and $b_1 \simeq 4.85M_1$) in comparison to the black hole case [see Figs. 3.17(a) and 3.17(d)]. Additionally, the density plot [see Fig. 3.17(b)] and the local density plot [see Fig. 3.17(c)] of the ATW show that the direct emission originates at the periphery of the black disk, while the narrow lensing band is confined within the black disk. Comparing to the black hole case shown in Figs. 3.17(e) and 3.17(f), the two additional photon rings appear closer to the center of the black disk in the shadow of the ATW.

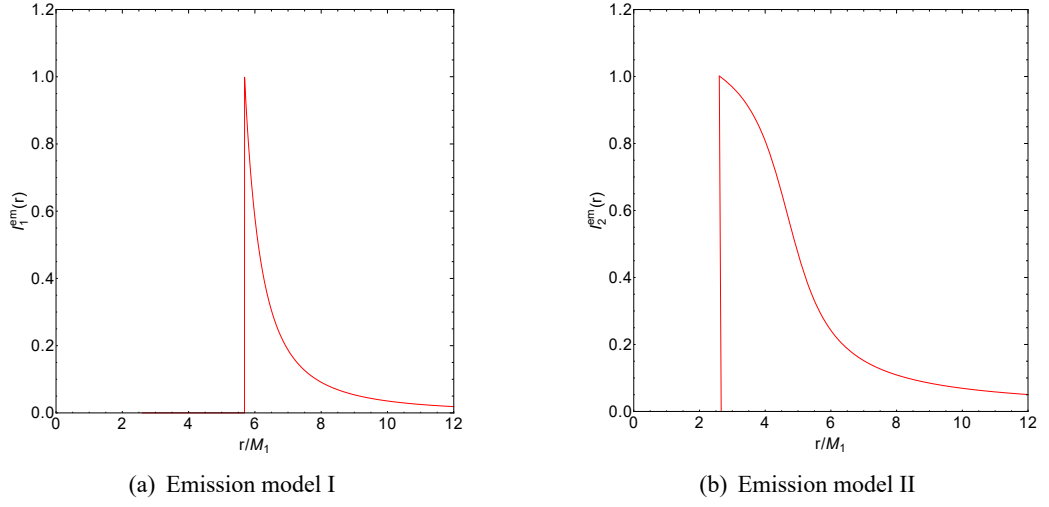


Figure 3.16: The radiation fuctions of emission models I and II for the thin accretion disk.

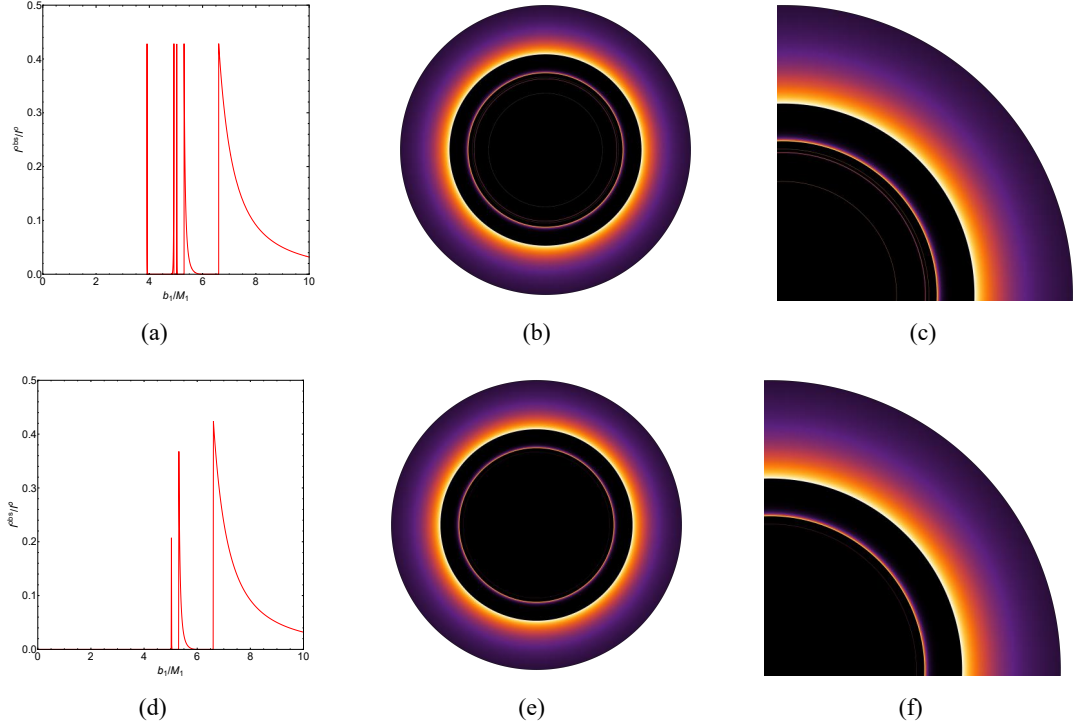


Figure 3.17: **Emission Model I**– Observed intensities (the left panels), density plots (the middle panels), and local density plots (the right panels) of the ATW (the top panels) and the black hole (the bottom panels). We set $\gamma = -0.2$, $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

In emission model II, the radiation function is expressed as

$$I_2^{\text{em}}(r) = \begin{cases} 0 & r < r_h, \\ \frac{\frac{\pi}{2} - \tan^{-1}(r - (r_{\text{isco}} - 1))}{\frac{\pi}{2} - \tan^{-1}(r_{\text{ph}})} & r \geq r_h, \end{cases} \quad (3.66)$$

where r_h is the event horizon radius. Therefore, no radiation is emitted within the region smaller than the event horizon radius. The radiation function (3.66) is plotted in Fig. 3.16(b). From Fig. 3.16, it is evident that emission model II has a more gradual decrease in the radiation function compared to emission model I. With the radiation function, we plot the observed intensity, density plot, and local density plot of the ATW in the top panels of Fig. 3.18. From Fig. 3.18(a), there is an overlap between the areas of the direct emission, lensing band, and photon rings of the ATW. The direct emission initiates at approximately $b_1 \simeq 2.76M_1$, while the photon rings are encompassed within the lensing band, giving rise to a luminous ring structure with multiple layers [see Figs. 3.18(b) and 3.18(c)]. For a black hole with the same mass parameter and radiation function, we plot the observational appearance of the black hole in the bottom panels of Fig. 3.18. Comparing emission models I and II, we find that there exists an additional lensing band between the critical curves $Zb_{c_2} \simeq 3.89M_1$ and $b_{c_1} \simeq 5.01M_1$ in emission model II. The result implies that the new second transfer function for emission model II contributes to the observed intensity of the ATW.

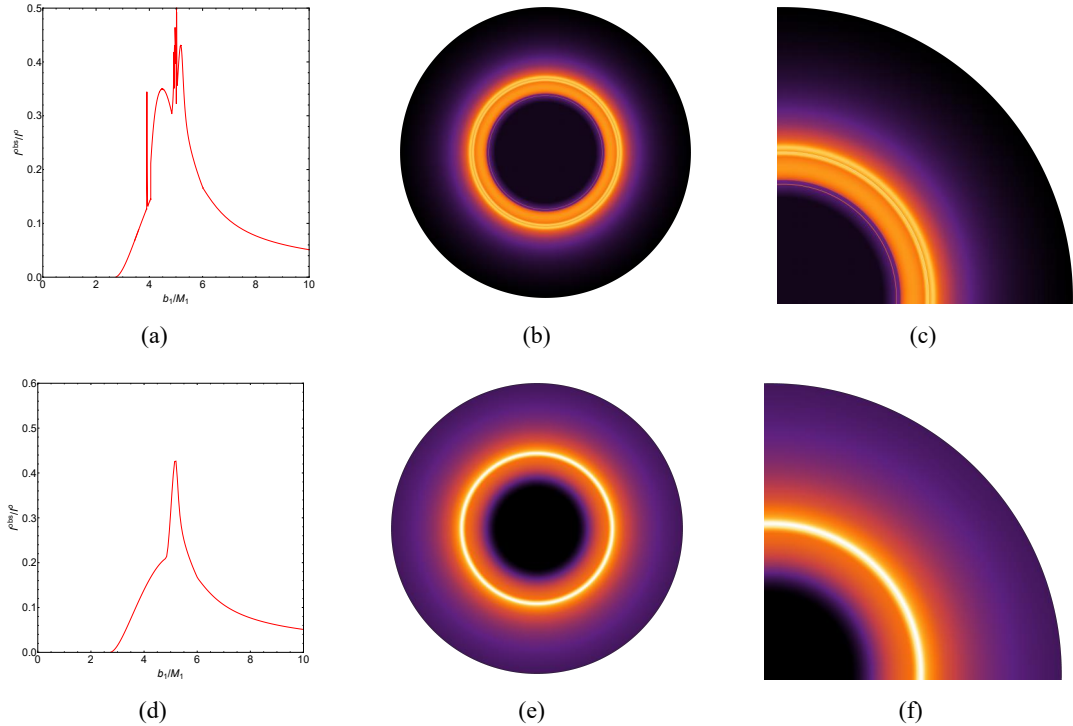


Figure 3.18: **Emission Model II**– Observed intensities (the left panels), density plots (the middle panels) and local density plots (the right panels) of the ATW (the top panels) and the black hole (the bottom panels). We set $\gamma = -0.2$, $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

To further study the impact of the parameter γ on the observational appearance of the ATW in Horndeski theory, we set the parameter $\gamma = 0$ and use two emission models to plot the observed intensity, density plot, and local density plot of the ATW in Fig. 3.19.

For the observed intensity of the ATW under emission model I [see Fig. 3.19(a)], it can be found that the direct emission appears near the critical curve $b_1 \simeq 6.93M_1$ with an initial intensity of 0.44. The region of the lensing band is $5.49M_1 < b_1 < 5.99M_1$. The photon rings appear near the critical curves $b_1 \simeq 3.60M_1$, $b_1 \simeq 4.99M_1$, and $b_1 \simeq 5.22M_1$. For the case under emission model II [see Fig. 3.19(d)], one can find that the direct emission appears at the critical curve $b_1 \simeq 2.87M_1$, and an additional lensing band appears between the critical curves $Zb_{c_2} \simeq 3.75M_1$ and $b_{c_1} \simeq 5.20M_1$. By comparing Fig. 3.17, 3.18, and 3.19, it can be found that the positions of these light bands (including the direct emission, lensing ring, and photon ring) become larger with the parameter γ . The results suggest that for the observational appearance of the ATW in Horndeski theory, as the parameter γ increases, the size of these light bands will also increase.

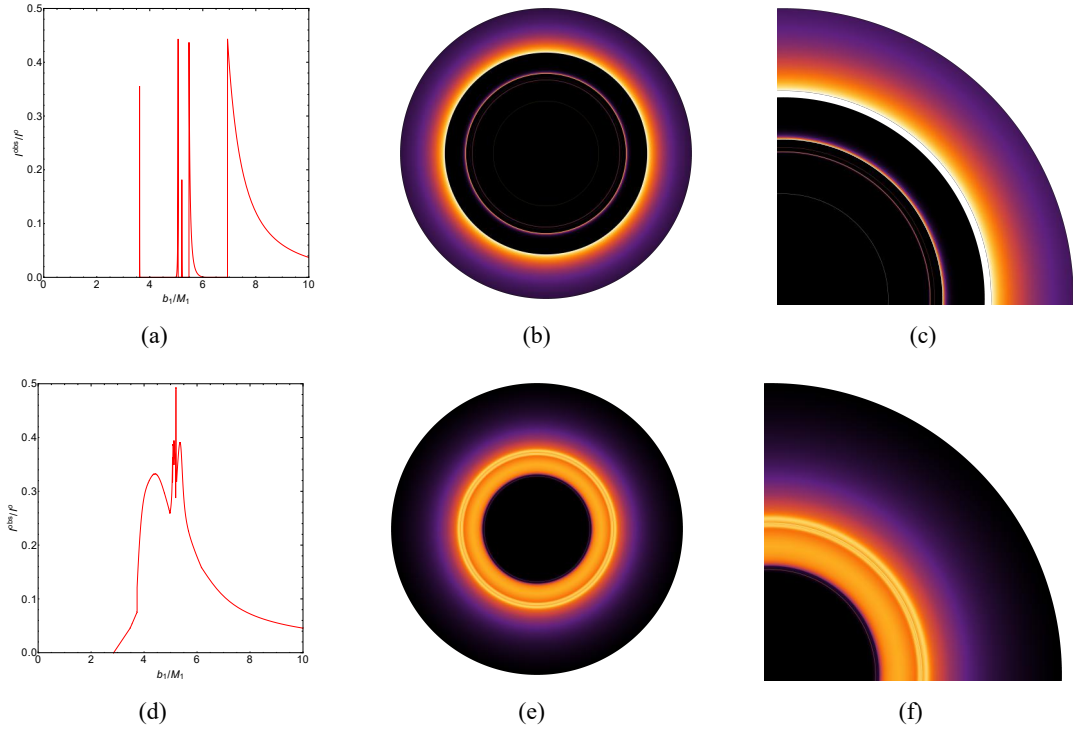


Figure 3.19: We use emission model I (the top panels) and emission model II (the bottom panels) to plot observed intensities (the left panels), density plots (the middle panels), and local density plots (the right panels) of the ATW in Horndeski theory. We set $\gamma = 0$, $M_1 = 1$, $M_2 = 1.2$, and $R = 2.6$.

3.6 Response function for AdS black holes in Horndeski theory

To begin our analysis, we consider the action of Horndeski gravity with a negative cosmological constant Λ , expressed as^[155]:

$$S = \int d^4x \sqrt{-g} (\mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5 - 2\Lambda), \quad (3.67)$$

with

$$\mathcal{L}_2 = G_2(X), \quad (3.68)$$

$$\mathcal{L}_3 = -G_3(X)\square\phi, \quad (3.69)$$

$$\mathcal{L}_4 = G_4(X)\mathcal{R} + G_{4X} [(\square\phi)^2 - (\nabla_\mu\nabla_\nu\phi)^2], \quad (3.70)$$

$$\mathcal{L}_5 = G_5(X)G_{\mu\nu}\nabla^\mu\nabla^\nu\phi - \frac{G_{5X}}{6} [(\square\phi)^3 - 3\square\phi(\nabla_\mu\nabla_\nu\phi)^2 + 2(\nabla_\mu\nabla_\nu\phi)^3]. \quad (3.71)$$

In this context, $\mathcal{R}$ denotes the Ricci scalar, and $G_{\mu\nu}$ represents the Einstein tensor. The scalar field is ϕ , with its kinetic term defined as $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$. The functions $G_i = G_i(\phi, X)$ depend on both ϕ and X , where the subscript X indicates differentiation with respect to X . Building upon the work in Ref.^[169], we consider a specific case of the Horndeski action (3.67), characterized by:

$$G_2 = \eta X, \quad G_4 = \zeta + \beta\sqrt{-X}, \quad G_3 = G_5 = 0. \quad (3.72)$$

By setting $\zeta = \frac{1}{2}$ and considering η and β as dimensionless parameters, the action in Eq. (3.67) reduces to

$$S = \int d^4x \sqrt{-g} \left\{ \left[\frac{1}{2} + \beta\sqrt{(\partial\phi)^2/2} \right] \mathcal{R} - \frac{\eta}{2}(\partial\phi)^2 - \frac{\beta}{\sqrt{2}(\partial\phi)^2} [(\square\phi)^2 - (\nabla_\mu\nabla_\nu\phi)^2] - 2\Lambda \right\}. \quad (3.73)$$

A spherically symmetric AdS black hole solution in Horndeski theory can be derived from Eq. (3.73) and is given by^[169]:

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2), \quad (3.74)$$

in which

$$f(r) = 1 - \frac{2M}{r} - \frac{\gamma}{r^2} + \frac{r^2}{l^2}. \quad (3.75)$$

To simplify the expressions in Eq. (3.75), we introduce the parameter $\gamma = \beta^2/\eta$. Here, $l = \sqrt{-3\Lambda}$ denotes the radius of the AdS spacetime, and the constant M acts as an integration constant corresponding to the mass of the black hole. By setting $f(r_h) = 0$, we determine the radius r_h of the black hole's outer event horizon. Consequently, the black hole mass can be re-expressed in terms of the event horizon radius r_h as

$$M = \frac{r_h^2 + r_h^4 - \gamma}{2r_h}, \quad (3.76)$$

where we have adopted the unit $l = 1$ in Eq. (3.76) for simplicity.

To develop a holographic model for AdS black holes in Horndeski theory, we reformulate the metric function by introducing a new variable $\xi = 1/r$. This substitution leads to the following expression:

$$f(r) = \frac{1}{\xi^2} f(\xi). \quad (3.77)$$

Furthermore, the Hawking temperature of the black hole is

$$T = \frac{3 + \xi_h^2 + \xi_h^4 \gamma}{4\pi \xi_h}, \quad (3.78)$$

where ξ_h is the event horizon of the black hole. It is important to note that $\eta \geq 0$ ensures a canonical scalar field kinetic term. If $\eta < 0$, the field becomes ghost-like and $\gamma = \beta^2/\eta$ is negative, resulting in a negative black hole temperature, which is not physically interpretable. Despite this, we present results for $\gamma < 0$ as a mathematical probe of parameter sensitivity, without implying these are physically realizable black holes.

Moreover, one can express the metric as follows:

$$ds^2 = \frac{1}{\xi^2} \left[-f(\xi) dt^2 + \frac{1}{f(\xi)} d\xi^2 + d\theta^2 + \sin^2 \theta d\varphi^2 \right], \quad (3.79)$$

where $f(\xi)$ is the metric function redefined in terms of ξ . To simplify calculations near the event horizon and to analyze wave propagation more effectively, we adopt the ingoing Eddington–Finkelstein coordinates $(v, \xi, \theta, \varphi)$, defined by

$$v = t + \xi_* = t - \int \frac{d\xi}{f(\xi)}. \quad (3.80)$$

In these new coordinates, the metric (3.79) transforms to

$$ds^2 = \frac{1}{\xi^2} \left[-f(\xi) dv^2 - 2 dv d\xi + d\theta^2 + \sin^2 \theta d\varphi^2 \right]. \quad (3.81)$$

Now we consider an evolution of the scalar field ϕ , which is governed by the Klein-Gordon equation,

$$\square \phi(v, \xi, \theta, \varphi) = 0. \quad (3.82)$$

Combining Eqs. (3.81) and (3.82), we arrive at

$$\xi^2 f(\xi) \partial_\xi^2 \phi + [\xi^2 f'(\xi) - 2\xi f(\xi)] \partial_\xi \phi - 2\xi^2 \partial_v \partial_\xi \phi + 2\xi \partial_v \phi + \xi^2 D_S^2 \phi = 0, \quad (3.83)$$

where $f'(\xi) = \partial_\xi f(\xi)$, and D_S^2 denotes the scalar Laplacian on the unit two-sphere S^2 . Following Ref.^[179], the asymptotic solution of Eq. (3.83) near the AdS boundary is given

by

$$\phi(v, \xi, \theta, \varphi) = \mathcal{J}_O(v, \theta, \varphi) + \xi \partial_v \mathcal{J}_O(v, \theta, \varphi) + \frac{1}{2} \xi^2 D_S^2 \mathcal{J}_O(v, \theta, \varphi) + \xi^3 \langle O \rangle + \mathcal{O}(\xi^4), \quad (3.84)$$

where $\mathcal{J}_O(v, \theta, \varphi)$ is the scalar source term, and $\langle O \rangle$ is the corresponding response function. For a monochromatic, axisymmetric Gaussian wave packet source situated at the South Pole of the AdS boundary, we have^[180]

$$\mathcal{J}_O(v, \theta) = e^{i\omega v} \cdot \frac{1}{2\pi\sigma^2} \exp \left[-\frac{(\pi - \theta)^2}{2\sigma^2} \right] = e^{i\omega v} \sum_{l=0}^{\infty} C_{l0} Y_{l0}(\theta), \quad (3.85)$$

where

$$C_{l0} = (-1)^l \sqrt{\frac{l+1/2}{2\pi}} \exp \left[-\frac{1}{2}(l+1/2)^2 \sigma^2 \right]. \quad (3.86)$$

Here, σ characterizes the width of the Gaussian wave packet, Y_{l0} are the spherical harmonics, and ω is the angular frequency of the monochromatic wave source, determining the temporal oscillation of the wave. In the limit $\sigma \ll \pi$, the scalar field $\phi(v, \xi, \theta, \varphi)$ can be expanded as

$$\phi(v, \xi, \theta, \varphi) = e^{i\omega v} \sum_{l=0}^{\infty} c_{l0} U_l(\xi) Y_{l0}(\theta, \varphi), \quad (3.87)$$

and the response function can be expressed as

$$\langle O \rangle = e^{i\omega v} \sum_{l=0}^{\infty} \langle O \rangle_l Y_{l0}(\theta). \quad (3.88)$$

Substituting Eq. (3.87) into the field equation, we find

$$\xi^2 f(\xi) U_l'' + [\xi^2 f'(\xi) - 2\xi f(\xi) + 2i\omega \xi^2] U_l' + [-2i\omega \xi - l(l+1)\xi^2] U_l = 0, \quad (3.89)$$

with the asymptotic behavior of U_l near the boundary given by

$$\lim_{\xi \rightarrow 0} U_l = 1 - i\omega \xi + \frac{1}{2} [-l(l+1)] \xi^2 + \langle O \rangle_l \xi^3 + \mathcal{O}(\xi^4). \quad (3.90)$$

The function U_l satisfies two boundary conditions. The first is at the horizon $\xi = \xi_h$:

$$[\xi_h^2 f'(\xi_h) + 2i\omega \xi_h^2] U_l' - [2i\omega \xi_h + l(l+1)\xi_h^2] U_l = 0, \quad (3.91)$$

and the second is the AdS boundary condition at $\xi = 0$:

$$U_l(0) = 1. \quad (3.92)$$

By imposing the boundary conditions (3.91) and (3.92), the differential equation (3.89) for U_l can be solved. Subsequently, the total response function $\langle O \rangle$ can be determined using Eqs. (3.88) and (3.90).

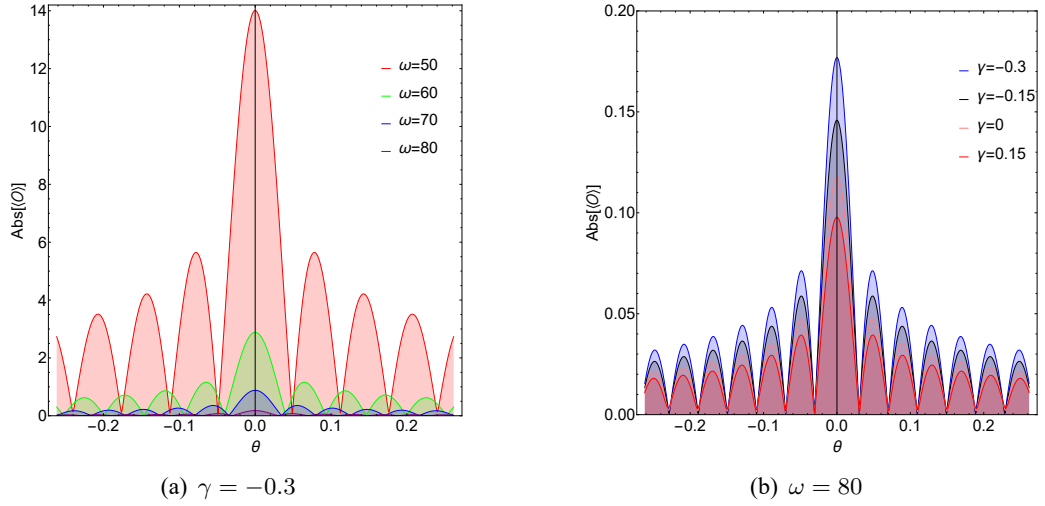


Figure 3.20: Effects of different source frequencies $\omega = 50, 60, 70, 80$ and parameters $\gamma = -0.3, -0.15, 0, 0.15$ on the response function. In this analysis, we set $\xi_h = 1$.

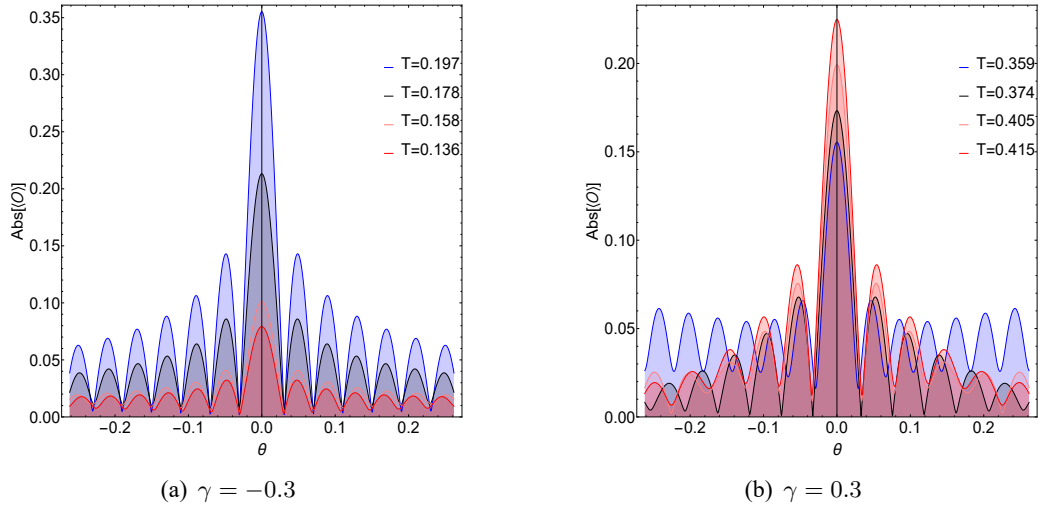


Figure 3.21: Effects of different temperatures on the response function. In this analysis, we set $\omega = 80$.

In Fig. 3.20, we display the amplitude of the response function as a function of source frequency ω for various values of the parameter γ . Note that while $\gamma < 0$ leads

to a ghost field and negative-temperature solutions, which are not physically meaningful, we still include these cases in the present and subsequent discussions as a theoretical extension. This approach allows us to explore how the image properties depend on the parameter space. Fig. 3.20(a) demonstrates that as the source frequency ω increases, both the amplitude and the period of the response function decrease. In contrast, Fig. 3.20(b) shows that the amplitude of the response function diminishes with increasing values of the parameter γ . Furthermore, Fig. 3.21 explores the relationship between the amplitude of the response function and the temperature T for both positive and negative values of γ . The findings suggest that in both scenarios, the amplitude of the response function increases with rising temperature.

3.7 Holographic rings of AdS black holes in Horndeski theory

In this section, we introduce the optical system shown in Fig. 3.22(a) to simulate the imaging of the Einstein ring and subsequently explore its wave characteristics. The system consists of a spherical screen and an infinitely thin convex lens, with the focal length f of the lens being much greater than its size d . The observation center is set at $(\theta_{obs}, 0)$, marked in blue in Fig. 3.22(a). This optical system is positioned at the observation center, with the lens at $\vec{x} = (x, y, 0)$ and the screen at $\vec{x}_S = (x_S, y_S, z_S)$. The response function is first transformed into a transmitted wave $\Psi_s(\vec{x})$ by the convex lens, which is then projected onto the screen and converted into an observable wave $\Psi_{sc}(\vec{x}_S)$, as depicted in Fig. 3.22(b). Therefore, one can obtain^[180]

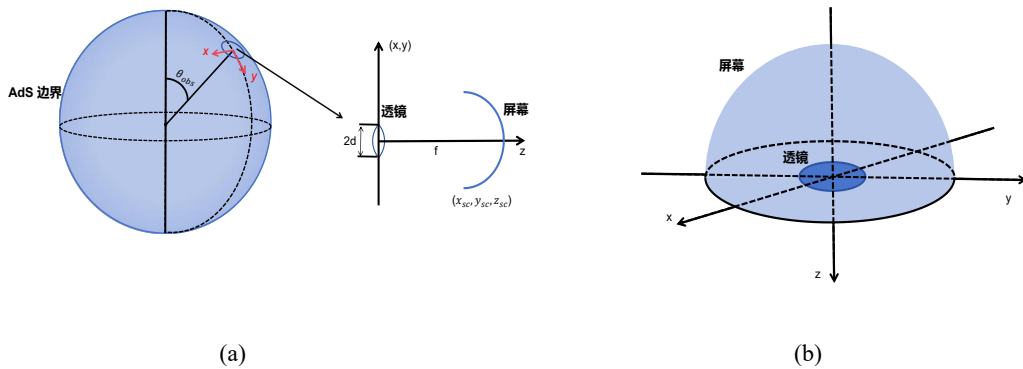


Figure 3.22: The optical system

$$\Psi_{sc}(\vec{x}_S) = \int_{|\vec{x}| < d} d^2x \Psi_s(\vec{x}) e^{-i\omega L} = \int_{|\vec{x}| < d} d^2x e^{-i\omega \frac{|\vec{x}|^2}{2f}} \Psi_p(\vec{x}) e^{-i\omega L}, \quad (3.93)$$

in which L represents the distance between $\vec{x}$ and $\vec{x}_S$, while f is defined by

$$f^2 = x_S^2 + y_S^2 + z_S^2. \quad (3.94)$$

By further combining the expression

$$L = \sqrt{(x_S - x)^2 + (y_S - y)^2 + z_S^2} \simeq f - \frac{\vec{x}_S \cdot \vec{x}}{f} + \frac{|\vec{x}|^2}{2f}, \quad (3.95)$$

where the Fresnel approximation $f \gg |\vec{x}|$, Eq. (3.93) can be expressed as

$$\Psi_{sc}(\vec{x}_S) \propto \int_{|\vec{x}| < d} dx^2 \Psi_p(\vec{x}) \varpi(\vec{x}) e^{-\frac{i\omega}{f} \vec{x} \cdot \vec{x}_S}, \quad (3.96)$$

where the window function is

$$\varpi(\vec{x}) = \begin{cases} 1, & 0 \leq |\vec{x}| \leq d \\ 0, & |\vec{x}| > d \end{cases}. \quad (3.97)$$

Eq. (3.96) demonstrate that the incident wave can be transformed into the observed wave on the screen via a Fourier transformation, thereby delineating the profiles of the double black holes on the observer's screen. In Fig. 3.23, we present holographic Einstein images at various observation angles θ and different values of the parameter γ . The results reveal that the holographic Einstein images exhibit significant variations depending on the observation angle and the value of γ . When the observation angle is $\theta = 0^\circ$, with the observer positioned at the north pole of the AdS boundary, Figs. 3.23(a), 3.23(d), and 3.23(g) display the holographic Einstein images as a series of axially symmetric concentric rings, with the brightest ring located at the center, forming a Poisson-like spot. Notably, the brightness of these rings remains relatively unchanged as the parameter γ increases. As the observation angle increases to $\theta = 45^\circ$, Figs. 3.23(b), 3.23(e), and 3.23(h) show that the holographic Einstein images transition into two arcs with varying brightness, where the left arc appears dimmer than the right one. As γ increases, the brightness of the left arc gradually intensifies. Finally, when the observation angle reaches $\theta = 90^\circ$, Figs. 3.23(c), 3.23(f), and 3.23(i) indicate that the two arcs further evolve into two small bright spots. As the parameter γ increases, the brightness of the left spot also gradually enhances. These findings underscore that both the observation angle θ and the parameter γ have a significant influence on the morphology and brightness of the holographic Einstein images. This suggests that the spacetime characteristics of the black hole are effectively encoded in these holographic images, offering a potential tool for probing the underlying

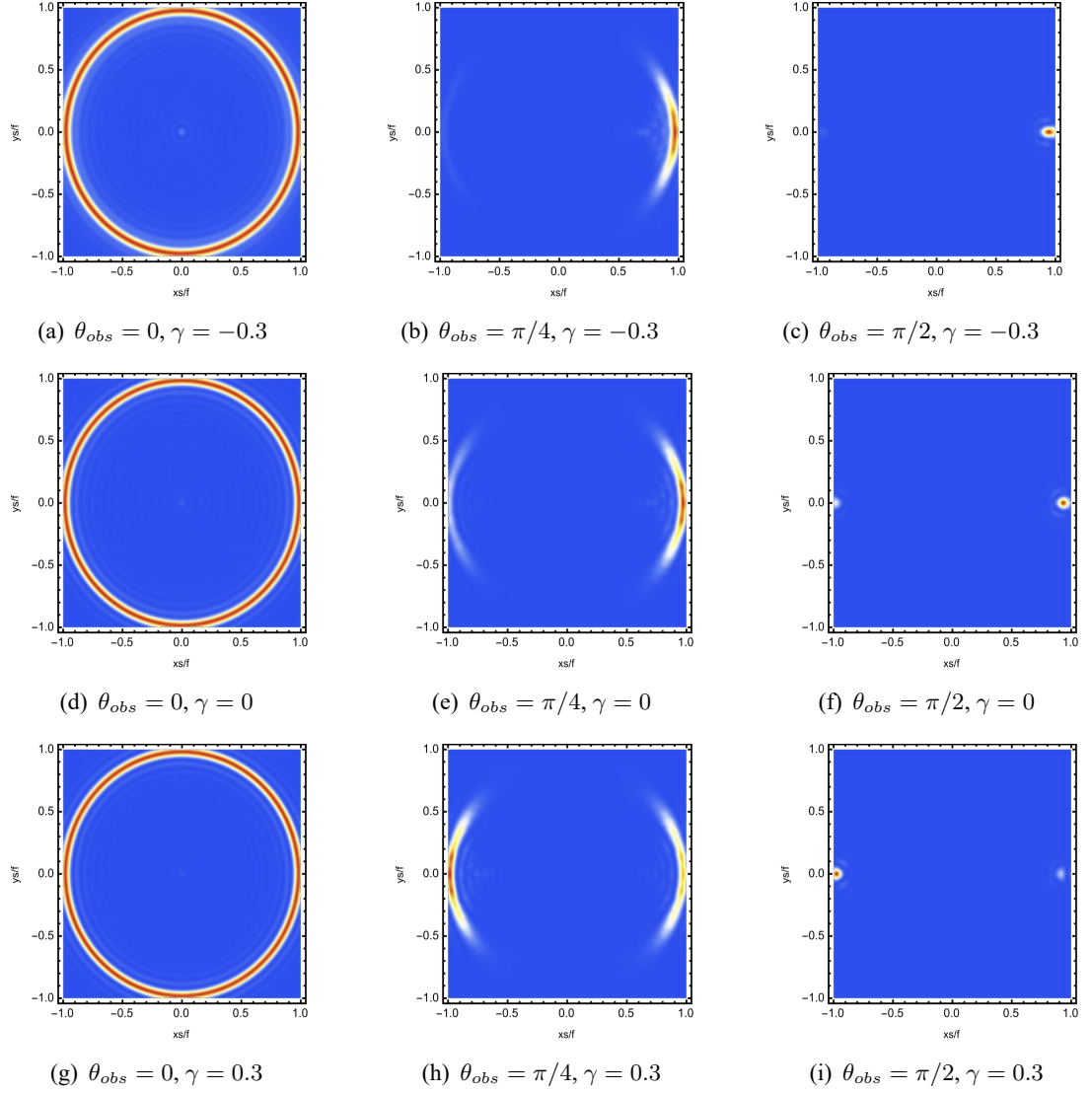


Figure 3.23: Holographic images at different observation angles θ_{obs} and parameter values γ for the AdS black hole in Horndeski theory. In this analysis, we set $\xi_h = 1$ and $\omega = 80$.

geometry of black holes through observational data.

In Fig. 3.24, we explore the effect of varying frequencies on the holographic Einstein images generated by the wave source. Figs. 3.24(a), 3.24(b), and 3.24(c) demonstrate that the rings become progressively sharper as the frequency increases. This sharpening is expected, as the geometric optics approximation becomes increasingly accurate at higher frequencies, allowing for finer image details to be captured. Additionally, Figs. 3.24(d), 3.24(e), and 3.24(f) depict the corresponding lensing response function for the holographic Einstein images. The results suggest that as the frequency increases, there is a noticeable reduction in the brightness of the lensing response. This decrease in brightness may be attributed to the diminishing diffraction effects at higher frequencies, which leads to a more concentrated but less intense image. Overall, these findings highlight the critical role

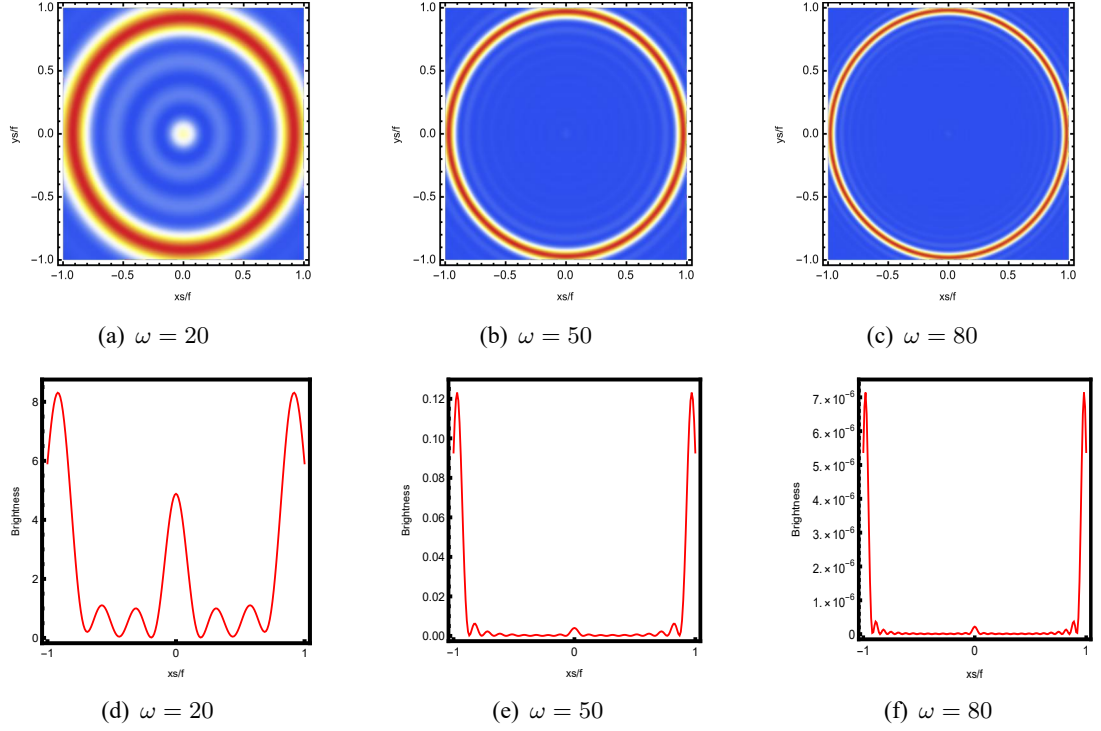


Figure 3.24: Holographic images and the corresponding lensed response brightness at different source frequencies $\omega = 20, 50, 80$. In this analysis, we set $\gamma = -0.3$, $\xi_h = 1$, and $\theta_{obs} = 0$.

that frequency plays in determining the resolution and brightness of holographic Einstein images, with higher frequencies enhancing the precision of the observed features while concurrently reducing the overall intensity of the lensing response.

Additionally, we examined the impact of horizon temperature on the holographic Einstein images for the parameter $\gamma = -0.3$, as illustrated in Fig. 3.25. In Figs. 3.25(a) and 3.25(d), when the horizon radius is $\xi_h = 4$ and the horizon temperature is $T = -1.14989$, the holographic Einstein image is characterized by a series of axially symmetric bright rings and a central bright spot. The brightest ring is more luminous and positioned farther from the central spot. As shown in Figs. 3.25(b) and 3.25(e), when the horizon radius is increased to $\xi_h = 6$ and the temperature decreases to $T = -4.63937$, the radii of these bright rings reduce, causing the brightest ring to move closer to the central spot. Upon further increasing the horizon radius to $\xi_h = 8$, as depicted in Figs. 3.25(c) and 3.25(f), with the horizon temperature now at $T = -11.5566$, the radii of the bright rings continue to shrink, and the brightness of the brightest ring diminishes, becoming less intense than that of the central spot. These findings suggest that as the horizon radius ξ increases, leading to a decrease in horizon temperature, the radii of the bright rings progressively diminish, while the central spot becomes relatively brighter compared to the rings. This behavior underscores the intricate relationship between horizon temperature and the morphology of

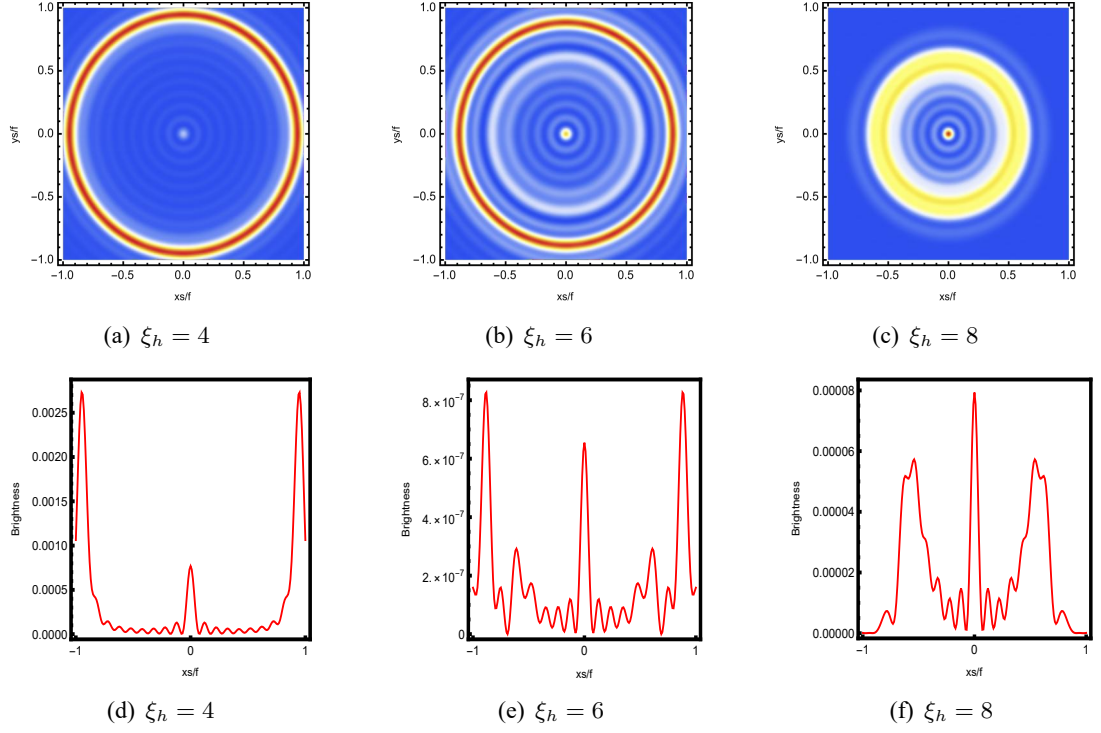


Figure 3.25: Holographic images and the corresponding lensed response brightness at different temperature. In this analysis, we set $\gamma = -0.3$, $\omega = 50$, and $\theta_{obs} = 0$.

holographic Einstein images, indicating that changes in thermodynamic properties of the black hole could be directly reflected in the observed image structure. We also examined the effect of horizon temperature on the holographic Einstein images for the parameter $\gamma = 0.3$, as illustrated in Fig. 3.26. As the radius of the event horizon increases from $\xi_h = 0.2$ ($T = 1.20977$) to $\xi_h = 1.2$ ($T = 0.33569$) and then to $\xi_h = 2.2$ ($T = 0.537787$), the ring gradually contracts. This result is consistent with the conclusions drawn from Fig. 3.25. However, we observe that the correlation between the ring radius and temperature is not very pronounced. This is because, when the parameter γ in Horndeski theory is positive, the black hole temperature is no longer a monotonic function of the event horizon, as illustrated in Fig. 3.27.

Analyzing the holographic Einstein ring solely through the lens of wave optics provides an incomplete perspective on the system's behavior. To achieve a more comprehensive understanding, we now approach the relationship between the photon ring and the holographic Einstein ring from the standpoint of geometric optics. In describing photon motion, two conserved quantities play a crucial role: the energy of the photon, $E = f(r)(\partial t / \partial \lambda)$, and its angular momentum, $\tilde{L} = r^2(\partial \varphi / \partial \lambda)$, where λ is the affine parameter. Considering photon motion in the equatorial plane ($\theta = \pi/2$), its four-velocity

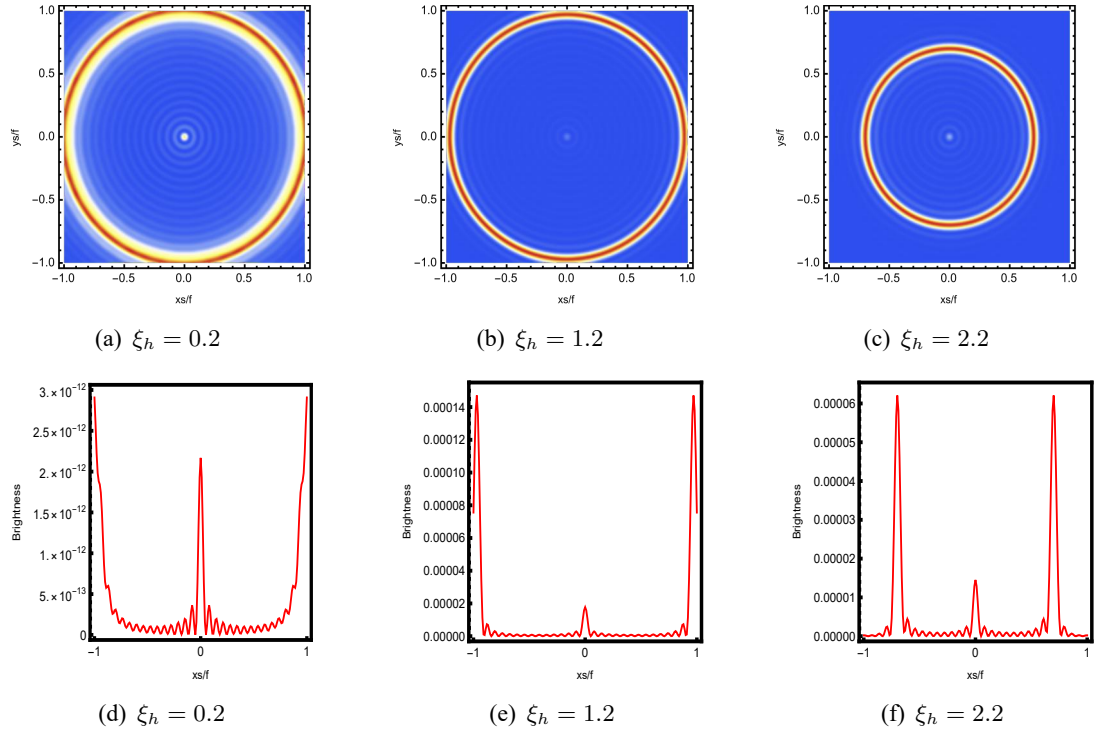


Figure 3.26: Holographic images and the corresponding lensed response brightness at different temperature. In this analysis, we set $\gamma = 0.3$, $\omega = 80$, and $\theta_{obs} = 0$.

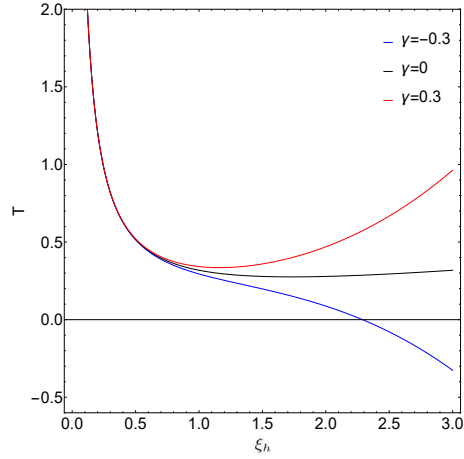


Figure 3.27: The temperature of the AdS black hole in Horndeski theory is represented by three lines: the blue line corresponds to $\gamma = -0.3$, the black line corresponds to $\gamma = 0$, and the red line corresponds to $\gamma = 0.3$.

vector $v^\alpha = (d/d\lambda)^\alpha$ satisfies the following condition:

$$-f(r) \left(\frac{dt}{d\lambda} \right)^2 + f(r)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + r^2 \sin^2 \theta \left(\frac{d\varphi}{d\lambda} \right)^2 = 0, \quad (3.98)$$

which simplifies to the equation

$$\dot{r}^2 = E - z(r)\tilde{L}, \quad (3.99)$$

where

$$z(r) = f(r)/r^2. \quad (3.100)$$

Building on Refs.^[179-181], we consider an incident angle defined by the boundary's normal vector θ_{in} and the direction $n^\zeta = \partial/\partial r^\zeta$. Under these conditions, we obtain

$$\cos \theta_{in} = \frac{g_{\alpha\beta} v^\alpha n^\beta}{|v||n|} \Big|_{r=\infty} = \sqrt{\frac{\dot{r}^2/f}{\dot{r}^2/f + \tilde{L}/r^2}} \Big|_{r=\infty}. \quad (3.101)$$

Therefore, we derive

$$\sin^2 \theta_{in} = 1 - \cos^2 \theta_{in} = \frac{\tilde{L}^2 z(r)}{\dot{r}^2 + \tilde{L}^2 z(r)} \Big|_{r \rightarrow \infty} = \frac{\tilde{L}^2}{E^2}. \quad (3.102)$$

This relationship indicates that the incident angle θ_{in} is connected to the photon's energy E and angular momentum $\tilde{L}$ as

$$\sin \theta_{in} = \frac{\tilde{L}}{E}. \quad (3.103)$$

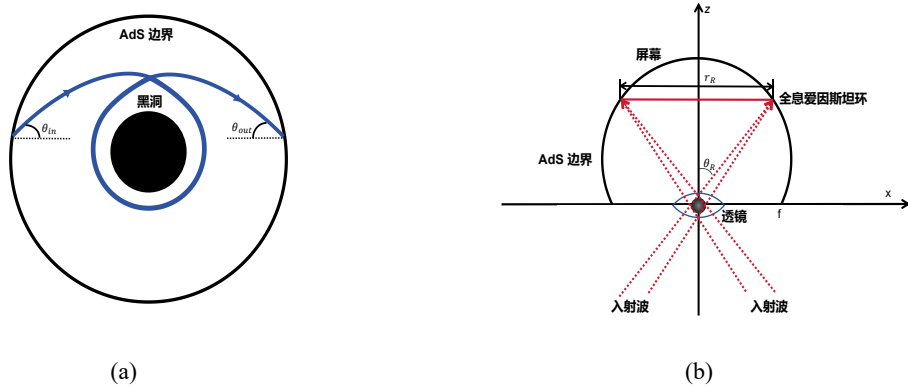


Figure 3.28: (a). The trajectory of an incident photon completes a full rotation around the black hole. (b). The relationship between the ring radius and the ring angular.

As shown in Fig. 3.28(a), when the photon's entry and exit points align with the black hole's center along a straight line, its trajectory forms a ring-shaped image. Upon reaching the photon ring, the photon enters a critical orbit around the black hole. We denote the

angular momentum of the photon in this critical state as L , and its orbital equation satisfies

$$\dot{r} = 0, \frac{dz}{dr} = 0. \quad (3.104)$$

We define the radius of the circle in Fig. 3.28(b) as r_R , leading to the relation

$$\sin \theta_R = \frac{r_R}{f}. \quad (3.105)$$

For sufficiently large angular momentum, we have $\sin \theta_R = \sin \theta_{\text{in}}$, and thus, we derive the relationship

$$\frac{r_R}{f} = \frac{L}{E}. \quad (3.106)$$

To validate Eq. (3.106), we present in Fig. 3.29 the Einstein ring radius under different event horizons. The results demonstrate that the incident angle obtained via geometric optics agrees well with the observational angle derived from wave optics across various parameters γ . Moreover, this conclusion holds regardless of the parameter γ in Horndeski theory.

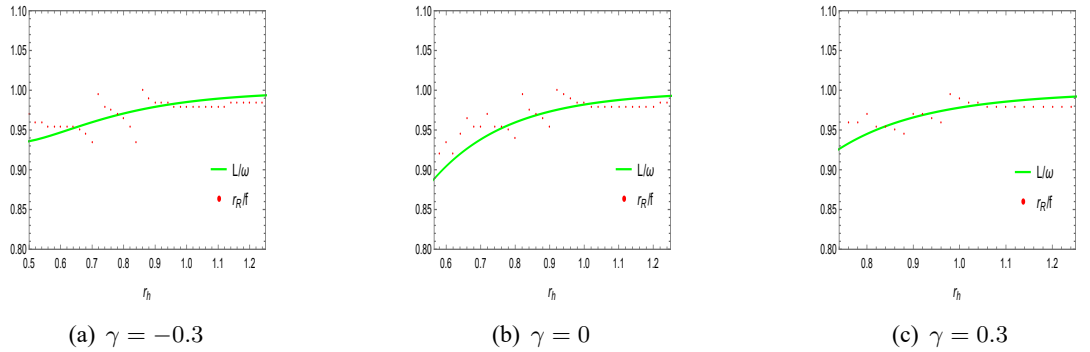


Figure 3.29: The Einstein ring radius is shown as a function of the event horizon under different values of the parameter γ .

3.8 Chapter Summary

In this chapter, we studied several complementary observational signatures of black holes and black-hole mimickers in Horndeski theory. The ray-tracing analysis of Horndeski black holes shows how the parameter γ affects the photon sphere, critical impact parameter, shadow radius, and optical appearance in thin-disk emission models. By comparing the predicted shadow diameter with EHT observations of M87* and Sgr A*, the parameter space of the theory can be constrained. We then analyzed asymmetric thin-shell wormholes and showed that their optical images may contain additional lensing bands and photon-ring groups, providing possible criteria for distinguishing them from

black holes. Finally, using wave optics and the AdS/CFT correspondence, we investigated holographic Einstein rings of AdS black holes in Horndeski theory and found good consistency between wave-optics and geometric-optics predictions for the relevant ring structures.

Chapter 4 Shadow Signatures and Dynamical Analysis of Black-Hole Phase Transitions

The existence of black-hole entropy shows that even the most extreme gravitational objects in the universe obey the second law of thermodynamics. The proportionality between black-hole entropy and horizon area is not only a beautiful union of geometry and thermodynamics, but also a clue to the thermodynamic nature of black holes.

— Jacob David Bekenstein

In this chapter, based on our works^[182-183], we study thermodynamic phase transitions of the Hayward-AdS black hole by using the shadow radius as an alternative to the event-horizon radius. This shadow-based formalism reveals how the magnetic charge affects the thermodynamic behavior and provides a possible observational probe of black-hole phase transitions. We then introduce the Gibbs free-energy landscape to analyze the probability evolution of black-hole states and investigate how a global monopole affects the dynamics of thermodynamic phase transitions.

4.1 Research Background

Black-hole thermodynamics has displayed striking similarities with ordinary thermodynamic systems since the seminal works of Bekenstein and Hawking^[25,38-41]. With the development of the AdS/CFT correspondence, AdS black-hole thermodynamics has become an important field of research^[42-43,46-52]. In extended phase space, the negative cosmological constant is interpreted as thermodynamic pressure, making it possible to compare black-hole critical behavior with that of ordinary fluids. The charged AdS black hole, for example, exhibits a van der Waals-like small/large black-hole phase transition^[44].

Recent studies have suggested that the shadow radius can be used as an observable thermodynamic parameter. For spherically symmetric AdS black holes, the shadow radius is closely related to the event-horizon radius and can therefore encode information about the phase structure^[184-186]. This motivates the study of “shadow thermodynamics,” in which black-hole shadows are used as indirect probes of black-hole phase transitions and microscopic structure.

Another important question is how black-hole phase transitions proceed dynamically. The Gibbs free-energy landscape treats the small and large black-hole phases as local minima separated by a free-energy barrier, and the Fokker-Planck equation describes the probability evolution induced by thermal fluctuations^[187-188]. This framework allows one

to compute dynamical quantities such as the mean first passage time and to understand how additional parameters, such as a global monopole, regulate the transition between competing phases.

4.2 Shadow of the Hayward-AdS black hole

In this section, we review the derivation of the shadow radius for the Hayward-AdS black hole. The metric of a static spherically symmetric black hole can be written as

$$ds^2 = -f(r)dt^2 + h(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (4.1)$$

where $f(r)$ and $h(r)$ represent the lapse functions associated with the radius parameter r . For a photon moving in such spacetime background of the metric (4.1), the corresponding Hamiltonian satisfies

$$2\mathcal{H} = g^{\mu\nu}p_\mu p_\nu = 0, \quad (4.2)$$

where $p_\mu = \frac{dx_\mu}{d\lambda}$ is the proper four-momentum of the photon and λ denotes the affine parameter. For convenience, we assume that the photon is moving on the equatorial plane and therefore we can fix the coordinate $\theta = \pi/2$ and $\dot{\theta} = 0$. Then, with Eq. (4.2) and the metric (4.1), one can obtain

$$-\frac{p_t^2}{2f(r)} + \frac{h(r)p_r^2}{2} + \frac{p_\phi^2}{2r^2} = 0, \quad (4.3)$$

where p_r is the radial momentum of the photon. For the geodesic motion in the context of such black hole background, there are two conserved quantities corresponding to the Killing vector fields $\frac{\partial}{\partial t}$ and $\frac{\partial}{\partial \phi}$, i.e.,

$$-E = p_t = \frac{\partial \mathcal{H}}{\partial \dot{t}}, \quad L = p_\phi = \frac{\partial \mathcal{H}}{\partial \dot{\phi}}, \quad (4.4)$$

where $E = -p_t$ represents the energy of the photon and $L = p_\phi$ represents the angular momentum of the photon. Note that the dot indicates the derivative with respect to the affine parameter λ . Then, one can get the equation of motion of the photon from Eq. (4.3),

$$\dot{t} = \frac{\partial \mathcal{H}}{\partial p_t} = -\frac{p_t}{f(r)}, \quad \dot{\phi} = \frac{\partial \mathcal{H}}{\partial p_\phi} = \frac{p_\phi}{r^2}, \quad \dot{r} = \frac{\partial \mathcal{H}}{\partial p_r} = p_r h(r). \quad (4.5)$$

From Eq. (4.2), the effective potential of the photon can be defined as

$$\dot{r}^2 + V_{eff}(r) = 0. \quad (4.6)$$

With Eqs. (4.3) and (4.5), the effective potential V_{eff} can be written as

$$V_{eff}(r) = h(r) \left(\frac{L^2}{r^2} - \frac{E^2}{f(r)} \right). \quad (4.7)$$

The circular orbit radius r_p of the photon needs to satisfy the following conditions,

$$V_{eff}(r_p) = 0, \quad \left. \frac{\partial V_{eff}(r)}{\partial r} \right|_{r=r_p} = 0. \quad (4.8)$$

The impact parameter μ_p of the photon can be deduced from Eqs. (4.7) and (4.8), which is given by

$$\mu_p = \frac{L}{E} = \left. \frac{r}{\sqrt{f(r)}} \right|_{r=r_p}. \quad (4.9)$$

Using Eq. (4.5), the orbit equation of the photon can be expressed as

$$\frac{dr}{d\phi} = \frac{\dot{r}}{\dot{\phi}} = \frac{r^2 h(r) p_r}{L}. \quad (4.10)$$

Combining with Eqs. (4.3) and (4.10), one can obtain

$$\frac{dr}{d\phi} = \pm r \sqrt{h(r) \left(\frac{r^2 E^2}{f(r) L^2} - 1 \right)}. \quad (4.11)$$

For the turning point ($r = \chi$) of the photon orbit, one has the mathematical constraint $\left. \frac{dr}{d\phi} \right|_{r=\chi} = 0$, which means that $\frac{E^2}{L^2} = \frac{f(\chi)}{\chi^2}$. Then Eq. (4.11) can be rewritten as

$$\frac{dr}{d\phi} = \pm r \sqrt{h(r) \left(\frac{r^2 f(\chi)}{f(r) \chi^2} - 1 \right)}. \quad (4.12)$$

Now, we study the black hole shadow observed by a static observer located at position r_o . For a light ray emitting from the observer and transmitting into the past with an angular ψ with respect to the radial direction, we have^[184-186,189],

$$\cot \psi = \frac{\sqrt{g_{rr}}}{\sqrt{g_{\phi\phi}}} \cdot \left. \frac{dr}{d\phi} \right|_{r=r_o} = \frac{1}{r \sqrt{h(r)}} \cdot \left. \frac{dr}{d\phi} \right|_{r=r_o}. \quad (4.13)$$

Combining with Eqs. (4.12) and (4.13), we get

$$\cot^2 \psi = \frac{r_o^2 f(\chi)}{f(r_o) \chi^2} - 1 \quad \text{or} \quad \sin^2 \psi = \frac{f(r_o) \chi^2}{r_o^2 f(\chi)}. \quad (4.14)$$

For the static observer located at position r_o , one can set $\chi = r_p$ to obtain the shadow radius r_s ,

$$r_s = r_o \sin \psi = \chi \sqrt{\frac{f(r_o)}{f(\chi)}} \Big|_{\chi=r_p}. \quad (4.15)$$

For the Hayward-AdS black hole, the coefficients in the metric (4.1) are given as^[190-191]

$$f(r) = h(r)^{-1} = 1 - \frac{2Mr^2}{r^3 + g^3} + \frac{8\pi Pr^2}{3}, \quad (4.16)$$

where M represents the black hole mass and g represents the magnetic charge. The thermodynamic pressure P is associated with the cosmological constant, namely, $P = -\frac{\Lambda}{8\pi} = \frac{3}{8\pi l^2}$. One can use Eqs. (4.8) and (4.16) to obtain the circular orbit radius of the photon, which is given as

$$r_p = M + \frac{M^2}{\left(M^3 - g^3 + \sqrt{g^6 - 2g^3 M^3}\right)^{1/3}} + \left(M^3 - g^3 + \sqrt{g^6 - 2g^3 M^3}\right)^{1/3}. \quad (4.17)$$

Finally, with Eqs. (4.15) and (4.16), the expression for the shadow radius of the Hayward-AdS black hole, for a static observer located at position r_o , can be directly written as

$$r_s = r_p \sqrt{\frac{f(r_o)}{f(r_p)}}, \quad (4.18)$$

where r_p is given by Eq. (4.17).

4.3 Thermodynamics of the Hayward-AdS black hole using shadow formalism

For the Hayward-AdS black hole, the mass M and temperature T are given by^[191]

$$M = \frac{g^3}{2r_h^2} + \frac{r_h}{2} + \frac{4}{3}P\pi(r_h^3 + g^3), \quad T = \frac{r_h^3 - 2g^3}{4\pi r_h(g^3 + r_h^3)} + \frac{2Pr_h^4}{g^3 + r_h^3}, \quad (4.19)$$

where r_h is the horizon radius of the Hayward-AdS black hole. Since the expression of r_h is extremely verbose, we do not display the exact form here. The equation of state of the Hayward-AdS black hole is given as

$$P = \frac{g^3}{4\pi r_h^5} - \frac{1}{8\pi r_h^2} + \frac{g^3 T}{2r_h^4} + \frac{T}{2r_h}. \quad (4.20)$$

To obtain the critical point of the thermodynamic phase transition for the Hayward-AdS black hole, one can take Eq. (4.20) into the critical condition $(\partial P / \partial r_h) = 0 = (\partial^2 P / \partial r_h^2)$,

which yields

$$P_c = \frac{3(57 - 23\sqrt{6})(14 + 6\sqrt{6})^{1/3}}{800g^2\pi} \simeq 0.002418g^{-2}, \quad (4.21)$$

$$r_c = (14 + 6\sqrt{6})^{1/3}g \simeq 3.061577g, \quad (4.22)$$

$$T_c = \frac{(5 - 2\sqrt{6})(7 + 3\sqrt{6})^{2/3}}{4 \times 2^{1/3}g\pi} \simeq 0.037675g^{-1}. \quad (4.23)$$

Substituting Eq. (4.19) into Eq. (4.17), one can rewrite the circular orbit radius of the photon as

$$r_p = \frac{g^3}{2r_h^2} + \frac{r_h}{2} + \frac{4}{3}(g^3 + r_h^3)P\pi + \frac{(3 + 8P\pi r_h^2)^2(g^3 + r_h^3)^2}{36r_h^4\xi^{1/3}} + \xi^{1/3}, \quad (4.24)$$

where

$$\xi = \frac{(3 + 8P\pi r_h^2)^3(g^3 + r_h^3)^3}{216r_h^6} + \sqrt{g^6 - \frac{g^3(3 + 8P\pi r_h^2)^3(g^3 + r_h^3)^3}{108r_h^6}} - g^3. \quad (4.25)$$

Combining with Eqs. (4.18) and (4.24), we plot the $r_s - r_h$ diagrams of the Hayward-AdS black hole in Fig. 4.1, where we have set $f(r_o = 100) = 1$ for the static observer^[184]. For different magnetic charge g and pressure P , the shadow radius r_s invariably increases monotonically with the horizon radius r_h . This result implies that the horizon radius r_h can be substituted with the shadow radius r_s to study the thermodynamics of the Hayward-AdS black hole.

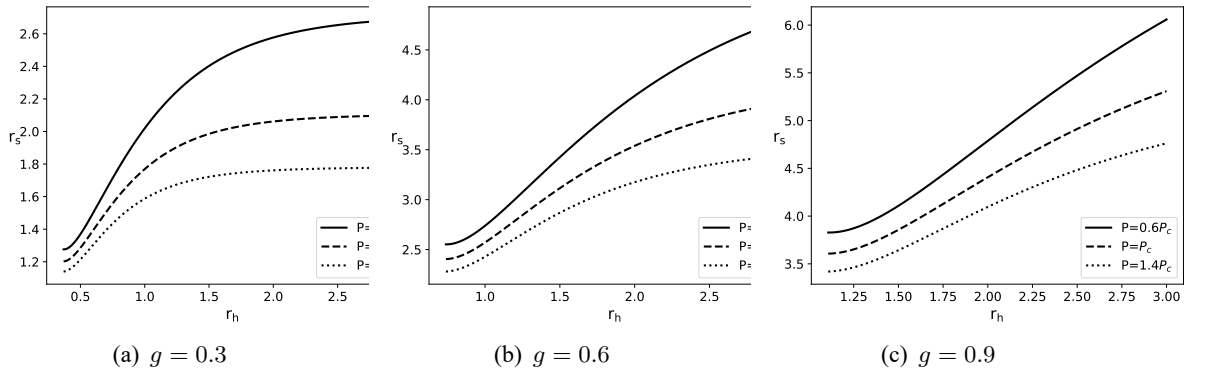


Figure 4.1: The shadow radius r_s versus the horizon radius r_h . The solid, dashed and dotted lines correspond to $P = 0.6P_c$, $P = P_c$ and $P = 1.4P_c$, respectively.

With Eq. (4.19), we plot the $T - r_h$ diagrams with three types of pressures based on the critical pressure P_c (see Figs. 4.2(a), 4.2(b) and 4.2(c) with the magnetic charge $g = 0.3, 0.6, 0.9$). It is found that when the pressure is less than the critical pressure

such as $P = 0.6P_c$, the corresponding temperature curve is not monotonous and there exists a vdW-like phase transition. The slopes of the small radius region ($r_h < r_{h1}$) and the large radius region ($r_h > r_{h2}$) are both positive, which correspond to the small black hole phase and the large black hole phase, respectively. The intermediate region ($r_{h1} < r_h < r_{h2}$) represents a coexistence region of the small and large black hole. The intermediate region is thermodynamically unstable and the corresponding temperature T_{co} is known as coexistence temperature. For the pressure equal to the critical pressure ($P = P_c$), there will be an unstable phase transition. As for the pressure larger than the critical pressure ($P = 1.4P_c$), no phase transition appears.

On the other hand, we can also study the thermodynamic phase transition of the Hayward-AdS black hole by replacing the horizon radius r_h with the shadow radius r_s . In Figs. 4.2(d), 4.2(e) and 4.2(f), we plot the $T - r_s$ diagrams with the parameters adopted in the $T - r_h$ diagrams. To observe the difference between the two kinds of diagrams, we also consider three sets of values of the magnetic charge g and three sets of values of the pressure P . The results show that the phase-transition process denoted by the shadow radius r_s is similar to the phase-transition process denoted by the horizon radius r_h . The critical point of the phase-transition process is still dependent on the pressure P . For $P < P_c$, there exists a vdW-like phase transition. The small ($r_s < r_{s1}$) and large ($r_s > r_{s2}$) radius regions correspond to the stable small and large black hole, respectively. The radius region $r_{s1} < r_s < r_{s2}$ corresponds to the unstable intermediate black hole. For $P = P_c$, there will be an unstable phase transition. When $P > P_c$, there is no phase transition. Moreover, from the whole Fig. 4.2, it can be seen that for the pressure $P = 0.6P_c$, as the magnetic charge g increases, the shadow radius r_s becomes larger while the coexistence temperature T_{co} becomes lower.

In order to analyze further the thermodynamic stability of the Hayward-AdS black hole, we investigate its heat capacity. The positive (negative) heat capacity corresponds to the thermodynamically stable (unstable) black hole phase. The heat capacity C_P of the Hayward-AdS black hole with a fixed pressure P and magnetic charge g is given by

$$C_P = T \left(\frac{dS}{dT} \right)_{P,g} = \frac{2\pi r_h^2 (g^3 + r_h^3) (-2g^3 + r_h^3 + 8P\pi r_h^5)}{2g^6 + r_h^6 (-1 + 8P\pi r_h^2) + 2g^3 r_h^3 (5 + 16P\pi r_h^2)}. \quad (4.26)$$

The $C_P - r_h$ diagrams of the Hayward-AdS black hole for different magnetic charge g and pressure P are shown in Figs. 4.3(a), 4.3(b) and 4.3(c). For the pressure $P = 0.6P_c$, $(\partial_{r_h} T)_P = 0$ has two roots r_{h1} and r_{h2} as shown in Fig. 4.2. These two roots indicate two divergences of the heat capacity C_P . The sign of C_P is changed at these two divergences. When the pressure is increased to $P = P_c$, these two roots coincide with each other, so

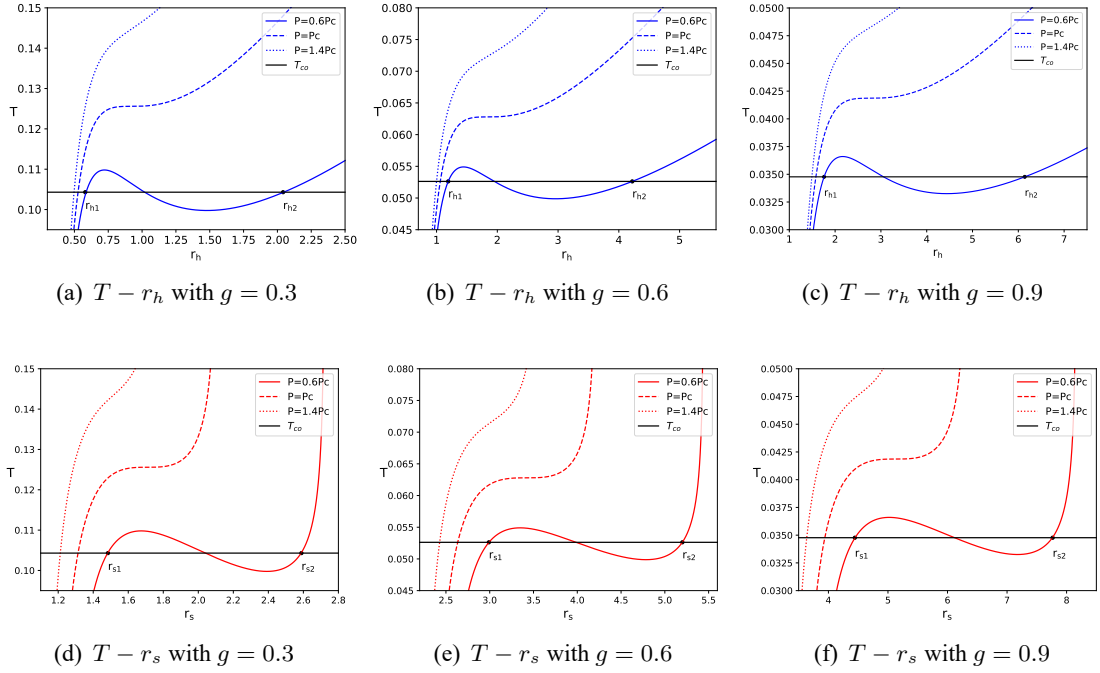


Figure 4.2: The temperature T versus r_h (4.2(a), 4.2(b), 4.2(c)) and r_s (4.2(d), 4.2(e), 4.2(f)) for the magnetic charge $g = 0.3, 0.6, 0.9$. T_{co} (black lines) is the coexistence temperature. The solid, dashed and dotted lines correspond to $P = 0.6P_c$, $P = P_c$ and $P = 1.4P_c$, respectively.

there is only one point of divergence in the heat capacity C_P . For the pressure $P = 1.4P_c$, it can be seen that the heat capacity C_P is a positive and continuous function of the horizon radius r_h .

Combining with Eqs. (4.18) and (4.26), we can also study the heat capacity of the Hayward-AdS black hole by replacing the horizon radius r_h with the shadow radius r_s . The $C_P - r_s$ diagrams of the Hayward-AdS black hole are shown in Figs. 4.3(d), 4.3(e) and 4.3(f). When the pressure is less than the critical pressure, the heat capacity C_P also diverges at two points. When the pressure is equal to the critical pressure, there is only one divergent behavior in the heat capacity C_P . For the pressure larger than the critical pressure, the heat capacity C_P is also a positive and continuous function of the shadow radius r_s .

To sum up, the behaviors of the heat capacity C_P in the $C_P - r_s$ diagrams are similar to those in the $C_P - r_h$ diagrams. In addition, Fig. 4.2 shows that for $P < P_c$, both the small ($r_s < r_{s1}$) and large ($r_s > r_{s2}$) black hole branches have positive slopes (i.e., positive heat capacities), which means that these two branches are thermodynamically stable. The slope of the intermediate black hole branch is negative, which represents a negative heat capacity, and thus it is thermodynamically unstable. These results are consistent with the

phenomena depicted in Fig. 4.3.

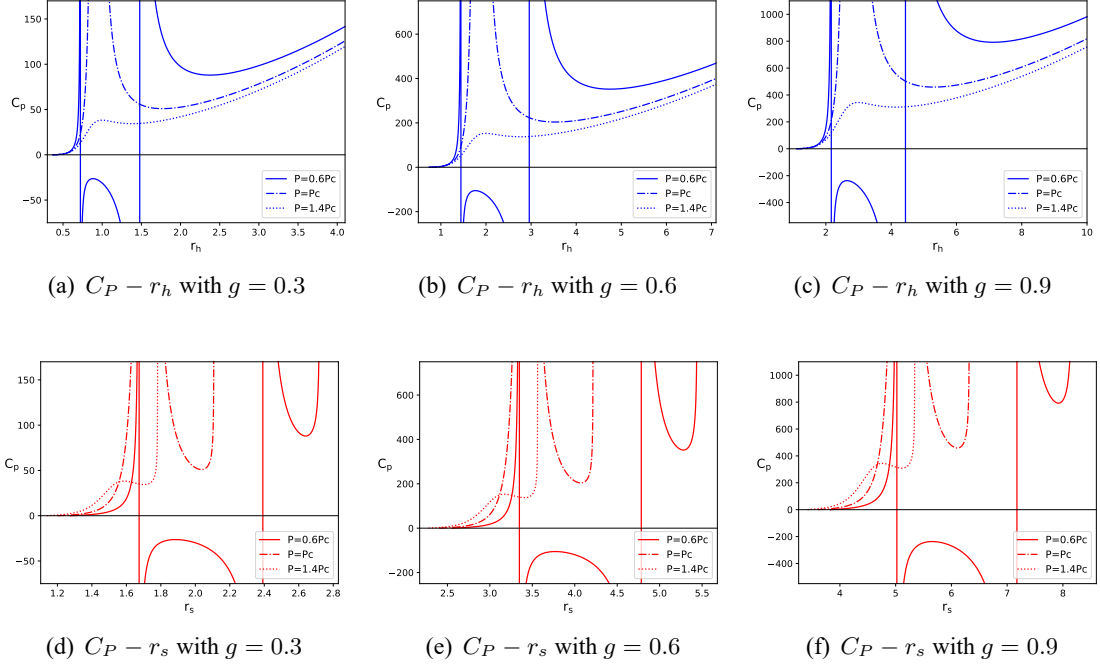


Figure 4.3: The heat capacity C_P versus r_h (4.3(a), 4.3(b), 4.3(c)) and r_s (4.3(d), 4.3(e), 4.3(f)) for the magnetic charge $g = 0.3, 0.6, 0.9$. The solid, dashed and dotted lines correspond to $P = 0.6P_c$, $P = P_c$ and $P = 1.4P_c$, respectively.

4.4 Thermal profile of the Hayward-AdS black hole

In order to show the phase structure of the Hayward-AdS black hole more intuitively, we calculate the boundary curve of the shadow of the Hayward-AdS black hole. For such a spherically symmetric black hole, the shape of the black hole shadow is circular for an arbitrary observer^[192]. Therefore, one can use a thermal profile in a two-dimensional plane to study the relation between the thermodynamic phase transition and shadow of the Hayward-AdS black hole. In this case, for the stereographic projection of the Hayward-AdS black hole located in the celestial coordinate onto the two-dimensional plane, the boundary curve of the shadow is given as follows^[193],

$$x = \lim_{r \rightarrow \infty} \left(-r^2 \sin \theta_0 \frac{d\phi}{dr} \right)_{\theta_0 \rightarrow \frac{\pi}{2}}, \quad y = \lim_{r \rightarrow \infty} \left(r^2 \frac{d\theta}{dr} \right)_{\theta_0 \rightarrow \frac{\pi}{2}}, \quad (4.27)$$

where x and y are the Cartesian coordinates with the inclination angle $\theta_0 \rightarrow \frac{\pi}{2}$.

The shadow contours of the Hayward-AdS black hole for different parameters are shown in Fig. 4.4, where we have set $M = 60$ and $f(r_o = 100) = 1$ for the static observer. The results show that the shadow radius consistently increases with the pressure.

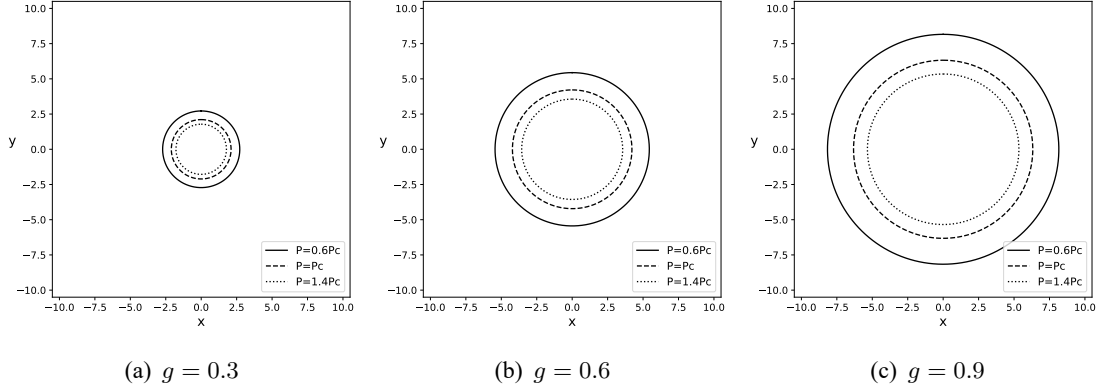


Figure 4.4: The boundary curve of the shadow of the Hayward-AdS black hole for the magnetic charge $g = 0.3, 0.6, 0.9$. The solid, dashed and dotted lines correspond to $P = 0.6P_c$, $P = P_c$ and $P = 1.4P_c$, respectively. We have set $M = 60$ and $r_o = 100$.

The solid lines denote $P < P_c$, which correspond to the largest shadow radius region. As we clarified earlier, in this case, there exists a vdW-like phase transition between the small black hole and large black hole. The dashed lines represent $P = P_c$, which is a critical point because there is an unstable phase transition. The dotted lines correspond to $P > P_c$ and the shadow represents a supercritical phase, for which no phase transition occurs. Moreover, for the same pressure, comparing the radii of the shadow contours with different magnetic charges, one can find that the shadow radius of the Hayward-AdS black hole increases with the magnetic charge g .

To analyze further, from multiple aspects, the relation between the phase structure and the shadow of the Hayward-AdS black hole, one can study the superposition of the temperature diagram (see Fig. 4.2) and the shadow cast diagram (see Fig. 4.4), which is called the thermal profile of the Hayward-AdS black hole (see Fig. 4.5). In order to do that, we need to fix the pressure of the Hayward-AdS black hole and then the shadow contour can be considered as a function of temperature for a static observer located at position $r_o = 100$. Fig. 4.5 illustrates the thermal profiles of the Hayward-AdS black hole for three sets of magnetic charges and three sets of pressures. Figs. 4.5(a), 4.5(d) and 4.5(g) represent $P < P_c$, which correspond to the solid lines in Fig. 4.4. It can be found that when the shadow radius is smaller than a certain value, the temperature increases from the center to the boundary of the shadow. Then, the temperature oscillates as the shadow radius increases, i.e., the N-type change trend^[194]. Figs. 4.5(b), 4.5(e) and 4.5(h) denote $P = P_c$, which correspond to the dashed lines in Fig. 4.4. There exists a critical region of thermodynamic instability, where the temperature remains constant. Figs. 4.5(c), 4.5(f) and 4.5(i) represent $P > P_c$, which correspond to the dotted lines in Fig. 4.4. One can find that the temperature keeps increasing from the center to the boundary of the shadow, which

is consistent with the result of the dotted lines in Fig. 4.2. In addition, comparing these diagrams, one can find that when the magnetic charge g is fixed, the temperature increases with the pressure while the region of the thermal profile decreases with the pressure.

Finally, for the Hayward-AdS black hole, we study the region where the N-type change trend occurs. For the case of $P < P_c$, the thermal profiles of the Hayward-AdS black hole with different magnetic charges from r_{s1} to r_{s2} are shown in Fig. 4.6. It can be found that the temperature increases first, then decreases, and finally increases again from the center to the boundary of the shadow, which is consistent with the result of the solid lines in Fig. 4.2. The results imply that when $P < P_c$, the thermal profile can be used to investigate the phase structure of the Hayward-AdS black hole.

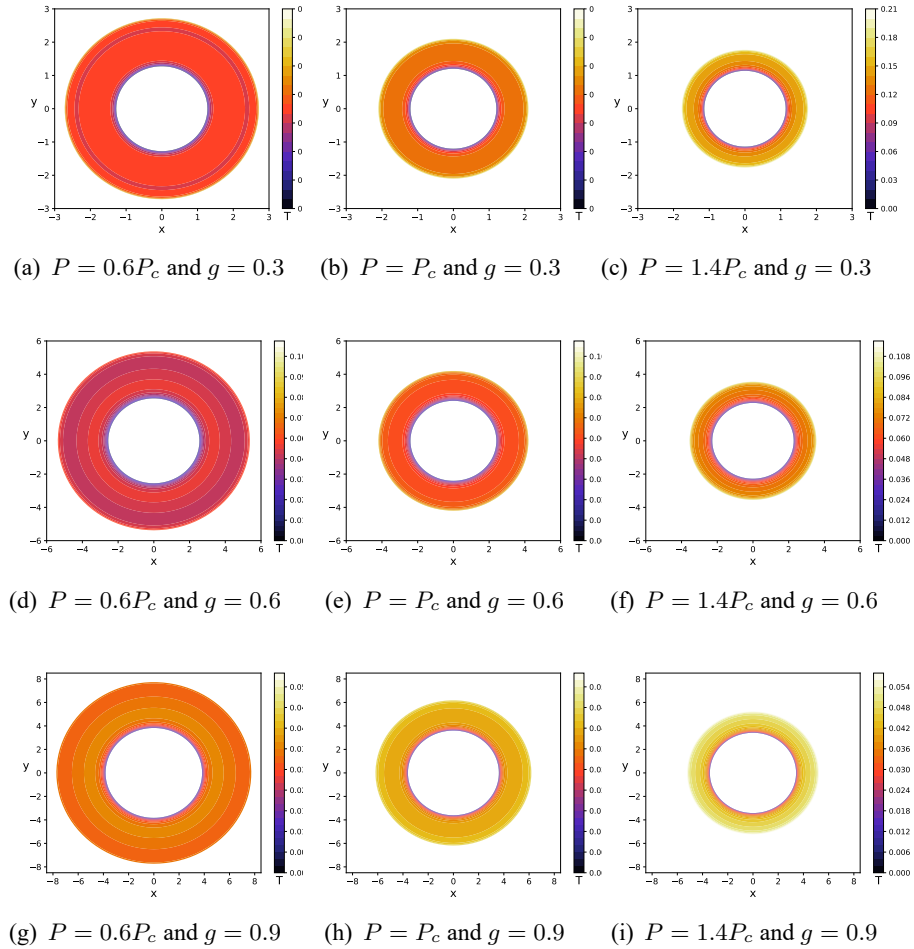


Figure 4.5: Thermal profiles of the Hayward-AdS black hole for the magnetic charge $g = 0.3, 0.6, 0.9$ and the pressure $P = 0.6P_c, P_c, 1.4P_c$. We have set $M = 60$ and $r_o = 100$.

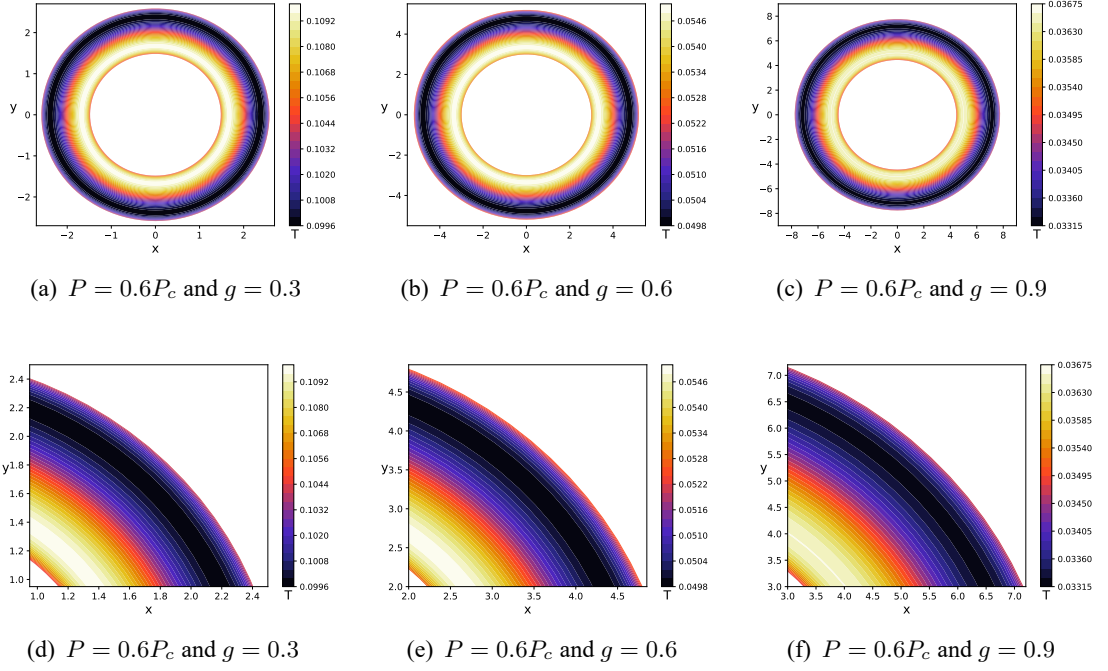


Figure 4.6: Thermal profiles of the region from r_{s1} to r_{s2} for the magnetic charge $g = 0.3, 0.6, 0.9$ and the pressure $P = 0.6P_c$. We have set $M = 60$ and $r_o = 100$.

4.5 Thermodynamics of the charged AdS black hole with global monopole

In this section, we present a brief overview of the charged AdS black hole with global monopole and its thermodynamics. The Lagrangian density of the charged AdS black hole with global monopole is^[195]

$$\mathcal{L} = R - 2\Lambda + \frac{1}{2}\partial_\mu\phi^a\partial^\mu\phi^{*a} - \frac{\gamma}{4}(\phi^a\phi^{*a} - \eta_0^2)^2, \quad (4.28)$$

where γ is a constant and η_0 is the energy scale of symmetry breaking. The parameter ϕ^a represents a triplet of the scalar field as

$$\phi^a = \eta_0 h(\tilde{r}) \frac{\tilde{x}^a}{\tilde{r}} \quad (4.29)$$

with the boundary conditions^[196-197]

$$\eta_0 h(0) = 0, \quad \lim_{\tilde{r} \rightarrow \infty} \eta_0 h(\tilde{r}) = \pm \eta_0. \quad (4.30)$$

Here x^a is the Cartesian coordinate and r is the spherical coordinate, which satisfy $\tilde{x}^a \tilde{x}^a = \tilde{r}^2$. The metric of the charged AdS black hole with global monopole is given by^[195]

$$ds^2 = -f(\tilde{r})d\tilde{t}^2 + \frac{1}{f(\tilde{r})}d\tilde{r}^2 + \tilde{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (4.31)$$

where the metric function $f(\tilde{r})$ is

$$f(\tilde{r}) = 1 - 8\pi\eta_0^2 - \frac{2\tilde{m}}{\tilde{r}} + \frac{\tilde{q}^2}{\tilde{r}^2} + \frac{\tilde{r}^2}{l^2}. \quad (4.32)$$

Here, $\tilde{m}$ is the black hole mass, $\tilde{q}$ is the black hole charge, and $l(= \sqrt{-\frac{\Lambda}{3}})$ is the AdS curvature radius. For convenience, we introduce the following coordinate transformation

$$\tilde{t} = (1 - 8\pi\eta_0^2)^{-1/2} t, \quad \tilde{r} = (1 - 8\pi\eta_0^2)^{1/2} r, \quad (4.33)$$

and new parameters

$$m = (1 - 8\pi\eta_0^2)^{-3/2} \tilde{m}, \quad q = (1 - 8\pi\eta_0^2)^{-1} \tilde{q}, \quad \eta^2 = 8\pi\eta_0^2. \quad (4.34)$$

Therefore, the metric (4.31) can be rewritten as

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + (1 - \eta^2) r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (4.35)$$

where

$$f(r) = 1 - \frac{2m}{r} + \frac{q^2}{r^2} + \frac{r^2}{l^2}. \quad (4.36)$$

It is noteworthy that the non-vanishing monopole parameter η will result in a solid angle deficit in Eq. (4.35). Therefore, the ADM mass and the electric charge of the black hole are modulated by the monopole parameter η :

$$M = (1 - \eta^2) m, \quad Q = (1 - \eta^2) q. \quad (4.37)$$

The event horizon r_h can be obtained by solving $f(r) = 0$, and then the ADM mass M can be rewritten as

$$M = \frac{Q^2}{2r_h - 2\eta^2 r_h} - \frac{1}{2} (\eta^2 - 1) r_h (1 + l^{-2} r_h^2). \quad (4.38)$$

The temperature and entropy of the black hole can be expressed as

$$T = \frac{1}{4\pi r_h} \left(1 + \frac{3r_h^2}{l^2} - \frac{Q^2}{(1-\eta^2)^2 r_h^2} \right), \quad S = \pi (1 - \eta^2) r_h^2. \quad (4.39)$$

Furthermore, in the extended phase space (i.e., $P = 3/8\pi l^2$), one can obtain

$$dM = TdS + \Phi dQ + VdP, \quad (4.40)$$

where

$$\Phi = \frac{q}{r_h}, \quad V = \frac{4}{3}\pi (1 - \eta^2) r_h^3. \quad (4.41)$$

The Smarr relation still holds as follows:

$$M = 2(TS - VP) + \Phi Q. \quad (4.42)$$

Note that M in Eq. (4.42) is now regarded as the enthalpy of the black hole.

Therefore, for the charged AdS black hole with global monopole, the equation of state is given as

$$P = \frac{T}{2r_h} - \frac{1}{8\pi r_h^2} + \frac{Q^2}{8\pi (1 - \eta^2) r_h^4}. \quad (4.43)$$

By solving the equations $(\partial_{r_h} P)_T = 0$ and $(\partial_{r_h, r_h} P)_T = 0$, one can obtain the critical point

$$P_c = \frac{(1 - \eta^2)^2}{96\pi Q^2}, \quad r_c = \frac{\sqrt{6}Q}{\sqrt{1 - 2\eta^2 + \eta^4}}, \quad T_c = \frac{\sqrt{(1 - \eta^2)^2}}{3\sqrt{6}\pi Q}. \quad (4.44)$$

In this paper, we set $Q = 1$ for convenience.

As shown in Fig. 4.7, the evolution of the black hole temperature over the black hole radius r_h is closely related to the value of the pressure P . When $P \geq P_c$, T is a monotonic function of the radius r_h , while when $P < P_c$ there is a local minimum temperature and a local maximum temperature. The local minimum and maximum of the black hole temperature T are determined by

$$\frac{\partial T}{\partial r_h} = 0, \quad (4.45)$$

which yields

$$r_{\min/\max} = \frac{1}{4} \sqrt{\frac{1 - \eta^2 \pm \sqrt{1 - 96P\pi Q^2 - 2\eta^2 + \eta^4}}{P\pi - P\pi\eta^2}}. \quad (4.46)$$

The positive (negative) sign in Eq. (4.46) corresponds to the black hole radius $r_{\min}$ ($r_{\max}$).

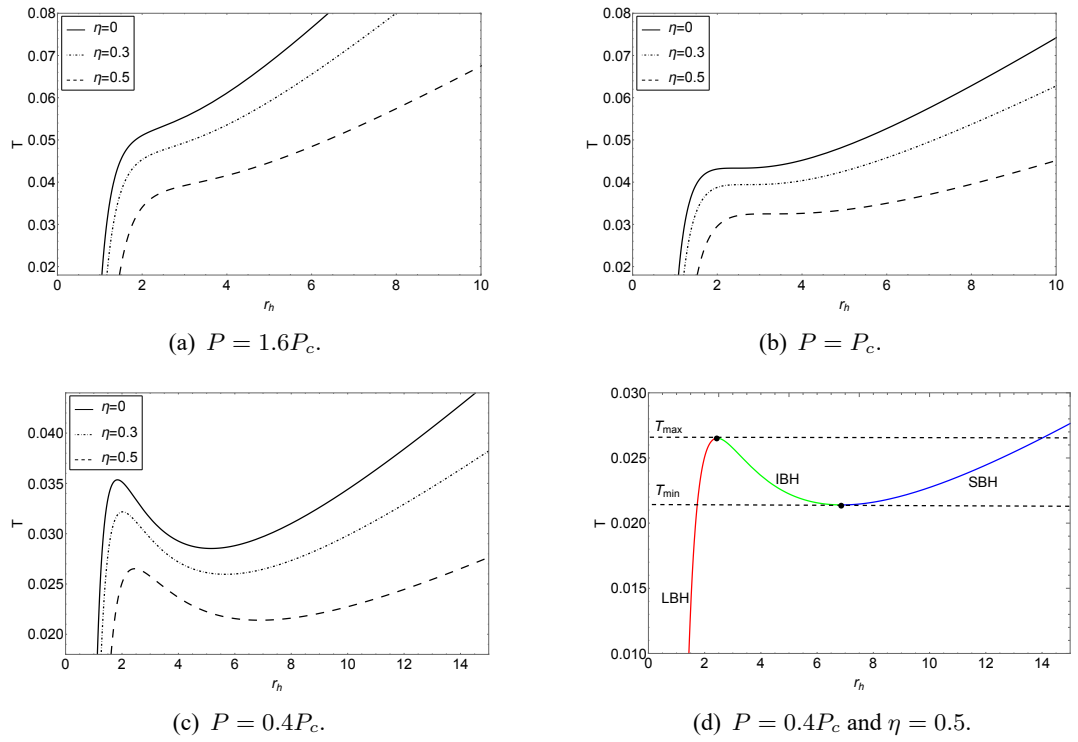


Figure 4.7: The temperature T as a function of the black hole radius r_h . The pressures in the first three panels are selected as (a) $P = 1.6P_c$; (b) $P = P_c$; (c) $P = 0.4P_c$. There are three sets of the monopole parameters ($\eta = 0, 0.3, 0.5$) for comparison. The last panel corresponds to the dashed line in figure (c).

By substituting Eq. (4.46) into Eq. (4.43), we get the local minimum and maximum of the black hole temperature:

$$T_{\min/\max} = \frac{-64P\pi Q^2 \pm 2(-1 + \eta^2) \left(1 - \eta^2 + \sqrt{-96P\pi Q^2 + (-1 + \eta^2)^2}\right)}{P\sqrt{\pi}(-1 + \eta^2)^2 \left(\frac{1 - \eta^2 + \sqrt{-96P\pi Q^2 + (-1 + \eta^2)^2}}{P - P\eta^2}\right)^{3/2}}, \quad (4.47)$$

where the positive (negative) sign in Eq. (4.47) corresponds to the local minimum $T_{\min}$ (the local maximum $T_{\max}$). When $T_{\min} < T < T_{\max}$, there are three branches: small black hole (SBH) state, intermediate black hole (IBH) state, and large black hole (LBH) state (see Figs. 4.7(c) and 4.7(d)). It is worth noting that the IBH state is not stable, and there exists a first-order SBH/LBH phase transition.

4.6 Dynamical properties of thermodynamic phase transition

4.6.1 Gibbs free energy

In the study of the black hole thermodynamics, Gibbs free energy is usually used to investigate the thermodynamic phase transition of black holes^[44,53,55-56,58,60,63-66,69,74]. For the charged AdS black hole with global monopole, the Gibbs free energy is given by

$$G = H - TS = \frac{1}{6}r_h (3 + 8P\pi r_h^2 - 6\pi r_h T) (1 - \eta^2) + \frac{Q^2}{2r_h (1 - \eta^2)}. \quad (4.48)$$

The $G - T$ diagram of the charged AdS black hole with global monopole is plotted in Fig. 4.8(a) with $\eta = 0.5$ and $P = 0.4P_c$, from which we can see there exist three branches of the black hole state and a swallowtail behavior. The blue, green, and red lines represent the SBH, IBH, and LBH branches, respectively. The temperature corresponding to the intersection (see the point A(C) in Fig. 4.8(a)) of the SBH branch and LBH branch is about $T_A = T_C = 0.022355$. The point B is in a state of the IBH, and the point A and C are in states of the SBH and LBH, respectively.

Recently, the concept of the Gibbs free energy landscape is introduced to study the dynamical properties of the thermodynamic phase transition of black holes^[187-188]. In the Gibbs free energy landscape, the Hawking temperature of the black hole is replaced by the temperature of the ensemble. Therefore, for the charged AdS black hole with global monopole, the Gibbs free energy G_L can be expressed as

$$G_L = H - T_E S = -\frac{1}{2}r_h (-1 + \eta^2) - \frac{4}{3}P\pi r_h^3 (-1 + \eta^2) + \pi r_h^2 T_E (-1 + \eta^2) + \frac{Q^2}{2r_h - 2r_h \eta^2}, \quad (4.49)$$

where T_E is the temperature of the ensemble. The $G_L - r_h$ diagram is plotted in Fig. 4.8(b)

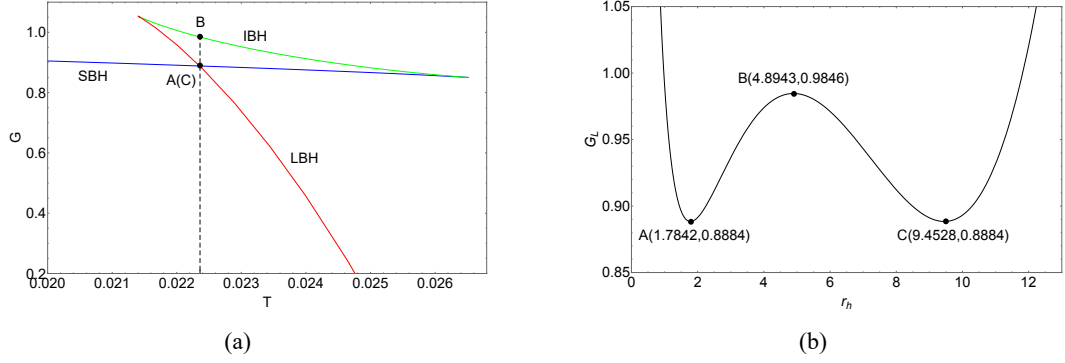


Figure 4.8: Gibbs free energy of the charged AdS black hole with global monopole. For all plots, we set $\eta = 0.5$ and $P = 0.4P_c$. The left panel is the $G - T$ diagram, and the right panel is the $G_L - r_h$ diagram with $T_E = 0.022355$.

with $\eta = 0.5$, $P = 0.4P_c$, and $T_E = 0.022355$. As can be seen, the $G_L - r_h$ diagram has a double-well shape and there exist two equivalent local minimums and a local maximum. Since the SBH and LBH correspond to the local minimums of the Gibbs free energy, they are stable. As for the IBH which has the local maximum of the Gibbs free energy, it is unstable.

4.6.2 Fokker-Planck equation and probability evolution

Recently, the Fokker-Planck equation has been proposed to study the probability evolution of the black hole phase transition, which is given by^[187-188,198-202]

$$\frac{\partial \rho(r_h, t)}{\partial t} = D \frac{\partial}{\partial r_h} \left\{ e^{-\beta G_L(r_h)} \frac{\partial}{\partial r_h} [e^{\beta G_L(r_h)} \rho(r_h, t)] \right\}, \quad (4.50)$$

where G_L is a function of the radius r_h modulated by the black hole temperature, $\rho(r_h, t)$ is the probability distribution of the black hole state in the ensemble, $\beta = \frac{1}{k_B T}$ presents the inverse temperature, and $D = \frac{k_B T}{\zeta}$ is the diffusion coefficient. The Boltzman constant k_B and the dissipation coefficient ζ as usual are set to 1.

Mathematically, one can calculate the Fokker-Planck equation by choosing specific boundary conditions. To study the probability evolution of the black hole phase transition, we use the reflecting and absorbing boundary conditions at $r_h = r_0$ to calculate the Fokker-Planck equation, which can be given as

$$\begin{aligned} \beta G'_L(T, P, r_0) \rho(r_0, t) + \rho'(r_0, t) &= 0, \\ \rho(r_0, t) &= 0. \end{aligned} \quad (4.51)$$

Here, the prime indicates the time derivative.

We first consider the reflecting boundary condition, for which there are two bound-

aries at $r_h = 0$ and $r_h = \infty$, respectively. The reflecting boundary condition preserves the normalization of the probability distribution. The Gaussian wave packet is selected as the initial wave packet, which is given by

$$\rho(r_h, 0) = \frac{1}{\sqrt{\pi}a} e^{-\frac{(r_h - r_s/m/l)^2}{a^2}}, \quad (4.52)$$

where the parameter a is set to 1 and the Gaussian wave packet is located at the SBH/IBH/LBH radius $r_s/m/l$.

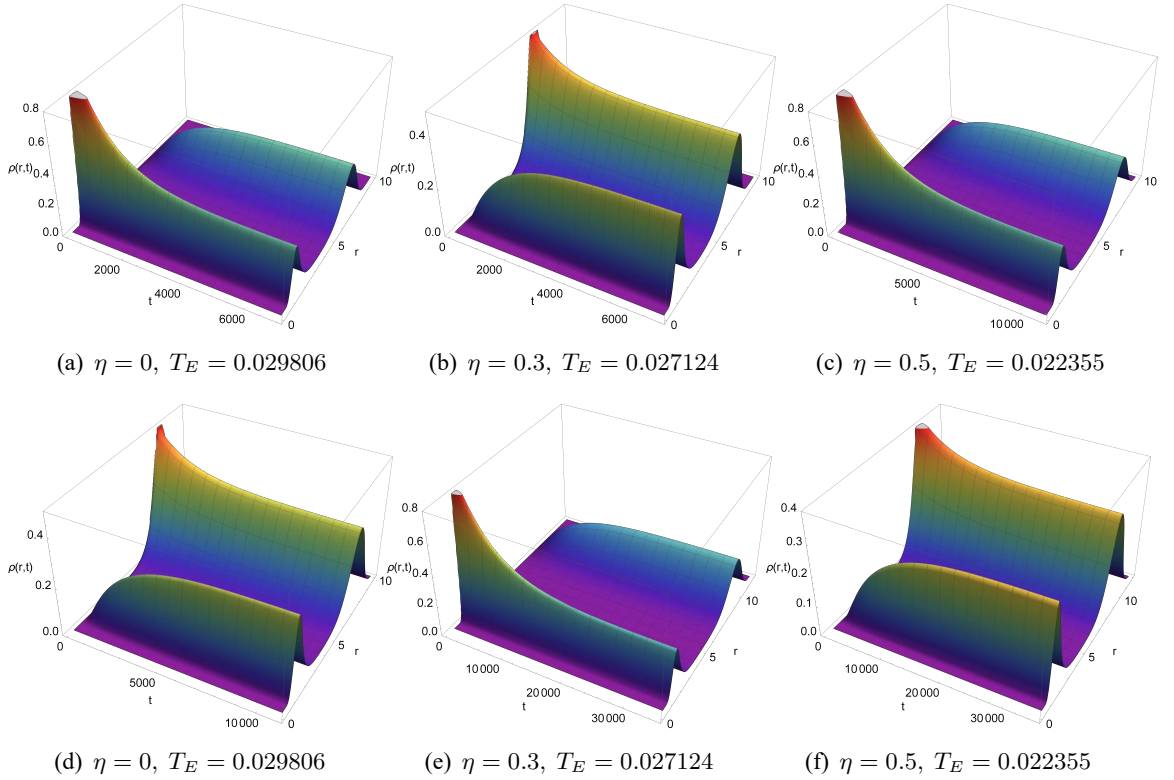


Figure 4.9: The distributions of probability $\rho(r_h, t)$ for the charged AdS black hole with global monopole. For the upper panels, the Gaussian wave packet is located at the SBH state. For the lower panels, the Gaussian wave packet is located at the LBH state.

In Fig. 4.9, we plot the probability distribution diagrams of the charged AdS black hole with global monopole via numerical calculation. Note that in order to avoid numerical instability we have replaced the boundaries at $r_h = 0$ and $r_h = \infty$ with $r_h = 0.1$ and $r_h = 12$, respectively. From Figs. 4.9(a), 4.9(b) and 4.9(c), one can find when the probability distribution evolves from the SBH state, the probability distribution of the SBH state decreases gradually, while the probability distribution of the LBH state increases from $\rho(r_l, t) = 0$. In Figs. 4.9(d), 4.9(e) and 4.9(f), we can observe a similar phenomenon for the case that the probability distribution evolves from the LBH state. The more intuitive descriptions of the time evolution of the probability distribution for

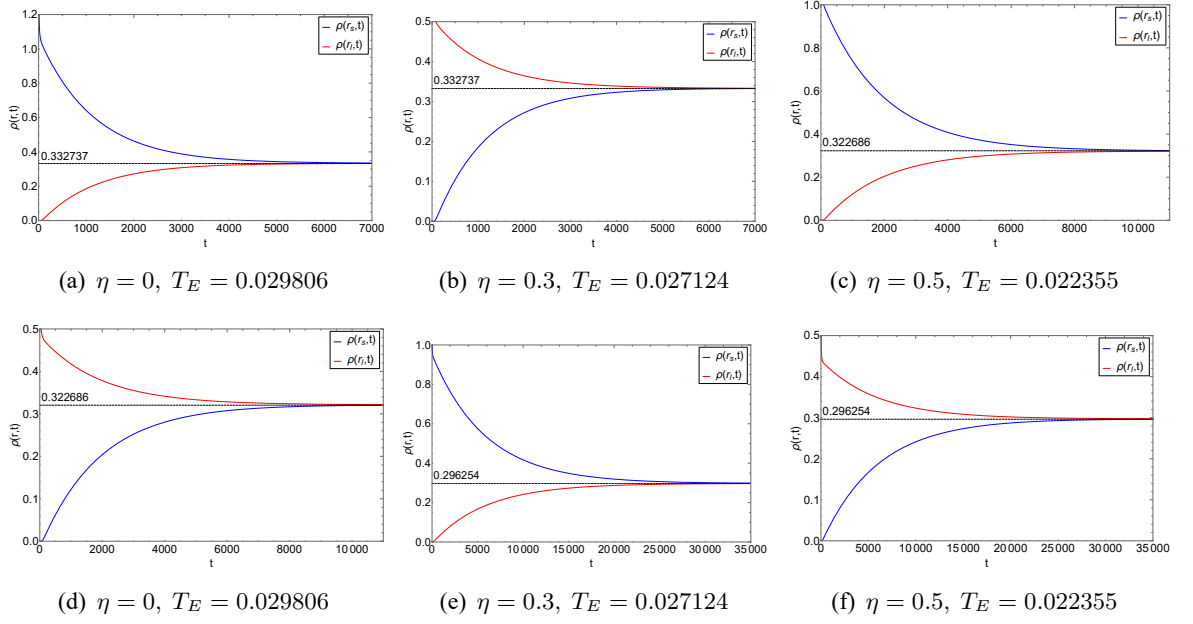


Figure 4.10: The distributions of probability $\rho(r_h, t)$ for given SBH and LBH radii. For the upper panels, the Gaussian wave packet is located at the SBH state. For the lower panels, the Gaussian wave packet is located at the LBH state.

the SBH and LBH states are shown in Fig. 4.10. When the Gibbs free energy of the SBH state is equal to that of the LBH state, the probability distribution of the black hole states is a steady-state distribution. In this case, the probability distributions of the SBH and LBH states are exactly equal. No matter what the value of the monopole parameter η is, the system will always reach a steady-state distribution after a period of time. Moreover, according to Fig. 4.10, it is also clear that the global monopole has effects on the thermodynamic phase transition of black holes. Comparing Figs. 4.10(a), 4.10(b) and 4.10(c), one can find the initial SBH state with global monopole (such as $\eta = 0.5$ and $\eta = 0.3$) evolves more slowly than the one without a global monopole ($\eta = 0$). For the initial LBH state with global monopole, one can obtain the same conclusion (see Figs. 4.10(d), 4.10(e) and 4.10(f)). Besides that, as the monopole parameter η increases, it takes longer for the system to reach equilibrium.

4.6.3 First passage process

In the black-hole phase transition, the process that the initial SBH (LBH) state reaches the IBH state for the first time is called the first passage process, and the time it takes is called the first passage time. In order to study the first passage process, we need to solve the Fokker-Planck equation not only with the reflecting boundary condition but also with the absorbing boundary condition $\rho(r_m, t) = 0$. Note that r_m represents the radius of the

IBH, which satisfies

$$G'_L(T, P, r)|_{r_h=r_m} = 0, \quad G''_L(T, P, r)|_{r_h=r_m} > 0. \quad (4.53)$$

One can suppose that there exists a perfect absorber placed at the coordinate $r_h = r_m$. When $t \rightarrow +\infty$, the probability $\Sigma(t)$ decreases and tends to zero.

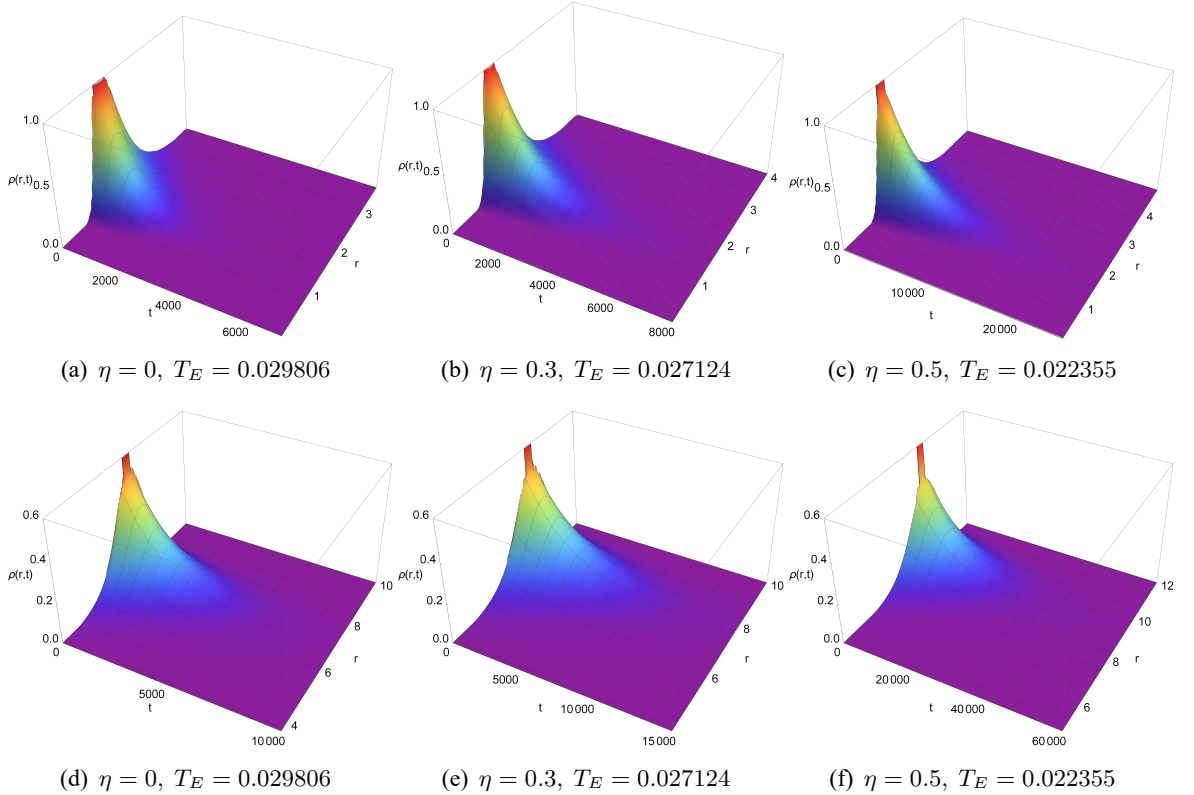


Figure 4.11: The distributions of probability $\rho(r_h, t)$ of the first passage process for the charged AdS black hole with global monopole. For the upper panels, the Gaussian wave packet is located at the SBH state. For the lower panels, the Gaussian wave packet is located at the LBH state.

Taking the reflecting and absorbing boundary conditions into the Fokker-Planck equation, we can solve the probability distribution $\rho(r_{s/l}, t)$ numerically. In the case of the transition from the LBH state to SBH state, the reflecting and absorbing boundary conditions are located at $r_h = 0$ and $r_h = r_m$, respectively. For the transition from the SBH state to the LBH state, the reflecting boundary condition should be located at $r_h = \infty$ instead of $r_h = 0$. In Fig. 4.11, we plot the time evolution of the probability distribution of the first passage process for the charged AdS black hole with global monopole. It is clear from Fig. 4.11 that the probability distribution of both the SBH and LBH states decays quickly. Moreover, by comparing Figs. 4.11(a), 4.11(b), and 4.11(c), we can conclude that the probability distribution of the SBH state with global monopole evolves

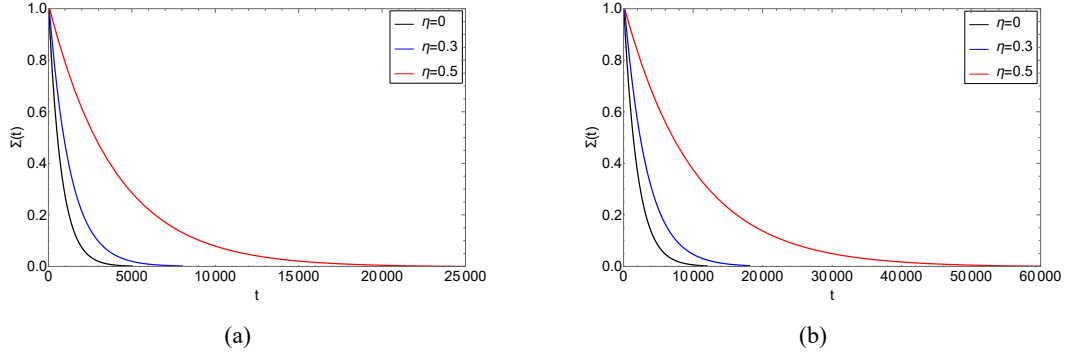


Figure 4.12: Time evolution of the probability $\Sigma(t)$. In figure (a), the Gaussian wave packet is located at the SBH state, and figure (b) denotes the LBH state.

more slowly than the one without a global monopole. And as the monopole parameter η increases, it also takes longer for the system to reach equilibrium. These effects of the global monopole on the probability distribution of the black hole state are consistent with the ones in Sec. 4.6.2 where we only consider the reflecting boundary condition.

Now, we define the probability that the present state of the black hole is still in the system at time t as follows:

$$\Sigma(t) = \int_0^{r_m} \rho(r_h, t) dr, \quad (4.54)$$

which also can be interpreted that the black hole at time t has not made the first passage process.

For the charged AdS black hole with global monopole, we plot the $\Sigma(t) - t$ diagrams in Fig. 4.12. One can see that $\Sigma(t)$ will vanish after a period of time, which means that the black hole has made the first passage process. It is clear that in the presence of the global monopole, the time evolution of the probability $\Sigma(t)$ is longer. As the global monopole parameter increases, the effect is more prominent.

Moreover, the distribution of the first passage time can be defined as^[188]

$$F_p(t) = -D \frac{\partial}{\partial r_h} \rho(r_h, t) \Big|_{r_h=r_m}. \quad (4.55)$$

We plot the $F_p(t) - t$ diagrams in Fig. 4.13. It can be found that for each η there is a single peak near the beginning time, which indicates that most of the first passage events are completed in a short time. It is clear that the impact of the global monopole on the first passage time is similar to the previous cases.

To study further the effects of the global monopole on the thermodynamic phase transition of the black hole, we calculate the mean first passage time, which is defined

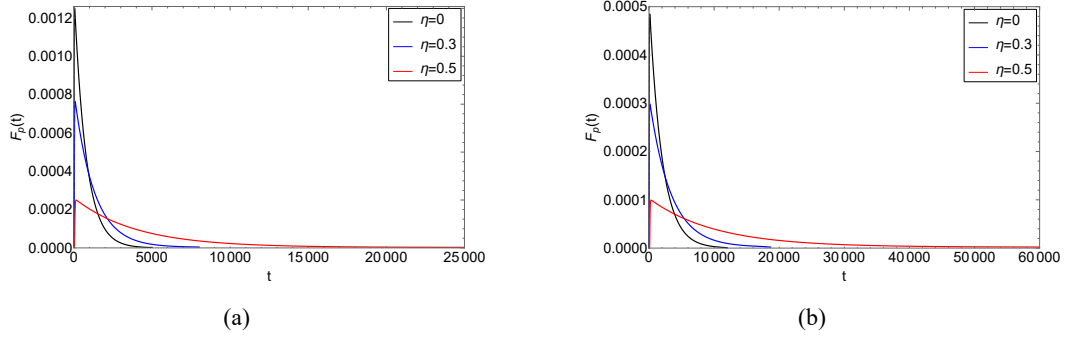


Figure 4.13: First passage time $F_p(t)$. In figure (a), the Gaussian wave packet is located at the SBH state, and figure (b) denotes the LBH state.

Table 4.1: The mean first passage time for different monopole parameters

η	0	0.1	0.2	0.3	0.4	0.5	0.6
$\langle t \rangle_{\text{SBH} \rightarrow \text{LBH}}$	774	845	958	1278	1937	3962	11290
$\langle t \rangle_{\text{LBH} \rightarrow \text{SBH}}$	1922	2086	2429	3108	5053	9933	24857

as^[188]

$$\langle t \rangle = \int_0^\infty t F_p(t) dt. \quad (4.56)$$

The values of the mean first passage time for different monopole parameters are listed in Tabel. 4.1. We have calculated seven sets of the values of the mean first passage time and the results show that the mean first passage time increases with the monopole parameter.

4.7 Chapter Summary

In this chapter, we first used the shadow formalism to analyze the thermodynamic phase transition of the Hayward-AdS black hole. The shadow radius is a monotonic function of the horizon radius and can therefore serve as an effective thermodynamic parameter. Increasing the magnetic charge enlarges the shadow radius while lowering the coexistence temperature. Thermal profiles constructed from the temperature and shadow radius show characteristic behavior below the critical temperature, supporting the use of black-hole shadows as probes of phase transitions. We then studied the charged AdS black hole with a global monopole through the Gibbs free-energy landscape and the Fokker-Planck equation. The global monopole slows the probability evolution of both the small and large black-hole states and increases the mean first passage time, demonstrating its regulatory effect on black-hole phase-transition dynamics.

Chapter 5 Conclusions and Outlook

Imagination is more important than knowledge. For knowledge is limited, whereas imagination embraces the entire world, stimulating progress and giving birth to evolution. Strictly speaking, imagination is a real factor in scientific research.

— Albert Einstein

5.1 Main Contributions and Innovations

This dissertation studies the observational appearances of black holes and wormholes in different spacetime backgrounds, holographic Einstein rings, black-hole thermodynamic phase transitions, and their underlying dynamical mechanisms. The main contributions and innovations are summarized as follows.

First, we systematically analyzed photon motion, shadow structure, and imaging features of black holes in Horndeski theory. We derived how the effective potential, photon-sphere radius, and shadow radius depend on the Horndeski parameter γ , and found that increasing γ enlarges the photon sphere and the apparent shadow size. By using EHT observations of M87* and Sgr A*, we constrained the Horndeski parameter to $-0.46 \leq \gamma \leq 0.31$ at the 1σ confidence level. We further performed ray-tracing simulations for black holes surrounded by thin accretion disks and classified photon trajectories into direct emission, lensing rings, and photon rings. The direct-emission component dominates the observed flux, while the lensing and photon rings contribute much less and are difficult to resolve after convolution with a $20 \mu\text{as}$ Gaussian filter. This study improves our understanding of Horndeski black-hole imaging and provides observational constraints on the theory.

Second, we investigated the observational appearance of asymmetric thin-shell wormholes in Horndeski theory. The wormhole connects two static spacetime regions with different mass parameters, M_1 and M_2 , with the observer located on the M_1 side. We calculated the effective potential, critical impact parameters, and photon-sphere radii for different values of γ . When the impact parameter lies between the two critical values, photons can be reflected by the M_2 region and return to the observer side, producing distinctive optical structures. For two representative thin-disk emission models, the wormhole images contain additional features absent from black-hole images, including a faint lensing band and photon-ring groups. These signatures provide possible criteria for distinguishing wormholes from black holes with high-resolution observations.

Third, using the AdS/CFT correspondence and wave optics, we studied holographic Einstein-ring imaging of AdS black holes in Horndeski theory. The size and brightness of

the holographic ring depend on the observer position, source configuration, event-horizon radius, temperature, and the Horndeski parameter γ . When the observer is located at the north pole of the AdS boundary and the source is placed at the south pole, the holographic image exhibits a bright ring related to the photon sphere, together with a central bright spot. Increasing the source frequency sharpens the ring while reducing its overall brightness, and increasing the black-hole temperature can shrink the ring and enhance the central spot. The parameter γ can distort the ring into arcs or bright spots. The agreement between wave-optics and geometric-optics results confirms the validity of the holographic imaging framework.

Fourth, in black-hole thermodynamics, we used the shadow radius as an observable parameter to study phase transitions of the Hayward-AdS black hole. The shadow radius is a monotonic function of the horizon radius and can therefore characterize the thermodynamic state. Increasing the magnetic charge enlarges the shadow radius and lowers the coexistence temperature of the small/large black-hole phase transition. By combining temperature-pressure diagrams with shadow-radius-temperature curves, we constructed thermal profiles for the Hayward-AdS black hole. Below the critical temperature, the temperature exhibits a characteristic N-shaped behavior as a function of the shadow radius, showing that the black-hole shadow can effectively encode thermodynamic phase-transition information.

Fifth, we studied the effect of a global monopole on the phase-transition dynamics of charged AdS black holes. We derived the thermodynamic quantities and phase-transition conditions, introduced the Gibbs free-energy landscape, and described the small and large black-hole phases as two local minima separated by a barrier. The Fokker-Planck equation was then used to model stochastic transitions driven by thermal fluctuations. We found that the global monopole parameter η significantly slows the transition dynamics: as η increases, the mean first passage time between small and large black-hole phases increases, indicating that the global monopole raises the free-energy barrier and enhances the stability of each phase. This result extends the study of black-hole phase transitions from equilibrium thermodynamics to nonequilibrium dynamics.

5.2 Outlook

Although this dissertation has obtained several useful results, many problems remain open.

First, one can extend the optical study of Horndeski theory to rotating black holes and dynamical wormholes, and include more complex emission models such as anisotropic radiation. Joint constraints from gravitational-wave data and EHT observations could further restrict the parameter γ and clarify whether future high-resolution images can identify

wormhole signatures.

Second, holographic imaging can be generalized to dynamical AdS black holes, rotating black holes, and more complicated source configurations. It would also be valuable to extend the framework to other modified gravity theories and develop a broader theory of holographic gravitational imaging.

Third, shadow thermodynamics can be applied to rotating Hayward-AdS black holes and to black holes in other modified gravity models. Combining shadow images with EHT data may make it possible to infer thermodynamic parameters observationally.

Fourth, the dynamical analysis of black-hole phase transitions can be extended to rotating or higher-dimensional AdS black holes. The effects of angular momentum and dimensionality on transition rates, free-energy barriers, and first passage times deserve systematic study. It would also be interesting to connect black-hole shadow features with the global monopole parameter and thus seek observational constraints on topological defects.

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Appendix

A Publications During the Doctoral Program

- [1] **Zhi Luo**, Hao Yu, Jin Li, Observational appearance and additional photon rings of the asymmetric thin-shell wormhole in Horndeski theory [J], Phys. Rev. D 109 (2024) 10, 104006.
- [2] **Zhi Luo**, Ke-Jian He, Jin Li, Holographic Einstein rings of AdS black holes in Horndeski theory [J], Chin. Phys. C 49 (2025) 3, 035108.
- [3] **Zhi Luo**, Hao Yu, Shuo Cao, Jin Li, Shadow thermodynamics of the Hayward-AdS black hole [J], Chin. Phys. C 47 (2023) 6, 065102.
- [4] **Zhi Luo**, Hao Yu, Jin Li, Effects of a global monopole on the thermodynamic phase transition of a charged AdS black hole [J], Chin. Phys. C 46 (2022) 12, 125101.
- [5] **Zhi Luo**, Jin Li, Gravitational perturbations of the Einstein-Euler-Heisenberg black hole [J], Chin. Phys. C 46 (2022) 8, 085107.
- [6] **Zhi Luo**, Jin Li, Ke-Jian He, Hao Yu, Shadows, Quasinormal Modes, and Optical Appearances of Black Holes in Horndeski Theory [J], arXiv:2406.00265, under review at Physics Letters B.
- [7] Hao Yu, Bao-Min Gu, **Zhi Luo**, Jin Li, Correspondence between energy conservation and energy-momentum tensor conservation in cosmology [J], Phys. Rev. D 105 (2022) 8, 083511.
- [8] Wan-Yuan Wu, **Zhi Luo**, Jin Li, Thermodynamic Phase Transition of AdS Black Holes in Massive Gravity on Free Energy Landscape [J], Int. J. Theor. Phys. 63 (2024) 7, 177.
- [9] Ke-Jian He, **Zhi Luo**, Sen Guo, Guo-Ping Li, Observational appearance and extra photon rings of an asymmetric thin-shell wormhole with a Bardeen profile [J], Chin. Phys. C 48 (2024) 6, 065105.
- [10] Ke-Jian He, Sen Guo, **Zhi Luo**, Guo-Ping Li, View of thermodynamic phase transition of the charged Gauss-Bonnet AdS black hole via the shadow [J], Chin. Phys. B 33 (2024) 4, 040403.

B Research Projects During the Doctoral Program

- [1] National Natural Science Foundation of China General Program, “Research on gravitational-wave data processing for compact binary coalescences using deep learning,” Grant No. 11873001, participant.
- [2] National Key Research and Development Program project, “Characteristics and signal identification of novel gravitational-wave sources such as cosmic strings and stochastic gravitational-wave backgrounds,” Grant No. 2021YFC2203004, participant.

C Academic Conferences Attended During the Doctoral Program

- [1] April 18–23, 2025, 2025 Annual Meeting and National Congress of the Chinese Physical Society Division of Gravitation and Relativistic Astrophysics; presentation titled “Shadows, Quasinormal Modes, and Optical Appearances of Black Holes in Horndeski Theory,” Kunming University of Science and Technology, Kunming.

- [2] April 19–24, 2024, 2024 Annual Meeting of the Chinese Physical Society Division of Gravitation and Relativistic Astrophysics and the 6th Galileo-Xu Guangqi International Conference; presentation titled “Observational appearance and additional photon rings of the asymmetric thin-shell wormhole in Horndeski theory,” University of South China, Hengyang.
- [3] October 13–15, 2023, 2023 Sichuan-Chongqing Workshop on Frontiers of Theoretical Physics, Southwest Jiaotong University, Chengdu.
- [4] April 21–25, 2023, 2023 Annual Meeting of the Chinese Physical Society Division of Gravitation and Relativistic Astrophysics, Chongqing University, Chongqing.

D Dissertation Dataset

Keywords		Security Level		CLC Number	
Black-hole shadow; asymmetric thin-shell wormhole; holographic Einstein ring; thermodynamic phase transition		Public		O41	
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She was still too young to know that life never gives anything for nothing, and that a price is always exacted for what fate bestows.

— Stefan Zweig

Time flows like water, and the years have passed quietly. After more than two decades of study, my long journey of education has finally reached its end. I once thought that writing a dissertation would be a relatively easy task; yet from the initial conception to the final draft, it has, almost imperceptibly, taken four or five months. Time has slipped away in silence, while every minute and every second has been filled with effort and perseverance, woven into every paragraph and every equation of this dissertation. As I sit in quiet reflection late at night and write these acknowledgements, I suddenly realize that the time for submitting my dissertation for review is already drawing near. Looking back on these years of study and research, countless memories return like a gentle breeze, still warm and vivid. Every moment of dedication and hard work has been deeply imprinted on my heart, becoming an indelible mark in my life.

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My doctoral life at Chongqing University was simple and pure. Apart from research, most of my time was spent moving between the dormitory, the office, the dining hall, and the sports field. Whenever I pause and look back, these ordinary fragments of daily life have long been engraved in my memory. In my spare time, I often ran on the sports field to strengthen my body and refresh my mind. In this era of rapidly developing information technology, I am also grateful to tools such as Doubao and ChatGPT. Like hidden sages, they quietly accompanied me and helped me solve many problems in research and study. I am likewise thankful to Genshin Impact, Naraka: Bladepoint, The Three-Body Problem, Final Fantasy VII, Bilibili, and other works and platforms related to ACG culture. They enriched my inner world and allowed me to find a pure land of freedom and dreams amid the demands of research.

Dawei, the founder of miHoYo, once said: “The world today is truly, extremely competitive. If the work you do is not something you genuinely love, you will not be able to compete with others, because there will always be a group of extraordinary people who work just as hard as you do, but at the same time love it deeply; even when they are challenged every day, they still feel happy. Such people are impossible to outcompete.” Seven years of research during my master’s and doctoral studies have made me understand more clearly that the path of scientific research may not be the one best suited to me. Nevertheless, every challenge and every trial in these years has become an irreplaceable treasure, shaping me into a more resilient person. Although my future work may no longer be closely connected with doctoral research, the modes of thinking and the rigorous attitude I have developed along the academic path will continue to accompany me and help me go further. I am grateful to everyone who has accompanied me on this journey, to the version of myself that persisted through every difficulty, and to the courage and strength that never allowed me to give up.

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Night of April 9, 2025 Written in Room EB408, Juyuan Doctoral Dormitory, Huxi
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